



David M. Himmelblau / James B. Riggs

Basic Principles and Calculations in Chemical Engineering

Fifth Edition

Prentice Hall International Series
in the Physical and Chemical
Engineering Sciences



HEAT, ENERGY, OR WORK EQUIVALENTS

(Btu)	kWh	hp-hr	Btu	calorie*	Joule
0.7376	2.778×10^{-4}	3.725×10^{-4}	9.484×10^{-4}	0.2390	1
723.3	2.724×10^{-3}	3.653×10^{-3}	9.296×10^{-3}	2.2438	9.80665
1	3.766×10^{-4}	5.0505×10^{-4}	1.285×10^{-3}	0.3241	1.356
2.655×10^3	1	1.341	3.4128×10^3	8.6057×10^3	3.6×10^3
1.98×10^3	0.7455	1	2.545×10^3	6.4162×10^3	2.6845×10^3
74.73	2.815×10^{-2}	3.774×10^{-2}	9.604×10^{-2}	24.213	1.0133×10^2
3.086×10^3	1.162×10^{-3}	1.558×10^{-3}	3.9657	1×10^3	4.184×10^3
778.16×10^3	2.930×10^{-4}	3.930×10^{-4}	1	2.52×10^3	1.055×10^3
3.086	1.162×10^{-3}	1.558×10^{-3}	3.97×10^{-3}		4.184

*The thermochemical calorie = 4.184 J; the IT calorie = 4.1867 J (see Sec. 4.1).

PRESSURE EQUIVALENTS

mm Hg	in. Hg	bar	atm	kPa
1	3.937×10^{-2}	1.333×10^{-3}	1.316×10^{-3}	0.1333
25.40	1	3.387×10^{-1}	3.342×10^{-2}	3.387
750.06	29.53	1	0.9869	100.0
760.0	29.92	1.013	1	101.3
7.502	0.2954	1.000×10^{-2}	9.872×10^{-3}	1

IDEAL GAS CONSTANT R

1.987 cal/(g mol)(K)
1.987 Btu/(lb mol)(°R)
10.73 (psia)(ft ³)/(lb mol)(°R)
8.314 (kPa)(m ³)/(kg mol)(K) = 8.314 J/(g mol)(K)
82.06 (cm ³)(atm)/(g mol)(K)
0.08206 L (atm)/(g mol)(K)
21.9 (in Hg)(ft ³)/(lb mol)(°R)
0.7302 (ft ³)(atm)/(lb mol)(°R)

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FIFTH EDITION

BASIC PRINCIPLES AND CALCULATIONS IN CHEMICAL ENGINEERING

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PREFACE

PURPOSE OF THE BOOK

This book is intended to serve as an introduction to the principles and techniques used in the field of chemical, petroleum, and environmental engineering. It lays a foundation of certain information and skills that can be repeatedly employed in subsequent courses as well as in professional life.

A good introductory book to chemical engineering principles and calculations should (1) explain the fundamental concepts in not too stilted language together with generous use of appropriate equations and diagrams; (2) provide sufficient examples with detailed solutions to clearly illustrate (1); (3) present ideas in small packages that are easily identified as part of a larger framework; (4) include tests and answers that enable the reader to evaluate his or her accomplishments; and (5) provide the instructor with a wide selection of problems and questions to evaluate student competence. All of these features have been built into the fifth edition.

I kept in mind four major objectives for a reader in preparing this fifth edition:

1. to **develop systematic problem solving skills**, enhance confidence, and generate careful work habits;
2. to **learn what material balances are** and how to apply them;
3. to **learn what energy balances are** and how to apply them;
4. to learn how to **deal with the complexity of big problems**.

In addition to accomplishing these goals, a reader is exposed to background information on units and measurements of **physical properties**, basic laws about the **behavior of gas, liquids, and solids**, and some **basic mathematical tools**. Although communication and programming skills are essential for both students and professionals, assimilation of material in this book does not require mastery of either.

Other objectives that an instructor may want to include in a course, such as information about professional activities, developing a professional attitude, establish-

ing personal goals, developing social awareness, and so on must be implemented by the instructor from other sources. Economic feasibility, a major factor in engineering decision making, costing, and optimization have been omitted because of lack of space. Nor has there been room for developing an appreciation of what processing equipment really looks like and how it works.

If this book is used as part of a scheduled course, the role of the teacher must be something more than just communicating the subject matter. The job of the teacher is to arouse emotional reactions of feeling good in connection with the content being conveyed. Creating positive feelings so that a student enjoys the subject makes a teacher effective.

SCOPE AND PARTS OF THE BOOK

The central themes of the book involve (1) learning how to formulate and solve (a) material balances, (b) energy balances, and (c) both simultaneously; (2) developing problem solving skills; and (3) becoming familiar with the use of units, physical properties, and the behavior of gases and liquids. At the front of each chapter is an information flow diagram that shows how the topics discussed in the chapter relate to the objective of being able to solve successfully problems involving material and energy balances. Figure P.1 shows the information flow among the chapters.

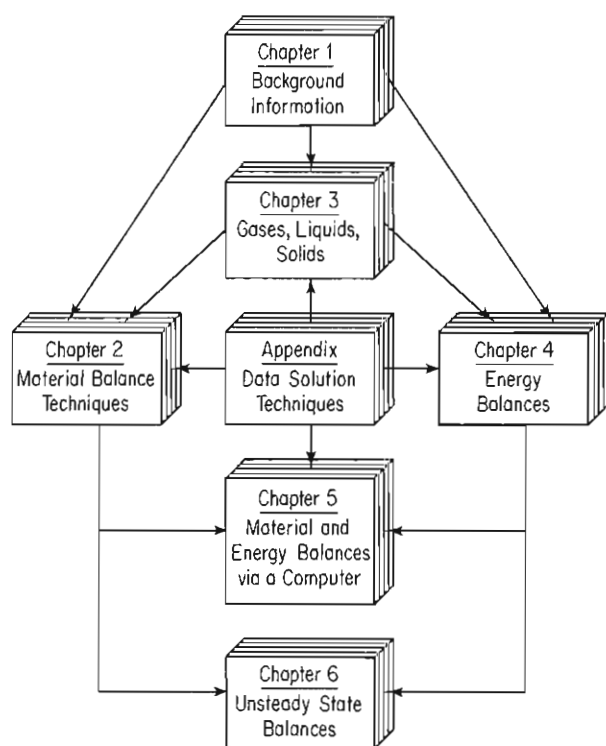


Figure P.1 The relationship of the chapters in this book.

GENERAL FEATURES OF THE BOOK

I have selected, arranged, and presented the material in this book with care based on past teaching experience. Some major features common to all the chapters are:

- The book is **self-contained** except for some homework problems that deliberately require outside information.
- The presentation is **detailed** enough so that reference to other books can be omitted.
- The **examples are simple** and concrete to make the book teachable and useful for self-instruction.
- The chapters are largely **independent** providing flexibility in teaching.
- The book has been reviewed for **readability**.
- The examples and homework problems support **good learning principles**.
- **Numerous illustrations** enhance learning.
- **Subheadings** clearly distinguish successive topics.
- **Thought problems** have been included for class discussion.
- A **table of contents** is listed **at the beginning of each chapter** to show the contents of the chapter.
- **Vital words and concepts** are in boldfaced type.
- What must be **memorized** is identified.
- At the end of each chapter **references** and numerous **supplementary references** are included.
- **Solutions** to about one-quarter of the **problems** in the problem sets are in the Appendix.
- **Data** for solving problems has been provided in tables and figures.

At the beginning of each section is a **list of objectives** to be achieved by the reader stated in such a way that attainment can be readily measured. Whether the objectives should go at the beginning or end of the section is a matter of controversy; that the text should have objectives is not. But we often present our objectives by such broad, fuzzy statements that neither the student nor the teacher can ascertain whether or not students have achieved them. (Unfortunately, this situation does not seem to inhibit the testing of students.) Each set of objectives is quite concrete and has a corresponding set of self-assessment questions and problems at the end of the respective section.

Piaget has argued that human intelligence proceeds in stages from the concrete to the abstract and that one of the big problems in teaching is that the teachers are formal reasoners (using abstraction) while many students are still concrete thinkers or at best in transition to formal operational thinking. I believe that this is true. Consequently, most topics are initiated with simple illustrations that illustrate the basic ideas. In this book the **topics are presented in order of easy assimilation** rather than in a strictly logical order. The organization is such that easy material is alternated with difficult material in order to give a “breather” after passing over each

hump. For example, discussion of unsteady-state (lumped) balances has been deferred until the final chapter because experience has shown that most students lack the mathematical and engineering maturity to absorb these problems simultaneously with the steady-state balances.

A principle of educational psychology is to reinforce the learning experience by providing detailed guided practice following each new principle. We all have found from experience that there is a vast difference between having a student understand a principle and in establishing his or her ability to apply it. For example, can you learn how to play the piano from a series of lectures? By the use of numerous detailed **examples** following each brief section of text, it is hoped that straightforward, orderly methods of procedure can be instilled along with some insight into the principles involved. Furthermore, the wide variety of **problems** at the end of each chapter, about one-fourth of which are accompanied by **answers**, offer practice in the application of the principles explained in the chapter.

After all these years a perplexing problem still remains for an author in preparing a new edition, namely, the extent to which SI units should be used. It is evident that the transition to SI is well under way in a large number of chemical engineering departments in the United States, although some resistance to the system still exists. I believe that SI is an important system of measurement that chemical engineers must be able to deal with, but feel that chemical engineering students must be familiar with a variety of systems for some years to come. Consequently, it is premature to convert an undergraduate text entirely to the SI system of units even though most journals have done so. As a compromise, a little more than one-half of the text, examples, and problems and most of the tables employ SI units. For convenience, some of the crucial tables, such as the steam tables, are presented in both American engineering and SI units.

Self-assessment tests have been included to provide readers with questions and answers that assist in them in appraising and developing their knowledge about a particular topic. Self-assessment is intended to be an educational experience for a student. The availability of answers to the self-assessment questions together with supplementary reading citations for further study is an inherent characteristic of self-assessment. To help the reader think about the concepts and decide whether to study further is one reason for having appraisal questions.

How should you as a reader use the self-assessment problems? Some readers will start with the questions. Others will read the problems. Still others will refer to the answers first. These approaches and others are all acceptable if at the end of the procedure you can say: "Yes, this has been a worthwhile experience" or "I have learned something." The self-assessment materials are not a timed exercise so work at your pace. I suggest that you first go through the questions and problems, and write down the responses you think are most appropriate for the questions that are easy for you. Compare your responses with the answers in Appendix A. In those cases where you differ, read the section again, examine the examples, try some of the homework problems at the end of the chapter, or look up the references if necessary. In those cases in which you agree, but you felt uncomfortable with the subject matter, and the subject is significant to you, be sure to solve some of the problems at the end of the chapter for which answers are provided. After doing the easy problems, tackle the more difficult ones.

Let me now mention some of the new features of the fifth edition, features that were not present in earlier editions.

NEW FEATURES IN THE BOOK

For the fifth edition I have added a number of new features (and deleted some old ones) described below that make both teaching and self-study easier.

In this edition special attention has been devoted to presenting a **sound strategy for solving material balance and energy balance** problems, one that can be used again and again as a framework for solving word problems. All the examples showing how to solve material and energy balances have been reformulated according to this strategy (see Table 2.1). In teaching I ask students to memorize the strategy and apply it in all their homework problems and exams. I discourage the use of self-devised heuristic algorithms, or "cookbook" methods pointing out that they may be successful for one class of problems but fail quite dismally for others. By this means a student is guided into forming generalized patterns of attack in problem solving that can be used successfully in connection with unfamiliar types of problems. The text is designed to acquaint the student with a sufficient number of fundamental concepts so that he or she can (1) continue with his or her training, and (2) start finding solutions to new types of problems on his or her own. It offers practice in finding out what the problem is, defining it, collecting data, analyzing and breaking down information, assembling the basic ideas into patterns, and, in effect, doing everything but testing the solution experimentally.

A major problem in any book is to what extent and in what manner should problems involving the preparation of or use of computer codes be introduced into the text. If computer techniques are to be integrated into the classroom successfully, it is wise to start early in the game, but on the other hand the preparation (as opposed to use) of computer programs assumes that the student will have had some instruction and competence in programming prior to or concurrently with, the use of this book. The selection of appropriate problems and the illustration of good computer habits, pointing out instances in which computer solutions are not appropriate as well as instances when they are, are important. A compromise has been reached. At the end of each chapter you will find a few problems calling for the **preparation of a complete computer code**. Other **problems** that are more appropriately **solved with the aid of a computer program**, particularly a library code, than with a calculator, are designated by an asterisk (*) after the problem number. I have found that the use of prepared computer programs reduces the tedium of trial-and-error solution techniques and regenerates interest in the learning process.

Simple FORTRAN programs have been prepared for the reader's use that solve linear and nonlinear equations, retrieve the properties of water and steam, and of air-water mixtures, calculate the vapor pressure of pure substances, calculate enthalpy changes from heat capacity equations, and so on. **A disk containing these codes will be found in a pocket in the back of the book.** (Readers are encouraged to use library codes when available, codes that may be more accurate and robust than the simple codes provided.) As a result, the portions of the book formerly treating

graphical integration, trial and error solutions, lever arm principles, and graphical solution methods has been drastically reduced.

Other new features are:

1. **Thought problems have been added** at the end of sections to challenge student problem solving skills and provide a basis for class discussion.
2. **Chapter 2 has been reorganized** to make it more understandable and teachable by separating the treatment of single subsystems and multiple subsystems, substantially reducing the emphasis on tie elements, and by employing a uniform strategy to solve all of the examples. Emphasis is now placed on **solving uncoupled vs. coupled material balance equations**.
3. Chapter 5 on the simultaneous solution of material and energy balance problems has been completely revised. It now focuses on **the use of flowsheeting codes to solve problems**. The problem set for Chapter 5 emphasizes the use of FLOWTRAN and PROCESS for complex problems, codes that have clear manuals and are available at most departments of chemical engineering.
4. In view of the interest in chemical engineering to expand the horizons of chemical engineering practice, a number of homework problems have been included in the areas of **biotechnology and solid state materials processing**.
5. A brief explanation of the **techniques for the solution of linear and nonlinear equations** has been added to the appendix.
6. An explanation of **least squares for data fitting** has been added to the appendix.
7. A discussion of the **accuracy and precision of data** has been added to Chapter 1.
8. How to use **spreadsheets and equation solving codes** for personal computers has been added to Chapter 2.
9. **Problem sets** have been changed and expanded.
10. The number of **worked-out examples** has been increased.

SUGGESTIONS FOR A COURSE

This book can be used in a variety of learning environments besides the traditional lectures, such as self-paced instruction, group study or discussion groups, and individual study. More topics have been included in the text than can be covered in one semester, so that an instructor has some choice as to pace of instruction and topics to include. In a lecture course for students with a background of having completed only freshman courses and no programming experience, Chapters 1, 2, 3, and 4 form the basis of an ample course. For more experienced students, if computer flowsheeting programs are available, perhaps Chapter 1 can be skimmed over and Chapter 5 included. Several case studies that are computer oriented are available to accompany Chapter 5 from various organizations and in publications such as the FLOWTRAN workbook. Chapter 6 will probably take at least one week (and probably two weeks) of time if you want to include it as well.

ACKNOWLEDGMENTS

I am indebted to many of my former teachers, colleagues and students who directly or indirectly helped me in preparing this book, in particular the present edition of it. Special thanks go to Kai Petersen and Keith Reinhardt for preparing a number of new homework problems. Professor Donald Woods was most helpful in providing information about sound techniques of problem solving. Far too many instructors using the text have contributed their corrections and suggestions for me to list them all by name. However, I do wish to express my appreciation for their kind assistance. Any further comments and suggestions for improvement of the book would be gratefully received.

David M. Himmelblau
Austin, Texas

INTRODUCTION TO ENGINEERING CALCULATIONS

1

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1.3	Conventions in Methods of Analysis and Measurement	22
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The chemical engineering profession encompasses a wide variety of activities and is engaged in resolving problems that occur in industry, government, and academia. Problems encountered by chemical engineers are found in design, operation, control, troubleshooting, research, and even politics, the latter because of environmental and economic concerns. Chemical engineers work in numerous areas besides petroleum refining and the chemical and petrochemical industries because their background and experience are easily portable and found useful. You will find chemical engineers solving problems in industries such as

- Drugs and pharmaceuticals
- Microelectronics
- Biotechnology
- Explosives and fireworks
- Fats and oils
- Fertilizer and agricultural chemicals
- Foods and beverages
- Leather tanning and finishing
- Lime and cement
- Man-made fibers
- Metallurgical and metal products
- Paints, varnishes, and pigments
- Pesticides and herbicides
- Plastic materials, synthetic resins

Rubber products
 Soap and toiletries
 Solid-state materials
 Stone, clay, glass, and ceramics
 Wood, pulp, paper, and board

Table 1.0 gives you some idea of the vast industrial capacity in the United States in which chemical engineers participate.

For you to learn how to appreciate and treat the problems that will arise in our modern technology, especially in the technology of the future, it is necessary to learn certain basic principles and practice their application. This text describes the principles of making material and energy balances and illustrates their application in a wide variety of ways.

We begin in this chapter with a review of certain background information. You have already encountered most of the concepts to be presented in basic chemistry and physics courses. Why, then, the need for a review? First, from experience we

TABLE 1.0 U.S. Production in 1986

Chemicals (in 10 ⁹ lb)		Fertilizers (in 10 ⁶ tons)	
Sulfuric acid	73.64	Nitrogen	10.4
Nitrogen	48.62	Phosphate (P ₂ O ₅)	8.6
Oxygen	33.03	Potash (K ₂ O)	1.0
Ethylene	32.81		
Lime	30.34	Pesticides and herbicides (in 10 ⁶ lb)	
Ammonia	28.01	Herbicides	756
Sodium hydroxide	22.01	Insecticides	370
Chlorine	20.98	Fungicides	109
Phosphoric acid	18.41		
Propylene	17.34	Plastics (in 10 ⁶ lb)	
Sodium carbonate	17.20	Thermosetting resins	5.8
Ethylene dichloride	14.53	(phenols, urea, polyesters, epoxies)	
Nitric acid	13.12	Thermoplastic resins	33.6
Urea	12.06	(polyethylene, polystyrene, polypropylene, copolymers)	
Ammonium nitrate	11.11		
Benzene	10.23	Synthetic fibers	
Ethylbenzene	8.92	Cellulosics (rayon, acetate)	0.6
Carbon dioxide	8.50	Polyester	3.3
Vinyl chloride	8.42	Nylon	2.5
Styrene	7.84	Olefinic and acrylic	3.0
Minerals (in 10 ⁶ tons)			
Phosphate rock	1.2		
Salt	36.8		
Sulfur	12.2		
Synthetic rubber (in metric tons)	4380		

have found it necessary to restate these familiar basic concepts in a somewhat more general and clearer fashion; second, you will need practice to develop your ability to analyze and work engineering problems. To read and understand the principles discussed in this chapter is relatively easy; to apply them to different unfamiliar situations is not. An engineer becomes competent in his or her profession by mastering the techniques developed by one's predecessors—thereafter comes the time to pioneer new ones.

The chapter begins with a discussion of units, dimensions, and conversion factors, and then goes on to review some terms you should already be acquainted with, such as:

- (a) Mole and mole fraction
- (b) Density and specific gravity
- (c) Measures of concentration
- (d) Temperature
- (e) Pressure

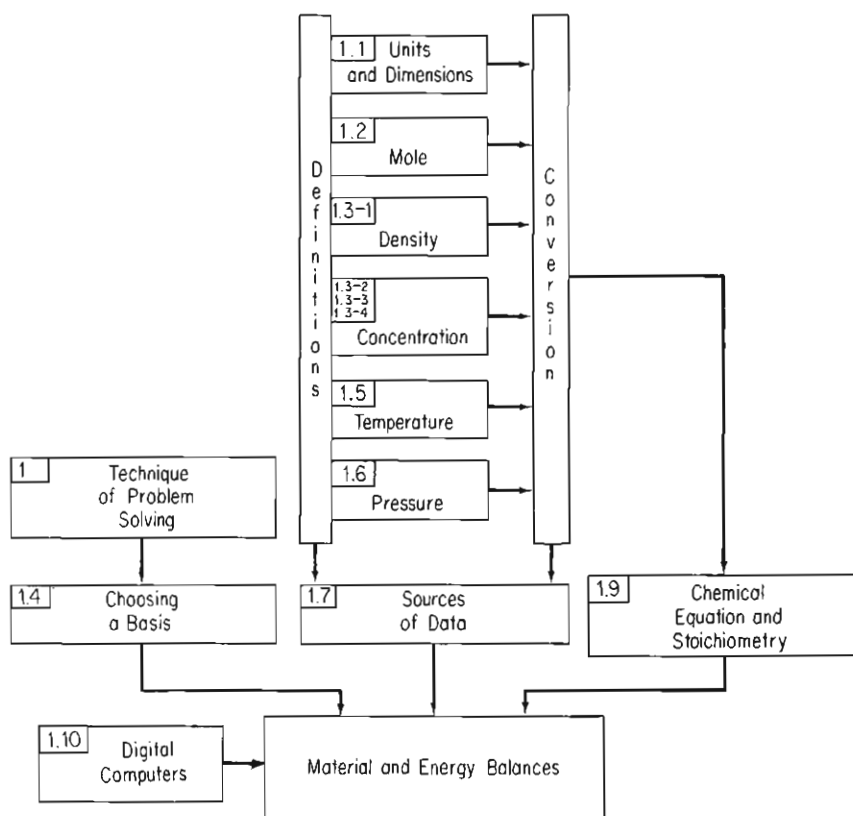


Figure 1.0 Hierarchy of topics to be studied in this chapter (section numbers are in the upper left-hand corner of the boxes).

It then provides some clues on “how to solve problems,” which should be of material aid in all the remaining portions of your career. Finally, the principles of stoichiometry are reviewed, and the technique of handling incomplete reactions is illustrated. Figure 1.0 shows the relation of the topics to be discussed to each other and to the ultimate goal of being able to solve problems involving both material and energy balances.

1.1 UNITS AND DIMENSIONS

Your objectives in studying this section are to be able to:

1. Add, subtract, multiply, and divide units associated with numbers.
2. Convert one set of units in a function or equation into another equivalent set for mass, length, area, volume, time, energy, and force.
3. Specify the basic and derived units in the SI and American engineering systems for mass, length, volume, density, and time, and their equivalences.
4. Explain the difference between weight and mass.
5. Define and know how to use the gravitational conversion factor g_c .
6. Apply the concepts of dimensional consistency to determine the units of any term in a function.

“Take care of your units and they will take care of you.”

At some time in every student's life comes the exasperating sensation of frustration in problem solving. Somehow, the answers or the calculations do not come out as expected. Often this outcome arises because of inexperience in the handling of units. The use of units or dimensions along with the numbers in your calculations requires more attention than you probably have been giving to your computations in the past, but paying attention will help avoid such annoying experiences. The proper use of dimensions in problem solving is not only sound from a logical viewpoint—it will also be helpful in guiding you along an appropriate path of analysis from what is at hand through what has to be done to the final solution.

1.1-1 Units and Dimensions

Dimensions are our basic concepts of measurement such as *length, time, mass, temperature*, and so on; **units** are the means of expressing the dimensions, such as *feet or centimeters* for length, or *hours or seconds* for time. Units are associated with some quantities you may have previously considered to be dimensionless. A good example is *molecular weight*, which is really the mass of one substance per mole of

that substance. This method of attaching units to all numbers which are not fundamentally dimensionless has the following very practical benefits:

- (a) It diminishes the possibility of inadvertent inversion of any portion of the calculation.
- (b) It reduces the calculation in many cases to simple ratios, which can be easily manipulated on a hand-held calculator.
- (c) It reduces the intermediate calculations and eliminates considerable time in problem solving.
- (d) It enables you to approach the problem logically rather than by remembering a formula and plugging numbers into the formula.
- (e) It demonstrates the physical meaning of the numbers you use.

Every freshman knows that what you get from adding apples to oranges is fruit salad! The rule for handling units is essentially quite simple: **treat the units as you would algebraic symbols. You can add, subtract, or equate numerical quantities only if the units of the quantities are the same.** Thus the operation

$$5 \text{ kilograms} + 3 \text{ joules}$$

is meaningless because the dimensions of the two terms are different. The numerical operation

$$10 \text{ pounds} + 5 \text{ grams}$$

can be performed (because the dimensions are the same, mass) *only* after the units are transformed to be the same, either pounds, or grams, or ounces, and so on. **In multiplication and division, you can multiply or divide different units, such as (10 centimeters ÷ 4 seconds) = 2.5 centimeters/second, but you cannot cancel them out unless they are the same.** For example, $3 \text{ m}^2/60 \text{ cm}$ must first be converted to $3 \text{ m}^2/0.6 \text{ m}$ and then to 5 m . The units contain a significant amount of information content that cannot be ignored. They also serve as guides in efficient problem solving, as you will see shortly.

EXAMPLE 1.1 Dimensions and Units

Add the following:

- (a) $1 \text{ foot} + 3 \text{ seconds}$
- (b) $1 \text{ horsepower} + 300 \text{ watts}$

Solution

The operation indicated by

$$1 \text{ ft} + 3 \text{ s}$$

has no meaning since the dimensions of the two terms are not the same. One foot has the dimensions of length, whereas 3 seconds has the dimensions of time. In the case of

$$1 \text{ hp} + 300 \text{ watts}$$

the dimensions are the same (energy per unit time) but the units are different. You must transform the two quantities into like units, such as horsepower, watts, or something else, before the addition can be carried out. Since $1 \text{ hp} = 746 \text{ watts}$,

$$746 \text{ watts} + 300 \text{ watts} = 1046 \text{ watts}$$

1.1-2 Conversion of Units and Conversion Factors

In this book, to help you follow the calculations, we will frequently make use of what is called the *dimensional equation*. It contains both units and numbers. One quantity is multiplied by a number of ratios termed **conversion factors** of equivalent values of combinations of time, distance, and so on, to arrive at the final desired answer. The ratios used are simple well-known values and thus the conversion itself should present no great problem. Examine Example 1.2.

EXAMPLE 1.2 Conversion of Units

If a plane travels at twice the speed of sound (assume that the speed of sound is 1100 ft/s), how fast is it going in miles per hour?

Solution

$$2 \left| \begin{array}{c|c} 1100 \text{ ft} & 1 \text{ mi} \\ \hline & 5280 \text{ ft} \end{array} \right| \begin{array}{c|c} 60 \text{ s} & 1 \text{ min} \\ \hline & 60 \text{ min} \\ & 1 \text{ hr} \end{array} = 1500 \frac{\text{mi}}{\text{hr}}$$

or

$$2 \left| \begin{array}{c|c} 1100 \text{ ft} & 60 \frac{\text{mi}}{\text{hr}} \\ \hline & 88 \frac{\text{ft}}{\text{s}} \end{array} \right| = 1500 \frac{\text{mi}}{\text{hr}}$$

Of course, it is possible to look up conversion ratios, which will enable the length of the calculation to be reduced; for instance, in Example 1.2 we could have used the conversion factor of 60 mi/hr equals 88 ft/s. However, it usually takes less time to use values you know than to look up shortcut conversion factors in a handbook. Common conversion ratios are listed on the inside front cover.

We have set up the dimensional equation with vertical lines to separate each ratio, and these lines retain the same meaning as an $\times$ or multiplication sign placed between each ratio. The dimensional equation will be retained in this form throughout most of this text to enable you to keep clearly in mind the significance of units in problem solving. It is recommended that you always write down the units next to the associated numerical value (unless the calculation is very simple) until you become quite familiar with the use of units and dimensions and can carry them in your head.

At any point in the dimensional equation you can determine the consolidated net units and see what conversions are still required. This may be carried out formally, as shown below by drawing slanted lines below the dimensional equation and

writing the consolidated units on these lines, or it may be done by eye, mentally canceling and accumulating the units, or, you can strike out pairs of units as you proceed:

$$\begin{array}{c|c|c|c}
 2 \times 1100 \text{ ft} & 1 \text{ mi} & 60 \text{ sec} & 60 \text{ min} \\
 \hline
 \text{s} & 5280 \text{ ft} & 1 \text{ min} & 1 \text{ hr} \\
 \hline
 & \frac{\text{ft}}{\text{s}} & \frac{\text{mi}}{\text{s}} & \frac{\text{mi}}{\text{min}}
 \end{array}$$

Consistent use of dimensional equations throughout your professional career will assist you in avoiding silly mistakes such as converting 10 centimeters to inches by multiplying by 2.54:

$$\frac{10 \text{ cm}}{1 \text{ in.}} \times 2.54 \frac{\text{cm}}{\text{in.}} \neq 25.4 \text{ cm} \quad \text{but instead} \quad = 25.4 \frac{\text{cm}^2}{\text{in.}}$$

Note how easily you discover that a blunder has occurred by including the units in the calculations.

Here is another example of the conversion of units.

EXAMPLE 1.3 Use of Units

Change $400 \text{ in.}^3/\text{day}$ to cm^3/min .

Solution

$$\frac{400 \text{ in.}^3}{\text{day}} \times \left(\frac{2.54 \text{ cm}}{1 \text{ in.}} \right)^3 \times \frac{1 \text{ day}}{24 \text{ hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 4.56 \frac{\text{cm}^3}{\text{min}}$$

In this example note that not only are the numbers raised to a power, but the units also are raised to the same power.

There shall be one measure of wine throughout our kingdom, and one measure of ale, and one measure of grain . . . and one breadth of cloth. . . . And of weights it shall be as of measures.

So reads the standard measures clause of the *Magna Carta* (June 1215). The standards mentioned were not substantially revised until the nineteenth century. When the American colonies separated from England, they retained, among other things, the weights and measures then in use. It is probable that at that time these were the most firmly established and widely used weights and measures in the world.

No such uniformity of weights and measures existed on the European continent. Weights and measures differed not only from country to country but even from town to town and from one trade to another. This lack of uniformity led the National Assembly of France during the French Revolution to enact a decree (May 8, 1790) that called upon the French Academy of Sciences to act in concert with the Royal Society of London to "deduce an invariable standard for all of the measures and all weights." Having already an adequate system of weights and measures, the English

did not participate in the French undertaking. The result of the French endeavor has evolved into what is known as the metric system.

Certainly in the American-British system there is ample room for confusion. The old question: "Which is heavier, a pound of feathers or a pound of gold?" indeed has a catch, one that is not obvious. The pound of feathers is heavier, because gold is measured in troy units in which $1 \text{ oz} = 3.110 \times 10^{-2} \text{ kg}$ and the feathers are measured in avoirdupois units, in which $1 \text{ oz} = 2.834 \times 10^{-2} \text{ kg}$. But the troy pound comprises only 12 rather than 16 oz!

The metric system became preferred by scientists of the nineteenth century, partly because it was intended to be an international system of measurement, partly because the units of measurement were theoretically supposed to be independently reproducible, and partly because of the simplicity of its decimal nature. Problems in the specification of the units for electricity and magnetism led to numerous international conferences to rectify inconsistencies and culminated in 1960 in the eleventh General Conference on Weights and Measures adopting the SI (Système International) system¹ of units. As of this date the United States is the last large country not employing or engaged in transforming to some form of the SI units. Tables 1.1 and 1.2 show the most common systems of units used by engineers in the last few decades. Note that the SI, the cgs, the fps (English absolute), and the British engineering systems all have three basically defined units and that the fourth unit is derived from these three defined units.

Many of the derived units in the SI system are given special names (see Table 1.2) honoring physicists and have corresponding symbols. The unit of force, for example, has the symbols $\text{kg} \cdot \text{m} \cdot \text{s}^{-2}$; for convenience, this combination has been given the name *newton* and the symbol N. Similarly, the unit of energy is the newton-meter,

$$\text{N} \cdot \text{m} \approx \text{m}^2 \cdot \text{kg} \cdot \text{s}^{-2}$$

with the short name *joule*, and symbol J. The unit of power is the *watt*, defined as one joule per second.

Only the American engineering system has four basically defined units. Consequently, in the American engineering system you have to use a conversion factor, g_c , a constant whose numerical value is not unity, to make the units come out properly. We can use Newton's law to see what the situation is with regard to conversion of units:

$$F = Cma \quad (1.1)$$

where² F = force

C = a constant whose numerical value and units depend on those selected for F , m , and a

m = mass

a = acceleration

¹ Note that the word "system" used in connection with SI units is redundant but commonly employed in English.

² A list of the nomenclature is included at the end of the book.

TABLE 1.1 Common Systems of Units

	Length	Time	Mass	Force	Energy*	Temperature	Remarks
<i>Absolute (dynamic) systems</i>							
Cgs	centimeter	second	gram	dyne*	erg, joule, or calorie	K, °C	Formerly common scientific
Fps (ft-lb-s or English absolute)	foot	second	pound	poundal*	ft poundal	°R, °F	
SI	meter	second	kilogram	newton*	joule	K, °C	Internationally adopted units for ordinary and scientific use
<i>Gravitational systems</i>							
British engineering	foot	second	slug*	pound weight	Btu(ft)(lb _r)	°R, °F	
American engineering	foot	second, hour	pound mass (lb _m)	pound force (lb _r)	Btu or (hp)(hr)	°R, °F	Used by chemical and petroleum engineers in the United States

*Unit derived from basic units; all energy units are derived.

TABLE 1.2 SI Units Encountered in this Book

Basic SI units			
Physical quantity	Name of unit	Symbol for unit*	
Length	metre, meter	m	
Mass	kilogramme, kilogram	kg	
Time	second	s	
Thermodynamic temperature	kelvin	K	
Amount of substance	mole	mol	
Derived SI units			
Physical quantity	Name of unit	Symbol for unit*	Definition of unit
Energy	joule	J	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$
Force	newton	N	$\text{kg} \cdot \text{m} \cdot \text{s}^{-2} \approx \text{J} \cdot \text{m}^{-1}$
Power	watt	W	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \approx \text{J} \cdot \text{s}^{-1}$
Frequency	hertz	Hz	cycle/s
Area	square meter		m^2
Volume	cubic meter		m^3
Density	kilogram per cubic meter		$\text{kg} \cdot \text{m}^{-3}$
Velocity	meter per second		$\text{m} \cdot \text{s}^{-1}$
Angular velocity	radian per second		$\text{rad} \cdot \text{s}^{-1}$
Acceleration	meter per second squared		$\text{m} \cdot \text{s}^{-2}$
Pressure	newton per square meter, pascal		$\text{N} \cdot \text{m}^{-2}$, Pa
Specific heat	joule per (kilogram · kelvin)		$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$
Alternative units			
Physical quantity	Allowable unit	Symbol for unit*	
Time	minute	min	
	hour	h	
	day	d	
	year	a	
Temperature	degree Celsius	°C	
Volume	litre, liter (dm^3)	L	
Mass	tonne, ton (Mg)	t	
	gram	g	
Pressure	bar (10^5Pa)	bar	

*Symbols for units do not take a plural form, but plural forms are used for the unabbreviated names.

In the cgs system the unit of force is defined as the dyne; hence if C is selected to be $C = 1 \text{ dyne}/(\text{g})(\text{cm})/\text{s}^2$, then when 1 g is accelerated at $1 \text{ cm}/\text{s}^2$

$$F = \frac{1 \text{ dyne}}{(\text{g})(\text{cm})/\text{s}^2} \left| \frac{1 \text{ g}}{1 \text{ g}} \right| \left| \frac{1 \text{ cm}}{\text{s}^2} \right| = 1 \text{ dyne}$$

Similarly, in the SI system in which the unit of force is defined to be the newton (N), if $C = 1 \text{ N}/(\text{kg})(\text{m})/\text{s}^2$, then when 1 kg is accelerated at $1 \text{ m}/\text{s}^2$,

$$F = \frac{1 \text{ N}}{(\text{kg})(\text{m})/\text{s}^2} \left| \frac{1 \text{ kg}}{1 \text{ kg}} \right| \left| \frac{1 \text{ m}}{\text{s}^2} \right| = 1 \text{ N}$$

However, in the American engineering system we ask that the numerical value of the force and the mass be essentially the same at the earth's surface. Hence, if a mass of 1 lb_m is accelerated at $g \text{ ft}/\text{s}^2$, where g is the acceleration of gravity (about $32.2 \text{ ft}/\text{s}^2$ depending on the location of the mass), we can make the force be 1 lb_f by choosing the proper numerical value and units for C :

$$F = (C) \frac{1 \text{ lb}_m}{(\text{lb}_m)(\text{ft})/\text{s}^2} \left| \frac{g \text{ ft}}{\text{s}^2} \right| = 1 \text{ lb}_f \quad (1.2)$$

Observe that for Eq. (1.2) to hold, the units of C have to be

$$C \approx \frac{\text{lb}_f}{\text{lb}_m \left(\frac{\text{ft}}{\text{s}^2} \right)}$$

A numerical value of $1/32.174$ has been chosen for the constant because 32.174 is the numerical value of the average acceleration of gravity (g) at sea level at 45° latitude when g is expressed in ft/s^2 . The acceleration of gravity, you may recall, varies a few tenths of 1% from place to place on the surface of the earth and changes considerably as you rise up from the surface as in a rocket. With this selection of units and with the number 32.174 employed in the denominator of the conversion factor,

$$F = \left(\frac{1 (\text{lb}_f)(\text{s}^2)}{32.174 (\text{lb}_m)(\text{ft})} \right) \left(\frac{1 \text{ lb}_m}{1 \text{ lb}_m} \right) \left(\frac{g \text{ ft}}{\text{s}^2} \right) = 1 \text{ lb}_f$$

The inverse of C is given the special symbol g_c :

$$g_c = 32.174 \frac{(\text{ft})(\text{lb}_m)}{(\text{s}^2)(\text{lb}_f)} \quad (1.3)$$

Division by g_c achieves exactly the same result as multiplication by C in Newton's law. You can see, therefore, that in the American engineering system we have the convenience that the numerical value of a pound mass is also that of a pound force if the numerical value of the ratio g/g_c is equal to 1, as it is approximately in most cases. A nonstandard usage sometimes encountered analogous to lb_f and lb_m is kg_f

and kg_m ; a conversion factor not equal to unity is involved in the transformation by an equation similar to Eq. (1.2).

Furthermore, a one pound mass is said to weigh one pound if the mass is in static equilibrium on the surface of earth. We can define **weight** as the opposed force required to support a mass. For the concept of weight for masses that are not stationary at earth's surface, or are affected by the earth's rotation (a factor of only 0.002 times the force exerted by gravity), or are located away from the earth's surface as in a rocket or satellite, consult your physics text.

To sum up, always keep in mind that the two quantities g and g_c are **not** the same. Also, **never forget that the pound (mass) and pound (force) are not the same units in the American engineering system** even though we speak of *pounds* to express force, weight, or mass. Nearly all teachers and writers in physics, engineering, and related fields in technical communications are careful to use the terms "mass," "force," and "weight" properly. On the other hand, in ordinary language most people, including scientists and engineers, omit the designation of "force" or "mass" associated with the pound or kilogram but pick up the meaning from the context of the statement. No one gets confused by the fact that a man is 6 feet tall but has only two feet. Similarly, you should interpret the statement that a bottle "weighs" 5 kg as meaning that the bottle has a mass of 5 kg and is attracted to the earth's surface with a force equal to

$$(5 \text{ kg})(9.80 \text{ m/s}^2) = 49(\text{kg})(\text{m})(\text{s}^{-2}) \quad \text{if } g = 9.80 \text{ m/s}^2$$

Perhaps one of these days people will speak of "massing" instead of "weighing" and "mass-minding" instead of "weight-watching," but probably not soon. **In this book, we will not subscript the symbol lb with m (for mass) or f (for force) unless it becomes essential to do so to avoid confusion. We will always mean by the unit lb without a subscript the quantity pound mass.**

EXAMPLE 1.4 Use of g_c

One hundred pounds of water is flowing through a pipe at the rate of 10.0 ft/s. What is the kinetic energy of this water in $(\text{ft})(\text{lb}_f)$?

Solution

$$\text{kinetic energy} = K = \frac{1}{2}mv^2$$

Assume that the 100 lb of water means the mass of the water.

$$K = \frac{1}{2} \left| \frac{100 \text{ lb}_m}{1} \right| \left| \frac{(10 \text{ ft})^2}{\text{s}^2} \right| \left| \frac{1}{32.174 \frac{(\text{ft})(\text{lb}_m)}{(\text{s}^2)(\text{lb}_f)}} \right| = 155 (\text{ft})(\text{lb}_f)$$

EXAMPLE 1.5 Use of g_c

What is the potential energy in $(\text{ft})(\text{lb}_f)$ of a 100-lb drum hanging 10 ft above the surface of the earth with reference to the surface of the earth?

Solution

$$\text{potential energy} = P = mgh$$

Assume that the 100 lb means 100 lb mass; g = acceleration of gravity = 32.2 ft/s^2 .

$$P = \frac{100 \text{ lb}_m}{1} \times \frac{32.2 \text{ ft}}{\text{s}^2} \times \frac{10 \text{ ft}}{1} \times \frac{1}{32.174 \frac{(\text{ft})(\text{lb}_m)}{(\text{s}^2)(\text{lb}_f)}} = 1001 (\text{ft})(\text{lb}_f)$$

Notice that in the ratio of g/g_c , or 32.2 ft/s^2 divided by $32.174 \frac{(\text{ft})(\text{lb}_m)}{(\text{s}^2)(\text{lb}_f)}$, the numerical values are almost equal. A good many people would solve the problem by saying that $100 \text{ lb} \times 10 \text{ ft} = 1000 (\text{ft})(\text{lb})$ without realizing that in effect they are canceling out the numbers in the g/g_c ratio.

EXAMPLE 1.6 Weight

What is the difference in the weight in newtons of a 100-kg rocket at height of 10 km above the surface of the earth, where $g = 9.76 \text{ m/s}^2$, as opposed to its weight on the surface of the earth, where $g = 9.80 \text{ m/s}^2$?

Solution

The weight in newtons can be computed in each case from Eq. (1.1) with $a = g$ if we ignore the tiny effect of centripetal acceleration resulting from the rotation of the earth (less than 0.3%):

$$\text{weight difference} = \frac{100 \text{ kg}}{1} \times \frac{(9.80 - 9.76) \text{ m}}{\text{s}^2} \times \frac{1 \text{ N}}{1 \frac{(\text{kg})(\text{m})}{\text{s}^2}} = 4.00 \text{ N}$$

Note that the concept of weight is not particularly useful in treating the dynamics of long-range ballistic missiles or of earth satellites because the earth is both round and rotating.

You should develop some facility in converting units from the SI system into the American engineering system, and vice versa, since these are the two sets of units in this text. Certainly you are familiar with the **common conversions** in both the American engineering and the SI systems. If you have forgotten, Table 1.3 lists a short selection of essential conversion factors. Memorize them. Common abbreviations and symbols also appear in this table.

The distinction between uppercase and lowercase letters should be followed, even if the symbol appears in applications where the other lettering is in uppercase style. Unit abbreviations have the same form for both singular and plural, and they are not followed by a period (except in the case of inches).

Other useful conversion factors are discussed in subsequent sections of this book.

One of the best features of the SI system is that (except for time) units and their multiples and submultiples are related by standard factors designated by the prefixes indicated in Table 1.4. Prefixes are not preferred for use in denominators (except for

TABLE 1.3 Basic Conversion Factors*

Dimension	American engineering	SI	Conversion: American engineering to SI
Length	12 in. = 1 ft 3 ft = 1 yd 5280 ft = 1 mi	10 mm [†] = 1 cm [†] 100 cm [†] = 1 m	1 in. = 2.54 cm 3.28 ft = 1 m
Volume	1 ft ³ = 7.48 gal	1000 cm ^{3†} = 1 L [†]	35.31 ft ³ = 1.00 m ³
Density	1 ft ³ H ₂ O = 62.4 lb _m	1 cm ^{3†} H ₂ O = 1 g 1 m ³ H ₂ O = 1000 kg	—
Mass	1 ton _m = 2000 lb _m [‡]	1000 g = 1 kg	1 lb = 0.454 kg
Time	1 min = 60 s 1 hr = 60 min	1 min [†] = 60 s 1 h [†] = 60 min [†]	—

*Some conversion factors in this table are approximate but have sufficient precision for engineering calculations.

[†]An acceptable but not preferred unit in the SI system.

[‡]2000 lb_m is the "short ton"; 2240 lb_m is the "long ton"; 1000 kg = 2204.6 lb_m is the "metric ton."

TABLE 1.4 SI Prefixes

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10 ¹⁸	exa	E	10 ⁻¹	deci*	d
10 ¹⁵	penta	P	10 ⁻²	centi*	c
10 ¹²	tera	T	10 ⁻³	milli	m
10 ⁹	giga	G	10 ⁻⁶	micro	μ
10 ⁶	mega	M	10 ⁻⁹	nano	n
10 ³	kilo	k	10 ⁻¹²	pico	p
10 ²	hecto*	h	10 ⁻¹⁵	femto	f
10 ¹	deka*	da	10 ⁻¹⁸	atto	a

*Avoid except for areas and volumes.

kg). Do not use double prefixes; that is, use nanometer, not millimicrometer. The strict use of these prefixes leads to some amusing combinations of noneuphonic sounds, such as nanonewton, nembujoule, and so forth. Also, some confusion is certain to arise because the prefix M can be confused with m as well as with M ≈ 1000 derived from the Roman numerical. When a compound unit is formed by multiplication of two or more other units, its symbol consists of the symbols for the separate units joined by a centered dot (e.g., N · m for newton meter). The dot may be omitted in the case of familiar units such as watt-hour (symbol Wh) if no confusion will result, or if the symbols are separated by exponents, as in N · m² kg⁻². Hy-

phens should not be used in symbols for compound units. Positive and negative exponents may be used with the symbols for units. If a compound unit is formed by division of one unit by another, its symbol consists of the symbols for the separate units either separated by a solidus or multiplied by using negative powers (e.g., m/s or $\text{m} \cdot \text{s}^{-1}$ for meters per second). **We do not use the center dot for multiplication in this text.** A dot can easily get confused with a period or missed entirely in handwritten calculations. Instead, we will use parentheses or vertical rules, whichever is more convenient, for multiplication. Also, the SI convention of leaving a space between groups of numbers such as 12 650 instead of inserting a comma, as in 12,650, will be ignored to avoid confusion in handwritten numbers.

Note that in the SI system the symbols are *not* abbreviations. A symbol such as dm has to be taken as a whole. When prefixes are used with symbols raised to a power, the prefix is also raised to the same power. You should interpret cm^3 as $(10^{-2} \cdot \text{m})^3 = 10^{-6}(\text{m}^3)$, not as $10^{-2}(\text{m}^3)$.

Also watch out for ambiguities such as:

$\text{N} \cdot \text{m}$ (newton $\times$ meter) vs. mN (millinewton)

$\text{m} \cdot \text{s}$ (meter $\times$ second) vs. ms (millisecond)

and the use of symbols for units jointly with mathematical symbols leading to such unusual formulas as

$N = 625 \text{ N}$ (the first N is the normal force)

$F = m (32 \text{ m} \cdot \text{s}^{-2})$ (the first m stands for mass)

1.1-3 Dimensional Consistency

Now that we have reviewed some background material concerning units and dimensions, we can immediately make use of this information in a very practical and important application. **A basic principle exists that equations must be dimensionally consistent.** What the principle requires is that each term in an equation must have the same net dimensions and units as every other term to which it is added or subtracted or equated. Consequently, dimensional considerations can be used to help identify the dimensions and units of terms or quantities in terms in an equation.

The concept of dimensional consistency can be illustrated by an equation that represents gas behavior and is known as the van der Waals equation, an equation to be discussed in more detail in Chap. 3:

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

Inspection of the equation shows that the constant a must have the units of $[(\text{pressure})(\text{volume})^2]$ in order for the expression in the first set of parentheses to be consistent throughout. If the units of pressure are atm and those of volume are cm^3 , a will have the units specifically of $[(\text{atm})(\text{cm})^6]$. Similarly, b must have the same units as V , or in this particular case the units of cm^3 . All equations must exhibit dimensional consistency.

EXAMPLE 1.7 Dimensional Consistency

Your handbook shows that microchip etching roughly follows the relation

$$d = 16.2 - 16.2e^{-0.021t} \quad t < 200$$

where d is the depth of the etch in microns (micrometers; μm) and t is the time of the etch in seconds. What are the units associated with the numbers 16.2 and 0.021? Convert the relation so that d becomes expressed in inches and t can be used in minutes.

Solution

Both values of 16.2 must have the units of microns. The exponential must be dimensionless so that 0.021 must have the units of 1/seconds.

$$\begin{aligned} d_{\text{in.}} &= \frac{16.2 \mu\text{m}}{10^6 \mu\text{m}} \left| \frac{1 \text{ m}}{39.37 \text{ in.}} \right| \left(1 - e^{\frac{0.021}{\text{s}} \left| \frac{60 \text{ s}}{1 \text{ min}} \right| t, \text{min}} \right) \\ &= 6.38 \times 10^{-4} (1 - e^{-1.26t/\text{min}}) \end{aligned}$$

Groups of symbols may be put together, either by theory or experiment, that have no net units. Such collections of variables or parameters are called **dimensionless** or **nondimensional groups**. One example is the Reynolds number (group) arising in fluid mechanics.

$$\text{Reynolds number} = \frac{Dv\rho}{\mu} = N_{\text{Re}}$$

where D is the pipe diameter, say in cm; v is the fluid velocity, say in cm/s; ρ is the fluid density, say in g/cm³; and μ is the viscosity, say in centipoise, units that can be converted to g/(cm)(s). Introducing the consistent set of units for D , v , ρ , and μ into $Dv\rho/\mu$, we find, as expected, that all the units cancel out.

$$\frac{\text{cm}}{\text{s}} \left| \frac{\text{cm}}{\text{s}} \right| \frac{\text{g}}{\text{cm}^3} \left| \frac{(\text{cm})(\text{s})}{\text{g}} \right|$$

Before proceeding to the next section in this chapter, we need to mention briefly some aspects of significant figures, accuracy, and precision of numbers. Measurements collected by process instruments can be expected to exhibit some random error and may also be biased. Resolution of the proper values to use in making material and energy balances falls into the province of statistics, and is not treated here. Refer to Box et al.³ for the appropriate techniques.

The significant figures in a number can be formally established by using scientific notation; that is, the rule that the first digit of all numbers is followed by a

³ G. E. P. Box, W. G. Hunter, and J. S. Hunter, *Statistics for Experimenters*, Wiley, New York, 1978.

decimal point and multiplied by a power of 10 (except for 10^0). Thus

$$4.61 \times 10^3$$

By this convention, the number of significant figures is one (for the first digit) plus the number of digits following the decimal point. Some examples are

Number	Number of significant digits
4.	1
4.0	2
4.0×10^2	2
4.000×10^2	4
3.14096×10^{-2}	6

We also need to mention something about the accuracy and precision of numbers as used in practice. *Accuracy* refers to how close a measured value is to its true value; *precision* refers to the degree of dispersion (or deviations) of measurements from their true values. In this book you will encounter integers such as 1, 2, 3 and so on, which are in some cases exact (2 reactors, 3 input streams) but in other cases are shortcut substitutes for presumed measurements in problem solving (3 moles, 10 kg). You can assume that 10 kg infers a reasonable number of significant figures in relation to the other values of parameters stated in an example or problem, such as $10 \text{ kg} \approx 10.00 \text{ kg}$. You will also occasionally encounter fractions such as $\frac{2}{3}$, which can also be treated as 0.6667 in relation to the accuracy of other values in a problem.

Feel free to round off parameters such as $\pi = 3.1416$, $\sqrt{2} = 1.414$, or Avogadro's number $N = 6.02 \times 10^{23}$. The conversion factors in Table 1.3 and other tables in this text have been rounded so that they can be memorized without too much difficulty. Also round-off your answers to problems even though in the intermediate calculations numbers are carried out to 10 or more digits in your computer, because the final answers can be no more accurate than the accuracy of the numbers introduced into the problem during its solution. For example, in this text for convenience we will use 273 K for the temperature equivalent to 0°C instead of 273.15 K, thus introducing a relative error of $0.15/273.15 = 0.00055$ into a temperature calculation (or an absolute error of 0.15). This is such a small error relative to the other known or presumed errors in your calculations that it can be neglected in almost all instances. Keep in mind, however, that in addition, subtraction, multiplication, and division, the errors you introduce propagate into the final answer. In addition and subtraction of several numbers, the maximum absolute error (M.A.E.) is equal to the sum of the M.A.E.'s of the individual values. In multiplication and division, the maximum relative error (M.R.E.) is the sum of the M.R.E.'s of the individual values. For powers and roots the M.R.E. of all the terms is the sum of the M.R.E.'s, each raised to the power of each term. Consequently, be cautious in truncating values of known accuracy too much. Carry along more digits than you need for your final result, and then round-off.

EXAMPLE 1.8 Propagation of Error

Given the maximum absolute errors in the pressures

$$p_1 = 2.0 \pm 0.3$$

$$p_2 = 8.0 \pm 0.5$$

$$p_3 = 4.0 \pm 0.1$$

what are the maximum absolute and relative errors in

$$p_4 = p_1 p_2 + p_3$$

and

$$p_5 = \frac{p_1 p_2}{p_3}$$

Solution

The fractional relative errors are:

$$p_1: \frac{0.3}{2.0} = 0.15$$

$$p_2: \frac{0.5}{8.0} = 0.063$$

$$p_3: \frac{0.1}{4.0} = 0.025$$

For p_4 :

$$\text{M.R.E. in } p_1 p_2 = 0.15 + 0.063 = 0.213 \quad \text{M.A.E. in } p_4 = 3.408 + 0.1 = 3.508$$

$$\text{M.A.E. in } p_1 p_2 = (0.213)(16) = 3.408 \quad \text{M.R.E. in } p_4 = \frac{3.508}{20} = 0.175$$

Because the errors are known to only one significant figure, we can truncate the relative error to 0.2. For p_5 , the relative error is

$$0.15 + 0.063 + 0.025 = 0.24$$

and the absolute error is $[p_5 = (2.0)(8.0)/4.0 = 4.0]$

$$(4.0)(0.24) = 0.95 \quad (\text{or } 1)$$

By way of information, the error in $p_1^5 = 32$ is

$$\text{M.R.E.: } (0.15)5 = 0.75$$

$$\text{M.A.E.: } (0.75)32 = 24!$$

In the next section we review the concept of a mole of a material.

Self-Assessment Test*

- Convert 20 gal/hr to m^3/s ; 50 lb/in.² to N/m^2 .
- On Phobos, the inner moon of Mars, the acceleration of gravity is $3.78 \text{ ft}/\text{sec}^2$. Suppose that an astronaut were walking around on Phobos, and this person plus space suit and equipment had an earth weight of 252 lb.
 - What is the mass in lb_m of the astronaut plus suit and equipment?
 - How much would the space-suited astronaut weigh in pounds on Phobos?
- Prepare a table in which the rows are: length, area, volume, mass, and time. Make two columns, one for the SI and the other for the American engineering systems of units. Fill in each row with the name of the unit, and in a third column, show the numerical equivalency (i.e., $1 \text{ ft} = 0.3048 \text{ m}$).
- An orifice meter is used to measure flow rate in pipes. The flows rate is related to the pressure drop by an equation of the form

$$u = c \sqrt{\frac{\Delta p}{\rho}}$$

where u = fluid velocity

Δp = pressure drop

ρ = density of the flowing fluid

c = constant of proportionality

What are the units of c in the SI system of units?

- What are the value and units of g_c ?
- In the SI system of units, the weight of a 180-lb man standing on the surface of the earth is approximately:
 - 801 N
 - 81.7 kg
 - Neither of these
 - Both of these
- The thermal conductivity k of a liquid metal is predicted via the empirical equation $k = A \exp(B/T)$, where k is in $\text{J}/(\text{s})(\text{m})(\text{K})$ and A and B are constants. What are the units of A ? Of B ?
- Fill in the following blanks:
 - _____ grams of prevention is worth _____ kilograms of cure.
 - A miss is as good as _____ kilometers.
 - Big sale on _____ liter hats!

Thought Problems

- Comment as to what is wrong with the following two statements from a textbook:
 - Weight is the product of mass times the force of gravity.
 - A 67-kg person on Earth will weigh only 11 kg on the moon.
- A state representative is proposing a bill in the legislature that declares: "No person shall discharge into the atmosphere any gaseous effluent containing radioactive materials." Would you be in favor of this bill? Is it feasible?

*Answers to the self-assessment test problems are given in Appendix A.

1.2 THE MOLE UNIT

Your objectives in studying this section are to be able to:

1. Define a kilogram mole, pound mole, and gram mole.
2. Convert from moles to mass and the reverse for any chemical compound given the molecular weight.
3. Calculate molecular weights from the molecular formula.

What is a mole? The best answer is that a **mole** is a certain number of molecules, atoms, electrons, or other specified types of particles.⁴ In particular, the 1969 International Committee on Weights and Measures approved the mole (symbol mol in the SI system) as being "the amount of a substance that contains as many elementary entities as there are atoms in 0.012 kg of carbon 12." Thus in the SI system the mole contains a different number of molecules than it does in the American engineering system. In the SI system a mole has about 6.023×10^{23} molecules; we shall call this a *gram mole* (symbol *g mol*) to avoid confusion even though in the SI system of units the official designation is simply mole (abbreviated mol). We can thereby hope to avoid the confusion that could occur with the American engineering system *pound mole* (abbreviated *lb mol*), which has $6.023 \times 10^{23} \times 453.6$ molecules. Thus a pound mole of a substance has more mass than does a gram mole of the substance.

Here is another way to look at the mole unit. To convert the number of moles to mass, we make use of the **molecular weight**—the mass per mole:

$$\text{the g mol} = \frac{\text{mass in g}}{\text{molecular weight}} \quad (1.4)$$

$$\text{the lb mol} = \frac{\text{mass in lb}}{\text{molecular weight}} \quad (1.5)$$

or

$$\text{mass in g} = (\text{mol. wt.})(\text{g mol}) \quad (1.6)$$

$$\text{mass in lb} = (\text{mol. wt.})(\text{lb mol}) \quad (1.7)$$

Furthermore, there is no reason why you cannot carry out computations in terms of ton moles, kilogram moles, or any corresponding units if they are defined analogously to Eqs. (1.4) and (1.5) even if they are not standard units. If you read

⁴For a discussion of the mole concept, refer to the series of articles in *J. Chem. Educ.*, v. 38, pp. 549–556 (1961), and to M. L. McGlashan, *Phys. Educ.*, p. 276 (July 1977).

about a unit such as a kilomole (kmol) without an associated mass specification, or a kg mol, assume that it refers to the SI system and 10^3 g mol.

Values of the molecular weights (relative molar masses) are built up from the tables of atomic weights based on an arbitrary scale of the relative masses of the elements. **Atomic weight** of an element is the mass of an atom based on the scale that assigns a mass of exactly 12 to the carbon isotope ^{12}C , whose nucleus contains 6 protons and 6 neutrons. The terms atomic "weight" and molecular "weight" are universally used by chemists and engineers instead of the more accurate terms atomic "mass" or molecular "mass." Since weighing was the original method for determining the comparative atomic masses, as long as they were calculated in a common gravitational field, the relative values obtained for the atomic "weights" were identical with those of the atomic "masses."

Appendix B lists the atomic weights of the elements. On this scale of atomic weights, hydrogen is 1.008, carbon is 12.01, and so on. (In most of our calculations we shall round these off to 1 and 12, respectively, for convenience.)

A compound is composed of more than one atom, and the molecular weight of the compound is nothing more than the sum of the weights of the atoms of which it is composed. Thus H_2O consists of 2 hydrogen atoms and 1 oxygen atom, and the molecular weight of water is $(2)(1.008) + 16.000 = 18.02$. These weights are all relative to the ^{12}C atom as 12.0000, and you can attach any unit of mass you desire to these weights; for example, H_2 can be 2.016 g/g mol, 2.016 lb/lb mol, 2.016 ton/ton mol, and so on. Gold (Au) is 196.97 oz/oz mol!

You can compute average molecular weights for mixtures of constant composition even though they are not chemically bonded if their compositions are known accurately. Later (Example 1.13) we show how to calculate the average molecular weight of air. Of course, for a material such as fuel oil or coal whose composition may not be exactly known, you cannot determine an exact molecular weight, although you might estimate an approximate average molecular weight good enough for engineering calculations. Keep in mind that the symbol lb refers to lb_m unless otherwise stated.

EXAMPLE 1.9 Use of Molecular Weights

If a bucket holds 2.00 lb of NaOH (mol. wt. = 40.0), how many

- Pound moles of NaOH does it contain?
- Gram moles of NaOH does it contain?

Solution

Basis: 2.00 lb of NaOH

$$(a) \quad \frac{2.00 \text{ lb NaOH}}{40.0 \text{ lb NaOH}} \left| \frac{1 \text{ lb mol NaOH}}{40.0 \text{ lb NaOH}} \right| = 0.050 \text{ lb mol NaOH}$$

$$(b_1) \quad \frac{2.00 \text{ lb NaOH}}{40.0 \text{ lb NaOH}} \left| \frac{1 \text{ lb mol NaOH}}{40.0 \text{ lb NaOH}} \right| \left| \frac{454 \text{ g mol}}{1 \text{ lb mol}} \right| = 22.7 \text{ g mol}$$

or

$$(b_2) \quad \frac{2.00 \text{ lb NaOH}}{1 \text{ lb}} \times \frac{454 \text{ g}}{1 \text{ lb}} \times \frac{1 \text{ g mol NaOH}}{40.0 \text{ g NaOH}} = 22.7 \text{ g mol}$$

EXAMPLE 1.10 Use of Molecular Weights

How many pounds of NaOH are in 7.50 g mol of NaOH?

Solution

Basis: 7.50 g mol of NaOH

$$\frac{7.50 \text{ g mol NaOH}}{454 \text{ g mol}} \times \frac{1 \text{ lb mol}}{454 \text{ g mol}} \times \frac{40.0 \text{ lb NaOH}}{1 \text{ lb mol NaOH}} = 0.661 \text{ lb NaOH}$$

Self-Assessment Test

1. What is the molecular weight of acetic acid (CH_3COOH)?
2. What is the difference between a kilogram mole and a pound mole?
3. Convert 39.8 kg of NaCl per 100 kg of water to kilogram moles of NaCl per kilogram mole of water.
4. How many pound moles of NaNO_3 are there in 100 lb?
5. One pound mole of CH_4 per minute is fed to a heat exchanger. How many kilograms is this per second?

Thought Problem

1. There is twice as much copper in 480 g of copper as there is in 240 g of copper, but is there twice as much copper in 480 g of copper as there is silver in 240 g of silver?

1.3 CONVENTIONS IN METHODS OF ANALYSIS AND MEASUREMENT

Your objectives in studying this section are to be able to:

1. Define density and specific gravity.
2. Calculate the density of a liquid or solid given its specific gravity and the reverse.
3. Look up the density or specific gravity of a liquid or solid in reference tables.
4. Interpret the meaning of specific gravity data taken from reference tables.
5. Specify the common reference material(s) used to determine the specific gravity of liquids and solids.

6. Convert the composition of a mixture from mole fraction (or percent) to mass (weight) fraction (or percent) and the reverse.
7. Transform a material from one measure of concentration to another, including mass/volume, moles/volume, ppm, and molarity.
8. Calculate the mass or number of moles of each component in a mixture given the percent (or fraction) composition, and the reverse, and compute the average molecular weight.
9. Convert a composition given in mass (weight) percent to mole percent, and the reverse.

There are certain definitions and conventions which we mention at this point since they will be used constantly throughout the book. If you memorize them now, you will immediately have a clearer perspective and save considerable trouble later.

1.3-1 Density

Density is the ratio of mass per unit volume, as, for example, kg/m^3 or lb/ft^3 . It has both a numerical value and units. To determine the density of a substance, you must find both its volume and its mass. If the substance is a solid, a common method to determine its volume is to displace a measured quantity of inert liquid. For example, a known weight of a material can be placed into a container of liquid of known weight and volume, and the final weight and volume of the combination measured. The density (or specific gravity) of a liquid is commonly measured with a hydrometer (a known weight and volume is dropped into the liquid and the depth to which it penetrates into the liquid is noted) or a Westphal balance (the weight of a known slug is compared in the unknown liquid with that in water). Gas densities are quite difficult to measure; one device used is the Edwards balance, which compares the weight of a bulb filled with air to the same bulb when filled with the unknown gas.

In most of your work using liquids and solids, density will not change very much with pressure, but for precise measurements for common substances you can always look up in a handbook the variation of density with pressure. The change in density with temperature is illustrated in Fig. 1.1 for liquid water and liquid ammo-

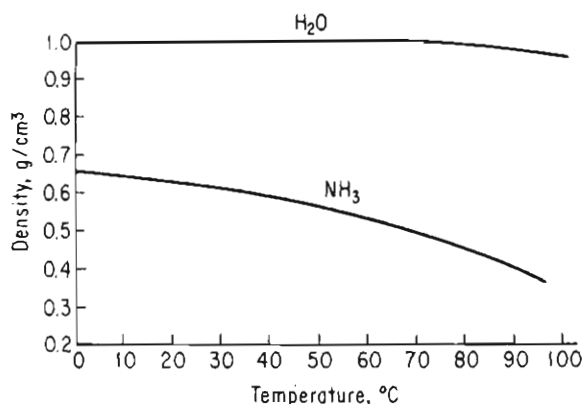


Figure 1.1 Densities of liquid H_2O and NH_3 as a function of temperature.

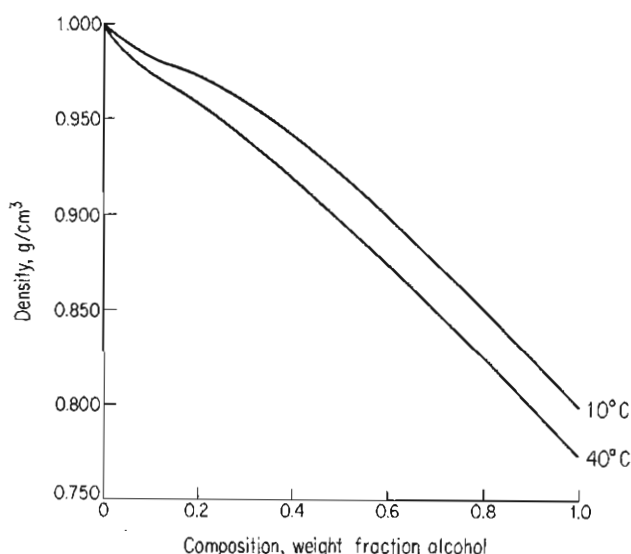


Figure 1.2 Density of a mixture of ethyl alcohol and water as a function of composition.

nia. Figure 1.2 illustrates how density also varies with composition. In the winter you may put antifreeze in your car radiator. The service station attendant checks the concentration of antifreeze by measuring the specific gravity and, in effect, the density of the radiator solution after it is mixed thoroughly. He has a little thermometer in his hydrometer kit in order to be able to read the density corrected for temperature.

1.3-2 Specific Gravity

Specific gravity is commonly thought of as a dimensionless ratio. Actually, it should be considered as the ratio of two densities—that of the substance of interest, A , to that of a reference substance. In symbols:

$$\text{sp gr} = \text{specific gravity} = \frac{(\text{lb/ft}^3)_A}{(\text{lb/ft}^3)_{\text{ref}}} = \frac{(\text{g/cm}^3)_A}{(\text{g/cm}^3)_{\text{ref}}} = \frac{(\text{kg/m}^3)_A}{(\text{kg/m}^3)_{\text{ref}}} \quad (1.8)$$

The reference substance for liquids and solids is normally water. Thus the specific gravity is the ratio of the density of the substance in question to the density of water. The specific gravity of gases frequently is referred to air, but may be referred to other gases, as discussed in more detail in Chap. 3. Liquid density can be considered to be nearly independent of pressure for most common calculations, but, as indicated in Fig. 1.1 it varies somewhat with temperature; therefore, to be very precise when referring to specific gravity, state the temperature at which each density is chosen. Thus

$$\text{sp gr} = 0.73 \frac{20^\circ}{4^\circ}$$

can be interpreted as follows: the specific gravity when the solution is at 20°C and the reference substance (water) is at 4°C is 0.73. Since the density of water at 4°C is

very close to 1.0000 g/cm³ in the SI system, the numerical values of the specific gravity and density in this system are essentially equal. Since densities in the American engineering system are expressed in lb/ft³ and the density of water is about 62.4 lb/ft³, it can be seen that the specific gravity and density values are not numerically equal in the American engineering system.

In the petroleum industry the specific gravity of petroleum products is usually reported in terms of a hydrometer scale called °API. The equation for the API scale is

$$^{\circ}\text{API} = \frac{141.5}{\text{sp gr}_{60^{\circ}}} - 131.5 \quad (1.9)$$

or

$$\text{sp gr}_{60^{\circ}} = \frac{141.5}{^{\circ}\text{API} + 131.5} \quad (1.10)$$

The volume and therefore the density of petroleum products vary with temperature, and the petroleum industry has established 60°F as the standard temperature for volume and API gravity. The °API is being phased out as SI units are accepted for densities.

There are many other systems of measuring density and specific gravity that are somewhat specialized; for example, the Baumé (°Be) and the Twaddell (°Tw) systems. Relationships among the various systems of density may be found in standard reference books.

1.3-3 Specific Volume = $\frac{1}{\rho}$

The specific volume of any compound is the inverse of the density, that is, the volume per unit mass or unit amount of material. Units of specific volume might be ft³/lb_m, ft³/lb mole, cm³/g, bbl/lb_m, m³/kg, or similar ratios.

1.3-4 Mole Fraction and Mass (Weight) Fraction

Mole fraction is simply the moles of a particular substance divided by the total number of moles present. This definition holds for gases, liquids, and solids. Similarly, the **mass (weight) fraction** is nothing more than the **mass (weight)** of the substance divided by the total mass (weight) of all substances present. Although the mass fraction is what is intended to be expressed, ordinary engineering usage employs the term **weight fraction**. Mathematically, these ideas can be expressed as

$$\text{mole fraction of } A = \frac{\text{moles of } A}{\text{total moles}} \quad (1.11)$$

$$\text{mass (weight) fraction of } A = \frac{\text{mass (weight) of } A}{\text{total mass (weight)}} \quad (1.12)$$

Mole percent and weight percent are the respective fractions times 100.

1.3-5 Analyses

The analyses of gases such as air, combustion products, and the like are usually on a dry basis—that is, water vapor is excluded from the analysis. Such an analysis, called an Orsat analysis, is explained in Sec. 2.3. If the gas acts as an ideal gas and its components are each measured by volume, you know (or will learn in Sec. 3.1) that volume percent under ordinary conditions is the same as mole percent. For example, consider the composition of air, which is approximately

21%	oxygen
79%	nitrogen
100%	total

This means that at room temperature and pressure, any sample of air will contain 21% oxygen by volume and also be 21 mole % oxygen.

In this book, the composition of gases will always be presumed to be given in mole percent or fraction unless otherwise stated.

Analysis of liquids and solids are usually given by mass (weight) percent or fraction, but occasionally by mole percent.

In this text, the analysis of liquids and solids will always be assumed to be weight percent unless otherwise stated.

EXAMPLE 1.11 Mole Fraction and Mass (Weight) Fraction

An industrial-strength drain cleaner contains 5.00 kg of water and 5.00 kg of NaOH. What are the mass (weight) fraction and mole fraction of each component in the bottle of water?

Solution

Basis: 10.0 kg of total solution

Component	kg	Weight fraction	Mol. wt.	kg mol	Mole fraction
H ₂ O	5.00	$\frac{5.00}{10.0} = 0.500$	18.0	0.278	$\frac{0.278}{0.403} = 0.690$
NaOH	5.00	$\frac{5.00}{10.0} = 0.500$	40.0	0.125	$\frac{0.125}{0.403} = 0.310$
Total	10.00	1.000		0.403	1.00

The kilogram moles are calculated as follows:

$$\frac{5.00 \text{ kg H}_2\text{O}}{18.0 \text{ kg H}_2\text{O}} \times \frac{1 \text{ kg mol H}_2\text{O}}{18.0 \text{ kg H}_2\text{O}} = 0.278 \text{ kg mol H}_2\text{O}$$

$$\frac{5.00 \text{ kg NaOH}}{40.0 \text{ kg NaOH}} \left| \frac{1 \text{ kg mol NaOH}}{40.0 \text{ kg NaOH}} \right| = 0.125 \text{ kg mol NaOH}$$

Adding these quantities together gives the total kilogram moles.

EXAMPLE 1.12 Density and Specific Gravity

If dibromopentane (DBP) has a specific gravity of 1.57, what is its density in (a) g/cm³? (b) lb_m/ft³? and (c) kg/m³?

Solution

No temperatures are cited for the dibromopentane or the reference compound (presumed to be water); hence we assume that the temperatures are the same and that water has a density of $1.00 \times 10^3 \text{ kg/m}^3$ (1.00 g/cm^3).

$$\begin{array}{l} \text{(a)} \quad \frac{1.57 \frac{\text{g DBP}}{\text{cm}^3}}{1.00 \frac{\text{g H}_2\text{O}}{\text{cm}^3}} \left| \frac{1.00 \frac{\text{g H}_2\text{O}}{\text{cm}^3}}{1.00 \frac{\text{g H}_2\text{O}}{\text{cm}^3}} \right| = 1.57 \frac{\text{g DBP}}{\text{cm}^3} \\ \text{(b)} \quad \frac{1.57 \frac{\text{lb}_m \text{DBP}}{\text{ft}^3}}{1.00 \frac{\text{lb}_m \text{H}_2\text{O}}{\text{ft}^3}} \left| \frac{62.4 \frac{\text{lb}_m \text{H}_2\text{O}}{\text{ft}^3}}{62.4 \frac{\text{lb}_m \text{H}_2\text{O}}{\text{ft}^3}} \right| = 97.97 \frac{\text{lb}_m \text{DBP}}{\text{ft}^3} \\ \text{(c)} \quad \frac{1.57 \frac{\text{g DBP}}{\text{cm}^3}}{1.00 \frac{\text{g H}_2\text{O}}{\text{cm}^3}} \left| \frac{(100 \text{ cm})^3}{(1 \text{ m})^3} \right| \left| \frac{1 \text{ kg}}{1000 \text{ g}} \right| = 1.57 \times 10^3 \frac{\text{kg DBP}}{\text{m}^3} \end{array}$$

or

$$\frac{1.57 \frac{\text{kg DBP}}{\text{m}^3}}{1.00 \frac{\text{kg H}_2\text{O}}{\text{m}^3}} \left| \frac{1.00 \times 10^3 \frac{\text{kg H}_2\text{O}}{\text{m}^3}}{1.00 \times 10^3 \frac{\text{kg H}_2\text{O}}{\text{m}^3}} \right| = 1.57 \times 10^3 \frac{\text{kg DBP}}{\text{m}^3}$$

Note how the units of specific gravity as used here clarify the calculations.

EXAMPLE 1.13 Average Molecular Weight of Air

What is the average molecular weight of air and its composition by weight percent?

Solution

Basis: 100 lb mol of air

Component	Moles = percent	Mol. wt.	lb	weight %
O ₂	21.0	32	672	23.17
N ₂ *	79.0	28.2	2228	76.83
Total	100		2900	100.00

*Includes Ar, CO₂, Kr, Ne, and Xe, and is called atmospheric nitrogen. The molecular weight is 28.2.

Table 1.5 lists the detailed composition of air.

The average molecular weight is $2900 \text{ lb}/100 \text{ lb mol} = 29.00$.

TABLE 1.5 Composition of Clean, Dry Air near Sea Level

Component	Percent by volume = mole percent
Nitrogen	78.084
Oxygen	20.9476
Argon	0.934
Carbon dioxide	0.0314
Neon	0.001818
Helium	0.000524
Methane	0.0002
Krypton	0.000114
Nitrous oxide	0.00005
Hydrogen	0.00005
Xenon	0.0000087
Ozone	
Summer	0–0.000007
Winter	0–0.000002
Ammonia	0–trace
Carbon monoxide	0–trace
Iodine	0–0.000001
Nitrogen dioxide	0–0.000002
Sulfur dioxide	0–0.0001

Do **not** attempt to get an average specific gravity or average density for a mixture of solids or liquids by multiplying the individual component specific gravities or densities by the respective mole fractions of the components in the mixture and summing the products. The proper way to use specific gravity is demonstrated in the next example.

EXAMPLE 1.14 Application of Specific Gravity

In the production of a drug having a molecular weight of 192, the exit stream from the reactor flows at the rate of 10.3 L/min. The drug concentration is 41.2% (in water), and the specific gravity of the solution is 1.025. Calculate the concentration of the drug (in kg/L) in the exit stream, and the flow rate of the drug in kg mol/min.

Solution

For the first part of the problem, we want to transform the mass fraction of 0.412 into mass per liter of the drug. Take 1.000 kg of exit solution as a basis for convenience. See Fig. E1.14.

Basis: 1.000 kg solution

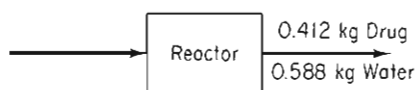


Figure E1.14

How do we get mass per volume (the density) from the given data which is in terms of mass per mass? Use the specific gravity of the solution.

$$\text{density of solution} = \frac{1.025 \frac{\text{g soln}}{\text{cm}^3}}{1.000 \frac{\text{g H}_2\text{O}}{\text{cm}^3}} \times \frac{1.000 \frac{\text{g H}_2\text{O}}{\text{cm}^3}}{1.000 \frac{\text{g H}_2\text{O}}{\text{cm}^3}} = 1.025 \frac{\text{g soln}}{\text{cm}^3}$$

Next

$$\frac{0.412 \text{ kg drug}}{1.000 \text{ kg soln}} \times \frac{1.025 \text{ g soln}}{1 \text{ cm}^3} \times \frac{1 \text{ kg}}{10^3 \text{ g}} \times \frac{10^3 \text{ cm}^3}{1 \text{ L}} = 0.422 \text{ kg drug/L soln}$$

To get the flow rate, we take a different basis, 1 minute.

Basis: 1 min $\equiv$ 10.3 L solution

$$\frac{10.3 \text{ L soln}}{1 \text{ min}} \times \frac{0.422 \text{ kg drug}}{1 \text{ L soln}} \times \frac{1 \text{ kg mol drug}}{192 \text{ kg drug}} = 0.0226 \text{ kg mol/min}$$

1.3-6 Concentrations

Concentration means the quantity of some solute per fixed amount of solvent, or solution, in a mixture of two or more components; for example:

- Mass per unit volume (lb_m of solute/ ft^3 , g of solute/L, lb_m of solute/bbl, kg of solute/ m^3).
- Moles per unit volume (lb mol of solute/ ft^3 , g mol of solute/liter, g mol of solute/ cm^3).
- Parts per million—a method of expressing the concentration of extremely dilute solutions. Ppm is equivalent to a weight fraction for solids and liquids because the total amount of material is of a much higher order of magnitude than the amount of solute; it is a mole fraction for gases. Why?
- Other methods of expressing concentration with which you should be familiar are molarity (g mole/liter) and normality (equivalents/liter).

A typical example of the use of some of these concentration measures is the set of guidelines by which the Environmental Protection Agency defined the extreme levels at which the five most common air pollutants could harm people over stated periods of time.

- Sulfur dioxide: $365 \mu\text{g}/\text{m}^3$ averaged over a 24-hr period
- Particulate matter: $260 \mu\text{g}/\text{m}^3$ averaged over a 24-hr period

- (c) *Carbon monoxide*: 10 mg/m^3 (9 ppm) when averaged over an 8-hr period;
 40 mg/m^3 (35 ppm) when averaged over 1 hr
 (d) *Nitrogen dioxide*: $100 \text{ } \mu\text{g/m}^3$ averaged over 1 year

It is important to remember that in an ideal solution, such as in gases or in a simple mixture of hydrocarbon liquids or compounds of like chemical nature, the volumes of the components may be added without great error to get the total volume of the mixture. For the so-called nonideal mixtures this rule does not hold, and the total volume of the mixture is bigger or smaller than the sum of the volumes of the pure components.

In Chap. 2 we will use **stream flows** and compositions in making material balances. To calculate the **mass flow rate, $\dot{m}$** , from a known **volumetric flow rate, q** , you multiply the volumetric flow rate by the mass concentration thus

$$\frac{q \text{ m}^3}{\text{s}} \times \frac{\rho \text{ kg}}{\text{m}^3} = \dot{m} \frac{\text{kg}}{\text{s}}$$

How would you calculate the volumetric flow rate from a known mass flow rate? From the volumetric flow rate you can calculate the **average velocity, v** , in a pipe if you know the area, A , of the pipe from the relation

$$q = Av$$

Self-Assessment Test

- Answer the following questions true (T) or false (F).
 - The density and specific gravity of mercury are the same.
 - The inverse of density is the specific volume.
 - Parts per million expresses a mole ratio.
 - Concentration of a component in a mixture does not depend on the amount of the mixture.
- A cubic centimeter of mercury has a mass of 13.6 g at the earth's surface. What is the density of the mercury?
- What is the approximate density of water at room temperature?
- For liquid HCN, a handbook gives: $\text{sp gr } 10^\circ\text{C}/4^\circ\text{C} = 1.2675$. What does this mean?
- For ethanol, a handbook gives: $\text{sp gr } 60^\circ\text{F}/60^\circ\text{F} = 0.79389$. What is the density of ethanol at 60°F ?
- Commercial sulfuric acid is 98% H_2SO_4 and 2% H_2O . What is the mole ratio of H_2SO_4 to H_2O ?
- A container holds 1.704 lb of HNO_3 /lb of H_2O and has a specific gravity of 1.382 at 20°C . Compute the composition in the following ways:
 - Weight percent HNO_3
 - Pounds HNO_3 per cubic foot of solution at 20°C
 - Molarity at 20°C

8. The specific gravity of steel is 7.9. What is the volume in cubic feet of a steel ingot weighing 4000 lb?
9. A solution contains 25% salt by weight in water. The solution has a density of 1.2 g/cm³. Express the composition as:
 - (a) Kilograms salt per kilogram of H₂O
 - (b) Pounds of salt per cubic foot of solution
10. A liquefied mixture of *n*-butane, *n*-pentane, and *n*-hexane has the following composition in percent:

<i>n</i> -C ₄ H ₁₀	50
<i>n</i> -C ₅ H ₁₂	30
<i>n</i> -C ₆ H ₁₄	20

For this mixture, calculate:

- (a) The weight fraction of each component
- (b) The mole fraction of each component
- (c) The mole percent of each component
- (d) The average molecular weight

Thought Problems

1. **"Drop in the Bucket.** Service Station Operator Balks At Paying for 'Shrinking' Gasoline" was a recent headline in the newspaper. The way the station operator sees it, when you pay for a gallon of gasoline, you should get a gallon of gasoline. He contends he's paying for thousands of gallons of gasoline each year he doesn't get. He's suing to try to settle the issue.

The issue is "shrinking" gasoline. According to the American Petroleum Institute, gasoline shrinks at the rate of six gallons per 10,000 gallons for every one-degree-Fahrenheit drop in temperature and expands at the same rate when the temperature rises. On a typical 8,000-gallon shipment from a refinery at, say, 75 degrees, that means a loss of 72 gallons when the gasoline gets stored in the station operator's 60-degree underground tanks and thereafter is sold to customers.

"It may seem like a small matter," he says, "but I'm a high-volume dealer, and I lose \$3,000 to \$4,000 a year to shrinkage." Major oil companies adjust their billing for temperature changes in big shipments to each other, but "We don't bill that way because changes in temperature in winter and summer balance each other out, so the dealer comes out pretty much even," an oil company spokesman says.

"Dealers Don't Agree" was the next heading. "Dealers say that's nonsense." Why do you think the dealers believe the lack of temperature billing is not fair?

2. The National Museum is considering buying a Maya plaque from Honduras that the seller claims to be jade. Jade is either jadite (sp gr 3.2 to 3.4) or nephrite (sp gr 3.0). What liquid would you recommend using to test whether or not the mask is jade?
3. A refinery tank that had contained gasoline was used for storing pentane. The tank overflowed when the level indicator said that it was only 85% full. The level indicator was a DP cell that measured weight of fluid. Can you explain to the operator what went wrong?

1.4 BASIS

Your objectives in studying this section are to be able to:

1. State the three questions useful in selecting a basis.
2. Apply the three questions to problems and select a suitable basis or sequences of bases.

Have you noted in previous examples that the word **basis** has appeared at the top of the computations? This concept of basis is vitally important both to your understanding of how to solve a problem and also to your solving it in the most expeditious manner. The basis is the reference chosen by you for the calculations you plan to make in any particular problem, and a proper choice of basis frequently makes the problem much easier to solve. The basis may be a period of time—for example, hours, or a given mass of material—such as 5 kg of CO_2 or some other convenient quantity. In selecting a sound basis (which in many problems is predetermined for you but in some problems is not so clear), you should ask yourself the following questions:

- (a) What do I have to start with?
- (b) What do I want to find out?
- (c) What is the most convenient basis to use?

These questions and their answers will suggest suitable bases. Sometimes, when a number of bases seem appropriate, you may find it is best to use a unit basis of 1 or 100 of something, as, for example, kilograms, hours, moles, cubic feet. For liquids and solids when a weight analysis is used, a convenient basis is often 1 or 100 lb or kg; similarly, 1 or 100 moles is often a good choice for a gas. The reason for these choices is that the fraction or percent automatically equals the number of pounds, kilograms, or moles, respectively, and one step in the calculations is saved.

EXAMPLE 1.15 Choosing a Basis

Aromatic hydrocarbons form 15 to 30% of the components of leaded fuels and as much as 40% of nonleaded gasoline. The carbon/hydrogen ratio helps to characterize the fuel components. If a fuel is 80% C and 20% H by weight, what is the C/H ratio in moles?

Solution

If a basis of 100 lb or kg of oil is selected, percent = pounds or kilograms.

Basis: 100 kg of oil (or 100 lb of oil)

Component	kg = percent or lb = percent	Mol. wt.	kg mol or lb mol
C	80	12.0	6.67
H	20	1.008	19.84
Total	100		

Consequently, the C/H ratio in moles is

$$C/H = \frac{6.67}{19.84} = 0.336$$

EXAMPLE 1.16 Choosing a Basis

Most processes for producing high-energy-content gas or gasoline from coal include some type of gasification step to make hydrogen or synthesis gas. Pressure gasification is preferred because of its greater yield of methane and higher rate of gasification.

Given that a 50.0-kg test run of gas averages 10.0% H₂, 40.0% CH₄, 30.0% CO, and 20.0% CO₂, what is the average molecular weight of the gas?

Solution

The obvious basis is 50.0 kg of gas ("what I have to start with"), but a little reflection will show that such a basis is of no use. You cannot multiply *mole percent* of this gas times kg and expect the answer to mean anything. Thus the next step is to choose a "convenient basis," which is 100 kg mol or lb mol of gas, and proceed as follows:

Basis: 100 kg mol or lb mol of gas

Component	percent = kg mol or lb mol	Mol. wt.	kg or lb
CO ₂	20.0	44.0	880
CO	30.0	28.0	840
CH ₄	40.0	16.04	642
H ₂	10.0	2.02	20
Total	100.0		2382

$$\text{average molecular weight} = \frac{2382 \text{ kg}}{100 \text{ kg mol}} = 23.8 \text{ kg/kg mol}$$

It is important that your basis be indicated near the beginning of the problem so that you will keep clearly in mind the real nature of your calculations and so that anyone checking your problem will be able to understand on what basis they are performed. If you change bases in the middle of the problem, a new basis should be

indicated at that time. Many of the problems that we shall encounter will be solved on one basis and then at the end will be shifted to another basis to give the desired answer. The significance of this type of manipulation will become considerably clearer as you accumulate more experience.

EXAMPLE 1.17 Changing Bases

A sample of medium-grade bituminous coal analysis is as follows:

Component	Percent
S	2
N	1
O	6
Ash	11
Water	3

The residuum is C and H in the mole ratio $H/C = 9$. Calculate the weight fraction composition of the coal with the ash and the moisture omitted.

Solution

Take as a basis 100 kg of coal, for then percent = kilograms.

Basis: 100 kg of coal

The sum of the S + N + O + ash + water is

$$2 + 1 + 6 + 11 + 3 = 23 \text{ kg}$$

Hence the C + H must be $100 - 23 = 77 \text{ kg}$.

To determine the kilograms of C and H, we have to select a new basis. Is 77 kg satisfactory? No. Why? Because the H/C ratio is in terms of moles, not weight (mass). Pick instead:

Basis: 100 kg mol of C + H

Component	Mole fraction	kg mol	Mol. wt.	kg
H	$\frac{9}{1+9} = 0.90$	90	1.008	90.7
C	$\frac{1}{1+9} = 0.10$	10	12	120
Total	1.00	100		210.7

Finally, to return to the original basis, we have

$$\text{H: } \frac{77 \text{ kg}}{210.7 \text{ kg total}} \times 90.7 \text{ kg H} = 33.15 \text{ kg H}$$

$$\text{C: } \frac{77 \text{ kg} \mid 120 \text{ kg C}}{210.7 \text{ kg total}} = 43.85 \text{ kg C}$$

and we can prepare a table summarizing the results.

Component	kg	Wt. fraction
C	43.85	0.51
H	33.15	0.39
S	2	0.02
N	1	0.01
O	6	0.07
Total	86.0	1.00

The ability to choose the basis that requires the fewest steps in solving a problem can only come with practice. You can quickly accumulate the necessary experience if, as you look at each problem illustrated in this text, you determine first in your own mind what the basis should be and then compare your choice with the selected basis. By this procedure you will quickly obtain the knack of choosing a sound basis.

Self-Assessment Test

1. What are the three questions you should ask yourself in selecting a basis?
2. What would be good initial bases to select in solving Problems 1.13, 1.18, 1.30, and 1.47?

1.5 TEMPERATURE

Your objectives in studying this section are to be able to:

1. Define temperature.
2. Explain the difference between absolute temperature and relative temperature.
3. Convert a temperature in any of the four scales ($^{\circ}\text{C}$, K, $^{\circ}\text{F}$, $^{\circ}\text{R}$) to any of the others.
4. Convert an expression involving units of temperature and temperature difference to other units of temperature and temperature difference.
5. Know the reference points for the four temperature scales.

Our concept of temperature no doubt originated with our physical sense of hot and cold. Temperature can be rigorously defined once you have an acquaintance with thermodynamics, but here we simply paraphrase Maxwell's definition:

The temperature of a body is a measure of its thermal state considered in reference to its power to transfer heat to other bodies.

Measurement of the thermal state can be accomplished through a wide variety of instruments, including

- (a) A thermometer containing a liquid such as mercury or alcohol.
- (b) The voltage produced by a junction of two dissimilar conductors that changes with temperature and is used as a measure of temperature (the *thermocouple*).
- (c) The property of changing electrical resistance with temperatures gives us a device known as the *thermistor*.
- (d) Two thin strips of metal bonded together at one end expand at different rates with change of temperature. These strips assist in the control of the flow of water in the radiator of an automobile and in the operation of air conditioners and heating systems.
- (e) High temperatures can be measured by devices called *pyrometers*, which note the radiant energy leaving a hot body.

Figure 1.3 illustrates the appropriate ranges for various temperature-measuring devices.

Temperature is normally measured in degrees **Fahrenheit** or **Celsius** (centigrade). The common scientific scale is the *Celsius* scale,⁵ where 0° is the ice point of water and 100° is the normal boiling point of water. In the early 1700s, Gabriel D. Fahrenheit (1686–1736), a glassblower by trade, was able to build mercury thermometers that gave temperature measurements in reasonable agreement with each other. The *Fahrenheit* scale is the one commonly used in everyday life in the United States. Its reference points are of more mysterious origin, but it is reported that the fixed starting point, or 0° on Fahrenheit's scale, was that produced by surrounding the bulb of the thermometer with a mixture of snow or ice and sal ammoniac; the highest temperature was that at which mercury began to boil. The distance between these two points was divided into 600 parts or degrees. By trial Fahrenheit found that the mercury stood at 32 of these divisions when water just began to freeze, the temperature of human blood was at 96 divisions, and the mercury was at 212 divisions when the thermometer was immersed in boiling water. In the SI system, temperature is measured in kelvin, a unit named after the famous Lord Kelvin (1824–1907). Note that in the SI system the degree symbol is suppressed (e.g., the boiling point of water is 373 K).

The Fahrenheit and Celsius scales are *relative* scales; that is, their zero points were arbitrarily fixed by their inventors. Quite often it is necessary to use *absolute*

⁵ As originally devised by Anders Celsius a Swedish astronomer (1701–1744) in 1742, the freezing point was designated as 100°. Officially, °C now stands for *degrees Celsius*.

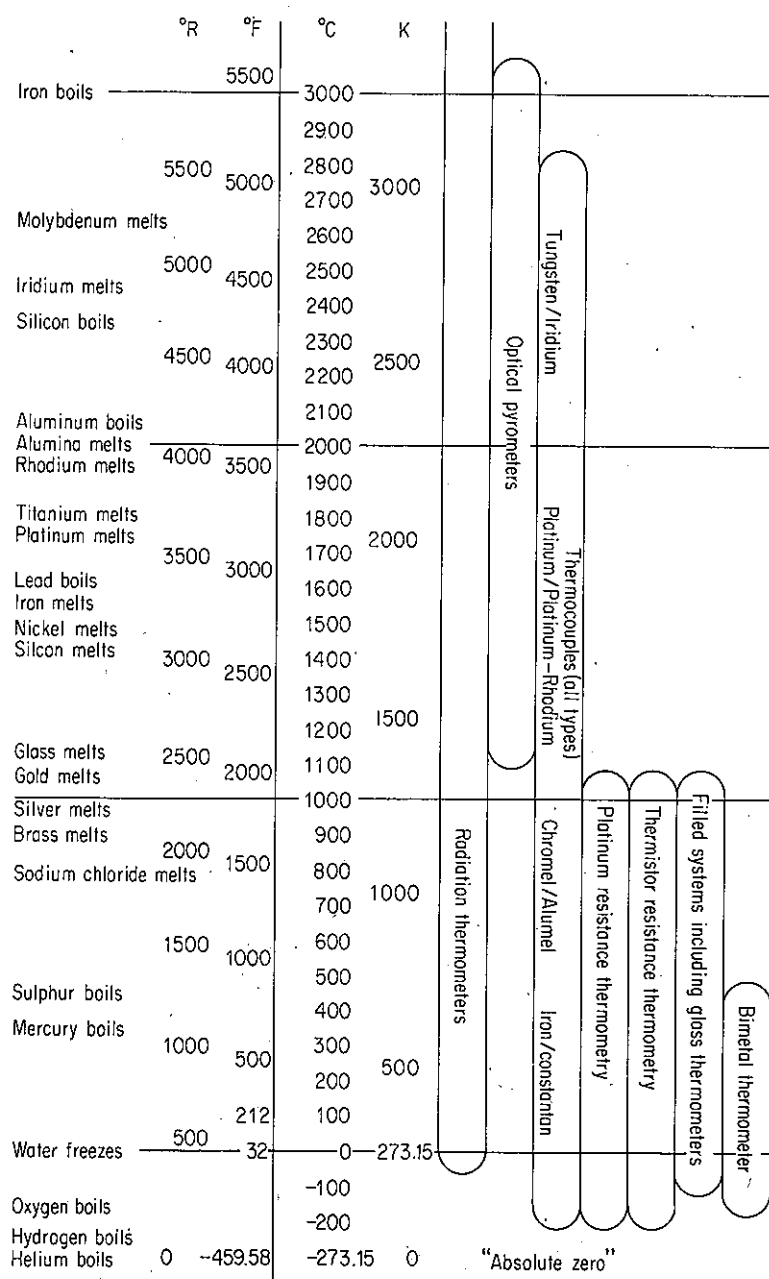


Figure 1.3 Temperature measuring instruments span the range from near absolute zero to beyond 3000 K. The chart indicates the preferred methods of thermal instrumentation for various temperature regions.

temperatures instead of relative temperatures. Absolute temperature scales have their zero point at the lowest possible temperature which we believe can exist. As you may know, this lowest temperature is related both to the ideal gas laws and to the laws of thermodynamics. The absolute scale which is based on degree units the size of those in the Celsius (centigrade) scale is called the *kelvin* scale; the absolute scale which corresponds to the Fahrenheit degree scale is called the *Rankine* scale in honor of W. J. M. Rankine (1820–1872), a Scottish engineer. The relations between relative temperature and absolute temperature are illustrated in Fig. 1.4. We shall round off absolute zero on the Rankine scale of -459.67° to -460°F ; similarly, -273.15°C will be rounded off to -273°C . In Fig. 1.4 all the values of the temperatures have been rounded off, but more significant figures can be used. 0°C and its equivalents are known as *standard conditions of temperature*.

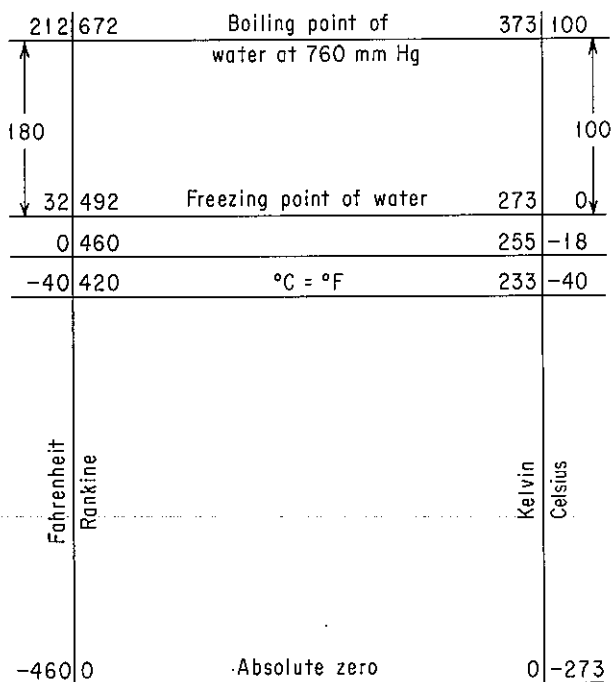


Figure 1.4 Temperature scales.

You should recognize that the unit degree (i.e., the unit temperature difference) on the kelvin–Celsius scale is not the same size as that on the Rankine–Fahrenheit scale. If we let $\Delta^\circ\text{F}$ represent the unit temperature difference in the Fahrenheit scale, $\Delta^\circ\text{R}$ be the unit temperature difference in the Rankine scale, and $\Delta^\circ\text{C}$ and ΔK be the analogous units in the other two scales, you should be aware that

$$\Delta^\circ\text{F} = \Delta^\circ\text{R} \quad (1.13)$$

$$\Delta^\circ\text{C} = \Delta\text{K} \quad (1.14)$$

Also, if you keep in mind that the $\Delta^\circ\text{C}$ is larger than the $\Delta^\circ\text{F}$,

$$\frac{\Delta^{\circ}\text{C}}{\Delta^{\circ}\text{F}} = 1.8 \quad \text{or} \quad \Delta^{\circ}\text{C} = 1.8\Delta^{\circ}\text{F} \quad (1.15)$$

$$\frac{\Delta\text{K}}{\Delta^{\circ}\text{R}} = 1.8 \quad \text{or} \quad \Delta\text{K} = 1.8\Delta^{\circ}\text{R} \quad (1.16)$$

Unfortunately, the symbols $\Delta^{\circ}\text{C}$, $\Delta^{\circ}\text{F}$, ΔK , and $\Delta^{\circ}\text{R}$ are not in standard usage because the Δ symbol becomes inconvenient, especially in typing. A few books try to maintain the difference between degrees of temperature ($^{\circ}\text{C}$, $^{\circ}\text{F}$, etc.) and the unit degree by assigning the unit degree the symbol C° , F° , and so on. But most journals and texts use the same symbol for the two different quantities. Consequently, **the proper meaning of the symbols $^{\circ}\text{C}$, $^{\circ}\text{F}$, K , and $^{\circ}\text{R}$, as either the temperature or the unit temperature difference, must be interpreted from the context of the equation or sentence being examined.**

You should learn how to convert one temperature to another with ease. The relations between $^{\circ}\text{R}$ and $^{\circ}\text{F}$ and between K and $^{\circ}\text{C}$ are, respectively,

$$T_{\text{R}} = T_{\text{F}} \left(\frac{1 \Delta^{\circ}\text{R}}{1 \Delta^{\circ}\text{F}} \right) + 460 \quad (1.17)$$

$$T_{\text{K}} = T_{\text{C}} \left(\frac{1 \Delta\text{K}}{1 \Delta^{\circ}\text{C}} \right) + 273 \quad (1.18)$$

Because the relative temperature scales do not have a common zero at the same temperature, as can be seen from Fig. 1.4, the relation between $^{\circ}\text{F}$ and $^{\circ}\text{C}$ is

$$T_{\text{F}} - 32 = T_{\text{C}} \left(\frac{1.8 \Delta^{\circ}\text{F}}{1 \Delta^{\circ}\text{C}} \right) \quad (1.19)$$

After you have used Eqs. (1.17)–(1.19) a bit, they will become so familiar that temperature conversion will become an automatic reflex. During your “learning period,” in case you forget them, just think of the appropriate scales side by side as in Fig. 1.4, and put down the values for the freezing and boiling points of water. Or, you can recover Eq. (1.19) by recognizing that the equation is linear (see Appendix M)

$$T_{\text{F}} = b_0 + b_1 T_{\text{C}} \quad (1.20)$$

Insert two known pairs of values of T_{F} and T_{C} in Eq. (1.20) and solve the resulting two equations together. For example, 0°C corresponds to 32°F , and 100°C corresponds to 212°F :

$$32 = b_0 + b_1(0) \quad (1.21a)$$

$$212 = b_0 + b_1(100) \quad (1.21b)$$

Solution of Eqs. (1.21a) and (1.21b) together yields

$$b_0 = 32.00 \quad b_1 = 1.800$$

so that $T_{\text{F}} = 32 + 1.8T_{\text{C}}$.

EXAMPLE 1.18 Temperature Conversion

Convert 100°C to (a) K, (b) °F, and (c) °R.

Solution

$$(a) \quad (100 + 273)^{\circ}\text{C} \frac{1 \Delta\text{K}}{1 \Delta^{\circ}\text{C}} = 373 \text{ K}$$

or with suppression of the Δ symbol,

$$(100 + 273)^{\circ}\text{C} \frac{1 \text{ K}}{1^{\circ}\text{C}} = 373 \text{ K}$$

$$(b) \quad (100^{\circ}\text{C}) \frac{1.8^{\circ}\text{F}}{1^{\circ}\text{C}} + 32^{\circ}\text{F} = 212^{\circ}\text{F}$$

$$(c) \quad (212 + 460)^{\circ}\text{F} \frac{1^{\circ}\text{R}}{1^{\circ}\text{F}} = 672^{\circ}\text{R}$$

or

$$(373 \text{ K}) \frac{1.8^{\circ}\text{R}}{1 \text{ K}} = 672^{\circ}\text{R}$$

The suppression of the Δ symbol perhaps makes the temperature relations more familiar looking.

EXAMPLE 1.19 Temperature Conversion

The thermal conductivity of aluminum at 32°F is 117 Btu/(hr)(ft²)(°F/ft). Find the equivalent value at 0°C in terms of Btu/(hr)(ft²)(K/ft).

Solution

Since 32°F is identical to 0°C, the value is already at the proper temperature. The “°F” in the denominator of the thermal conductivity actually stands for $\Delta^{\circ}\text{F}$, so that the equivalent value is

$$\frac{117 \text{ (Btu)(ft)}}{(\text{hr})(\text{ft}^2)(\Delta^{\circ}\text{F})} \left| \frac{1.8 \Delta^{\circ}\text{F}}{1 \Delta^{\circ}\text{C}} \right| \frac{1 \Delta^{\circ}\text{C}}{1 \Delta\text{K}} = 211 \text{ (Btu)/(hr)(ft}^2\text{)(K/ft)}$$

or with the Δ symbol suppressed,

$$\frac{117 \text{ (Btu)(ft)}}{(\text{hr})(\text{ft}^2)(^{\circ}\text{F})} \left| \frac{1.8^{\circ}\text{F}}{1^{\circ}\text{C}} \right| \frac{1^{\circ}\text{C}}{1 \text{ K}} = 211 \text{ (Btu)/(hr)(ft}^2\text{)(K/ft)}$$

EXAMPLE 1.20 Temperature Conversion

The heat capacity of sulfuric acid in a handbook has the units cal/(g mol)(°C) and is given by the relation

$$\text{heat capacity} = 33.25 + 3.727 \times 10^{-2}T$$

where T is expressed in $^{\circ}\text{C}$. Modify the formula so that the resulting expression gives units of $\text{Btu}/(\text{lb mol})(^{\circ}\text{R})$ and T is in $^{\circ}\text{R}$.

Solution

The units of $^{\circ}\text{C}$ in the denominator of the heat capacity are $\Delta^{\circ}\text{C}$, whereas the units of T are $^{\circ}\text{C}$. First, substitute the proper relation in the formula to convert T in $^{\circ}\text{C}$ to T in $^{\circ}\text{R}$, and then convert the units in the resulting expression to those requested.

$$\begin{aligned} \text{heat capacity} &= \left\{ 33.25 + 3.727 \times 10^{-2} \left[(T_{\text{R}} - 460 - 32) \frac{1}{1.8} \right] \right\} \frac{\text{cal}}{(\text{g mol})(^{\circ}\text{C})} \\ &\times \left| \frac{1 \text{ Btu}}{252 \text{ cal}} \right| \left| \frac{454 \text{ g mol}}{1 \text{ lb mol}} \right| \left| \frac{1^{\circ}\text{C}}{1.8^{\circ}\text{R}} \right| = 23.06 + 2.071 \times 10^{-2} T_{\text{R}} \end{aligned}$$

Self-Assessment Test

- What are the reference points of (a) the Celsius and (b) Fahrenheit scales?
- How do you convert a *temperature difference*, Δ , from Fahrenheit to Celsius?
- Is the unit temperature difference $\Delta^{\circ}\text{C}$ a larger interval than $\Delta^{\circ}\text{F}$? Is 10°C higher than 10°F ?
- In Appendix E, the heat capacity of sulfur is $C_p = 15.2 + 2.68T$, where C_p is in $\text{J}/(\text{g mol})(\text{K})$ and T is in K . Convert so that C_p is in $\text{cal}/(\text{g mol})(^{\circ}\text{F})$ with T in $^{\circ}\text{F}$.
- Complete the following table with the proper equivalent temperatures:

$^{\circ}\text{C}$	$^{\circ}\text{F}$	K	$^{\circ}\text{R}$
-40.0			
	77.0		
		698	
			69.8

- Suppose that you are given a tube partly filled with an unknown liquid and are asked to calibrate a scale on the tube in $^{\circ}\text{C}$. How would you proceed?
- Answer the following questions:
 - In relation to absolute zero, which is higher, 1°C , or 1°F ?
 - In relation to 0°C , which is higher, 1°C , or 1°F ?
 - Which is larger, $1\Delta^{\circ}\text{C}$ or $1\Delta^{\circ}\text{F}$?

Thought Problem

- In reading a report on the space shuttle you find the statement that "the maximum temperature on reentry is 1482.2°C . How many significant figures do you think are represented by this temperature?

1.6 PRESSURE

Your objectives in studying this section are to be able to:

1. Define pressure, atmospheric pressure, barometric pressure, standard pressure, and vacuum.
2. Explain the difference between absolute pressure and relative pressure (gauge pressure).
3. List four ways to measure pressure.
4. Convert from gauge pressure to absolute pressure and the reverse.
5. Convert a pressure measured in one set of units to another set, including kPa, mm Hg, in. H₂O, ft H₂O, atm, in. Hg, and psi using the standard atmosphere or density ratios of liquids.
6. Calculate the pressure from the density and height of a column of fluid.

Pressures, like temperatures, can be expressed by either absolute or relative scales. Pressure is defined as "normal force per unit area." Figure 1.5 shows a cylinder of water. Pressure is exerted on the top of the water by the atmosphere, and on the bottom of the cylinder by the water. If a hole were drilled in the side of the cylinder, you would have to exert a force against the water surface area by means of a plug to prevent the water from exiting the cylinder through the hole.

$$P = \frac{F}{A}$$

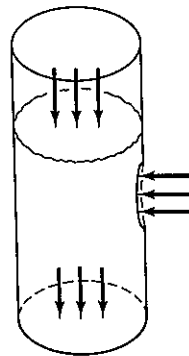


Figure 1.5 Pressure is the normal force per unit area. Arrows show the force exerted on the respective areas.

Figure 1.6 shows a column of mercury held in place by a sealing plate. Recall from physics that at the bottom of the column of mercury

$$p = \frac{F}{A} = \rho gh + p_0 \quad (1.22)$$

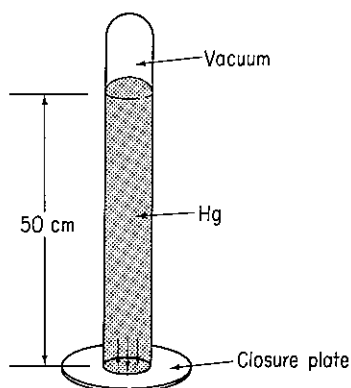


Figure 1.6 Pressure of a column of mercury.

where p = pressure at the bottom of the column of the fluid

F = force

A = area

ρ = density of the fluid

g = acceleration of gravity

h = height of the fluid column

p_0 = pressure at the top of the column of fluid

Suppose that the column of mercury has an area of 1 cm^2 and is 50 cm high. From Table D.1 we can find that the sp gr at 20°C and hence the density, essentially, of the Hg is 13.55 g/cm^3 . Thus the force exerted on the 1-cm^2 section of that plate by the column of mercury is

$$F = \frac{13.55 \text{ g}}{\text{cm}^3} \left| \frac{980 \text{ cm}}{\text{s}^2} \right| \left| \frac{50 \text{ cm}}{1} \right| \left| \frac{1 \text{ cm}^2}{1} \right| \left| \frac{1 \text{ kg}}{1000 \text{ g}} \right| \left| \frac{1 \text{ m}}{100 \text{ cm}} \right| \left| \frac{1 \text{ N}}{1 \text{ (kg)(m)}/\text{s}^2} \right|$$

$$= 6.64 \text{ N}$$

The pressure on the section of the plate covered by the mercury is

$$p = \frac{6.64 \text{ N}}{1 \text{ cm}^2} \left| \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^2 \right| = 6.64 \times 10^4 \frac{\text{N}}{\text{m}^2}$$

If we had started with units in the American engineering system, the pressure would be computed as

$$p = \frac{845.5 \text{ lb}_m}{1 \text{ ft}^3} \left| \frac{32.2 \text{ ft}}{\text{s}^2} \right| \left| \frac{50 \text{ cm}}{1} \right| \left| \frac{1 \text{ in.}}{2.54 \text{ cm}} \right| \left| \frac{1 \text{ ft}}{12 \text{ in.}} \right| \left| \frac{32.174 \text{ (ft)(lb}_m\text{)}}{(\text{s})^2 \text{ (lb}_f\text{)}} \right|$$

$$= 1388 \frac{\text{lb}_f}{\text{ft}^2}$$

Sometimes in engineering practice, a liquid column is referred to as **head of liquid**, the head being the height of the column of liquid. A **pressure can be interpreted as**

the head, h , of a hypothetical column of liquid, so that the pressure at the base of a column of liquid would be calculated, say in inches of mercury, as

$$p \text{ (in inches of Hg)} = P_0 \text{ (in inches of Hg)} + h \text{ (in inches of Hg)} \quad (1.23)$$

where p_0 is the gas pressure on the top of the column of mercury. Water, oil, or any other liquid could be substituted for Hg in Eq. (1.23). For example, suppose that you measure the pressure in a tank of water 2 ft down from the surface as 404.92 in. H_2O . What is the pressure 4 ft down from the surface? You can use Eq. (1.22) to calculate the pressure, but it is just as easy to use Eq. (1.23).

$$p = 404.92 \text{ in. H}_2\text{O} + 2 \text{ ft (12 in./ft)} = 428.92 \text{ in. H}_2\text{O} \quad (1.24)$$

Whether relative or absolute pressure is measured in a pressure-measuring device depends on the nature of the instrument used to make the measurements. For example, an open-end manometer (Fig. 1.7a) would measure a **relative pressure**, since the reference for the open end is the pressure of the atmosphere at the open end of the manometer. On the other hand, closing off the end of the manometer (Fig. 1.7b) and creating a vacuum in the end results in a measurement against a complete vacuum, or against "no pressure." This measurement is called **absolute pressure**. Since absolute pressure is based on a complete vacuum, a fixed reference point which is unchanged regardless of location or temperature or weather or other factors, absolute pressure then establishes a precise, invariable value which can be readily identified. Thus, the zero point for an absolute pressure scale corresponds to a perfect vacuum, whereas the zero point for a relative pressure scale usually corresponds to the pressure of the air which surrounds us at all times, and as you know, varies slightly.

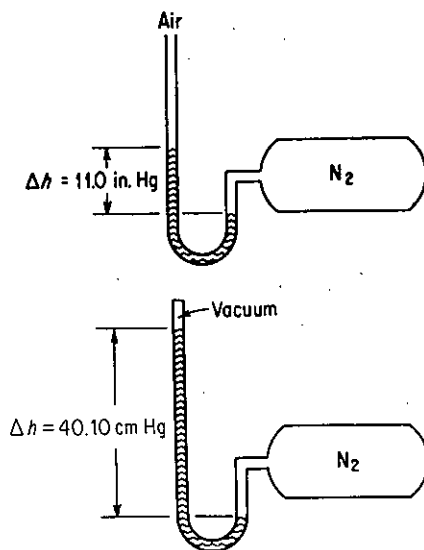


Figure 1.7 (a) Open-end manometer showing a pressure above atmospheric pressure. (b) Absolute pressure manometer.

If a mercury reading is set up as illustrated in Fig. 1.8, with the dish open to the atmosphere, the device is called a *barometer* and the reading of atmospheric pressure is termed **barometric pressure**.

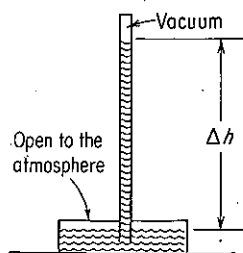


Figure 1.8 A barometer.

An understanding of the principle upon which a manometer operates will aid you in recognizing the nature of the pressure measurement taken from it. As shown in Fig. 1.7a for an open-end U-tube manometer, if the pressure measured for the N_2 is greater than atmospheric, the liquid is forced downward in the leg to which the pressure source is connected and upward in the open leg. Eventually, a point of hydrostatic balance is reached in which the manometer fluid stabilizes. The pressure exerted at the bottom of the U-tube in the part of the tube open to the atmosphere exactly balances the pressure exerted at the bottom of the U-tube in the part of the tube connected to the tank of N_2 . Furthermore, the difference between the height of fluid in the leg open to the atmosphere and the height of fluid in the leg attached to the tank is exactly equal to the difference between the pressure in the tank and atmospheric pressure (expressed in units of fluid column length). If some degree of vacuum instead of a high pressure existed in a tank in an arrangement similar to that shown in Fig. 1.7a, the fluid column would rise on the vacuum side. Again, the difference in pressure between the pressure source in the tank and atmospheric pressure would be measured by the difference in the height of the two legs of fluid. Water and mercury are commonly used indicating fluids for manometers; the readings thus can be expressed in "inches or cm of water," "inches or cm of mercury," "mm of fluid flowing," and so on. (In ordinary engineering calculations we ignore the vapor pressure of mercury and minor changes in the density of mercury due to temperature changes in making pressure measurements.)

Another type of common measuring device is the visual *Bourdon gauge* (Fig. 1.9), which normally (but not always) reads zero pressure when open to the atmosphere. The pressure-sensing device in the Bourdon gauge is a thin metal tube with an elliptical cross section closed at one end which has been bent into an arc. As the pressure increases at the open end of the tube, it tries to straighten out, and the movement of the tube is converted into a dial movement by gears and levers. Figure 1.9 also illustrates a diaphragm capsule gauge. Figure 1.10 indicates the pressure ranges for the various pressure-measuring devices.

Pressure scales may be temporarily somewhat more confusing than temperature scales since **the reference point or zero point for the relative pressure scales is not constant**, whereas in the temperature scales the boiling point or the freezing point of water is always a fixed value. However, you will become accustomed to this feature with practice.

The relationship between relative and absolute pressure is illustrated in Figs. 1.11 and 1.12 and is given by the following expression:

$$\text{gauge pressure} + \text{barometer pressure} = \text{absolute pressure} \quad (1.25)$$

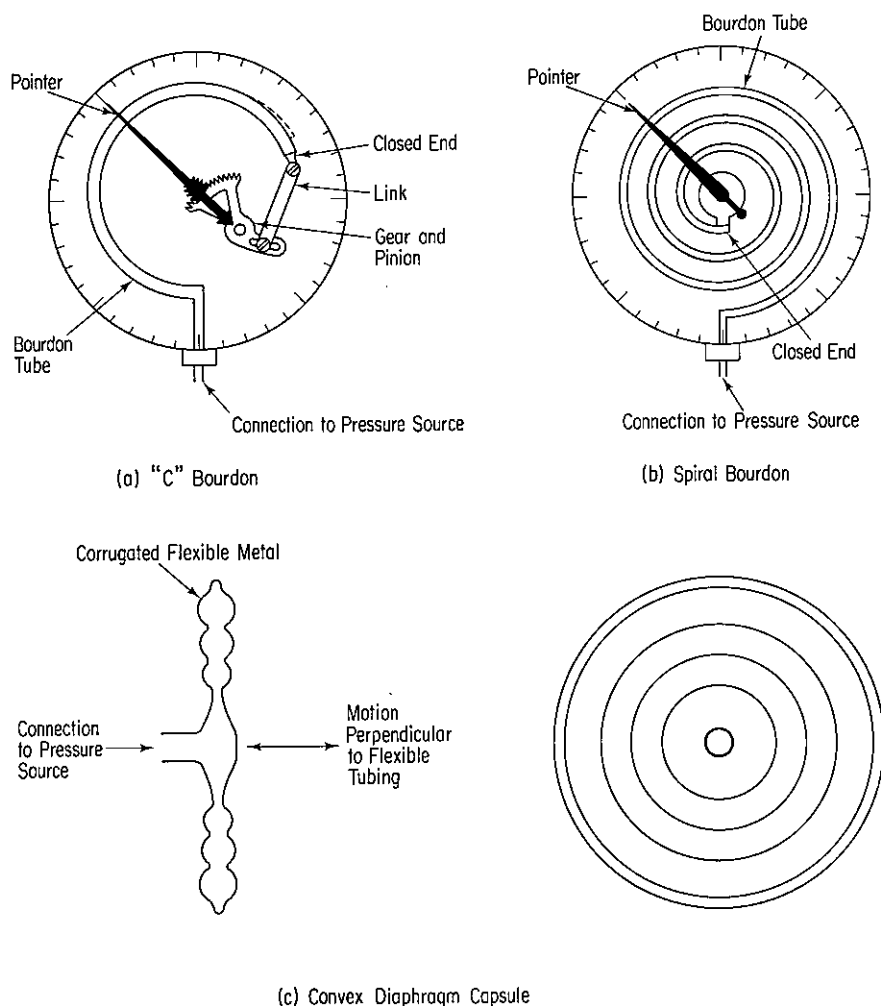


Figure 1.9 Bourdon and diaphragm pressure-measuring devices.

Equation (1.25) can be used only with consistent units. Note that you must add the atmospheric pressure (i.e., the barometric pressure) to the gauge, or relative pressure (or manometer reading if open on one end), in order to get the absolute pressure.

Another term applied in measuring pressure which is illustrated in Figs. 1.11 and 1.12 is **vacuum**. In effect, when you measure pressure as "inches of mercury vacuum," you reverse the usual direction of measurement and measure from the barometric pressure down to zero absolute pressure, in which case a perfect vacuum would be the highest vacuum that you could achieve. The vacuum system of measurement of pressure is commonly used in apparatus which operate at pressures less than atmospheric—such as a vacuum evaporator or vacuum filter. A pressure that is only slightly below barometric pressure may sometimes be expressed as a "draft"

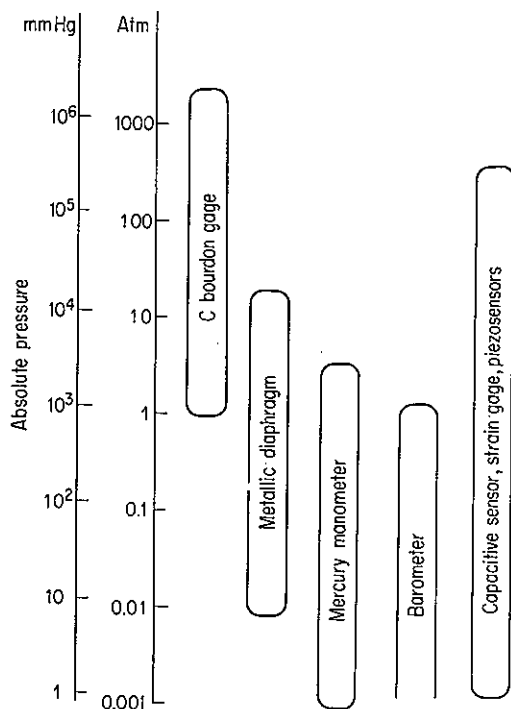


Figure 1.10 Ranges of application for pressure-measuring devices.

Pounds per square inch			Inches mercury		Pascals, newtons per square meter	
5.0	19.3	A pressure above atmospheric	39.3	10.2	0.34 × 10 ⁵	1.33 × 10 ⁵
0.4	14.7	Standard pressure	29.92	0.82	0.028 × 10 ⁵	1.013 × 10 ⁵
0.0	14.3	Barometric pressure	29.1	0 0.0	0.00	0.985 × 10 ⁵
-2.45	11.85	A pressure below atmospheric	24.1	-5.0 5.0	0.17 × 10 ⁵	0.82 × 10 ⁵
Gage pressure ↑ Absolute pressure			Absolute pressure ↑ Gage pressure ↓ Vacuum		Gage pressure ↑ Absolute pressure	
-14.3	0.0	Perfect vacuum	0	-29.1 29.1	-0.985 × 10 ⁵	0.00

Figure 1.11 Pressure comparisons when barometer reading is 29.1 in Hg.

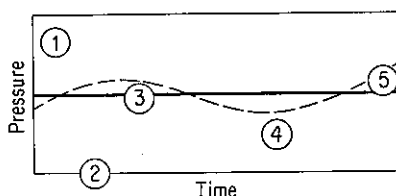


Figure 1.12 Pressure terminology. The standard atmosphere is shown by the heavy horizontal line. The dashed line illustrates the atmospheric (barometric) pressure, which changes from time to time. Point ① in the figure is a pressure of 19.3 psi referred to a complete vacuum or 5 psi referred to the barometric pressure; ② is the complete vacuum, ③ represents the standard atmosphere, and ④ illustrates a negative relative pressure or a pressure less than atmospheric. This type of measurement is described in the text as a vacuum type of measurement. Point ⑤ also indicates a vacuum measurement, but one that is equivalent to an absolute pressure above the standard atmosphere.

(which is identical to the vacuum system) in inches of water, as, for example, in the air supply to a furnace or a water cooling tower.

As to the units of pressure, Fig. 1.11 shows three common systems: pounds per square inch (psi), inches of mercury (in. Hg), and pascals. Pounds per square inch absolute is normally abbreviated "psia," and "psig" stands for "pounds per square inch gauge." For other units, be certain to carefully specify whether they are gauge or absolute; for example, state "300 kPa absolute" or "12 cm Hg gauge." Figure 1.11 compares the relative and absolute pressure scales in terms of the three systems of measurement when the barometric pressure is 29.1 in Hg (14.3 psia, 0.985×10^5 pascal). Other systems of expressing pressure exist; in fact, you will discover that there are as many different units of pressure as there are means of measuring pressure. Some of the other most frequently used systems are

- (a) Millimeters of mercury (mm Hg)
- (b) Feet of water (ft H₂O)
- (c) Atmospheres (atm)
- (d) Bars (bar): 100 kPa = 1 bar
- (e) Kilograms (force) per square centimeter (kg/cm²)—a common but theoretically prohibited measure⁶

To sum up our discussion of pressure and its measurement, you should now be acquainted with:

- (a) *Atmospheric pressure*—the pressure of the air and the atmosphere surrounding us which changes from day to day

⁶ Although the units of pressure in the SI system are pascal, you frequently find in common usage pressures denoted in the units of kg/cm² (i.e., kg_f/cm²), units that have to be multiplied by

$$9.80 \times 10^4 \left(\frac{\text{kg}_m(\text{m})}{(\text{kg}_f)(\text{s}^2)} \right) \left(\frac{\text{cm}^2}{\text{m}^2} \right) \text{ to get pascal.}$$

- (b) *Barometric pressure*—the same as atmospheric pressure, called “barometric pressure” because a barometer is used to measure atmospheric pressure
- (c) *Absolute pressure*—a measure of pressure referred to a complete vacuum, or zero pressure
- (d) *Gauge pressure*—pressure expressed as a quantity measured upward from atmospheric pressure (or some other reference pressure)
- (e) *Vacuum*—a method of expressing pressure measured downward from atmospheric pressure (or some other reference pressure)

You definitely must not confuse the standard atmosphere with atmospheric pressure. The *standard atmosphere* is defined as the pressure (in a standard gravitational field) equivalent to 1 atm or 760 mm Hg at 0°C or other equivalent value, whereas atmospheric pressure is a variable and must be obtained from a barometer each time you need it. The standard atmosphere may not equal the barometric pressure in any part of the world except perhaps at sea level on certain days, but it is extremely useful in converting from one system of pressure measurement to another (as well as being useful in several other ways to be considered later).

Expressed in various units, the *standard atmosphere* is equal to

- 1.000 atmospheres (atm)
- 33.91 feet of water (ft H₂O)
- 14.7 (14.696, more exactly) pounds per square inch absolute (psia)
- 29.92 (29.921, more exactly) inches of mercury (in. Hg)
- 760.0 millimeters of mercury (mm Hg)
- 1.013×10^5 pascal (Pa) or newtons per square meter (N/m²); or 101.3 kPa

You can convert from one set of pressure measurements to another by using pairs of standard atmospheres as conversion factors, as shown in the examples below. If pressures are measured by a height of a column of one liquid A, and p_0 is the same for both columns, you can convert the pressure to the height of another column of liquid B by the use of Eq. (1.22).

$$p_0 + \rho_A g h_A = p_0 + \rho_B g h_B$$

or

$$\frac{h_A}{h_B} = \frac{\rho_B}{\rho_A}$$

How can you arrange to get the same p_0 for each column? Will leaving the upper end of each of the columns open to the atmosphere accomplish this objective?

EXAMPLE 1.21 Pressure Conversion

Convert 35 psia to inches of mercury.

Solution

Use the ratio of 14.7 psia to 29.92 in. Hg, an identity, to carry out the conversion.

Basis: 35 psia

$$\frac{35 \text{ psia}}{1} \times \frac{29.92 \text{ in. Hg}}{\underbrace{14.7 \text{ psia}}_{\text{an identity}}} = 71.24 \text{ in. Hg}$$

EXAMPLE 1.22 Pressure Conversion

The density of the atmosphere decreases with increasing altitude. When the pressure is 340 mm Hg, how many inches of water is it? How many kilopascal?

Solution

Basis: 340 mm Hg

$$\frac{340 \text{ mm Hg}}{1} \times \frac{33.91 \text{ ft H}_2\text{O}}{760 \text{ mm Hg}} \times \frac{12 \text{ in.}}{1 \text{ ft}} = 182 \text{ in. H}_2\text{O}$$

$$\frac{340 \text{ mm Hg}}{1} \times \frac{101.3 \text{ kPa}}{760.0 \text{ mm Hg}} = 45.3 \text{ kPa}$$

EXAMPLE 1.23 Pressure Conversion

The pressure gauge on a tank of CO₂ used to fill soda-water bottles reads 51.0 psi. At the same time the barometer reads 28.0 in. Hg. What is the absolute pressure in the tank in psia? See Fig. E1.23.

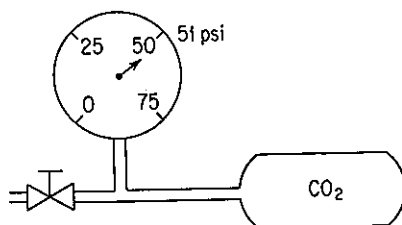


Figure E1.23

Solution

The pressure gauge is reading psig, not psia. From Eq. (1.25) the absolute pressure is the sum of the gauge pressure and the atmospheric (barometric) pressure expressed in the same units. We will change the atmospheric pressure to psia.

Basis: Barometric pressure = 28.0 in. Hg

$$\text{atmospheric pressure} = \frac{28.0 \text{ in. Hg}}{1} \times \frac{14.7 \text{ psia}}{29.92 \text{ in. Hg}} = 13.76 \text{ psia}$$

(Note: Atmospheric pressure does not equal 1 standard atm.) The absolute pressure in the tank is

$$51.0 + 13.76 = 64.8 \text{ psia}$$

In some instances the fluids in the legs of the manometer are not the same. Examine Fig. 1.13. When the columns of fluids are at equilibrium (it may take some time!) the relation between p_1 , p_2 and the heights of the various columns of fluid is

$$p_1 + \rho_1 d_1 g = p_2 + \rho_2 g d_2 + \rho_3 g d_3 \quad (1.26a)$$

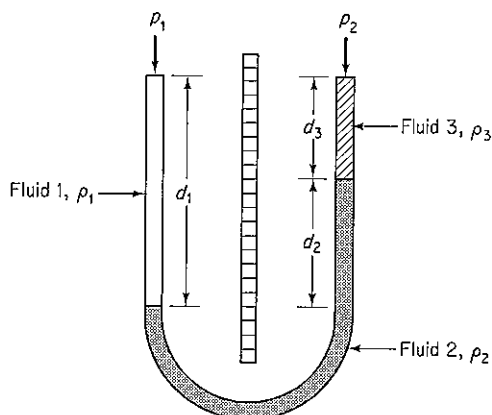


Figure 1.13 Manometer with three fluids.

Can you show for the case in which $\rho_1 = \rho_3 = \rho$ that the manometer expression reduces to

$$p_1 - p_2 = (\rho_2 - \rho) g d_2 \quad (1.26b)$$

Finally, suppose that fluids 1 and 3 are gases. Can you ignore the gas density ρ relative to the manometer fluid density? For what types of fluids?

EXAMPLE 1.24 Pressure Conversion

Air is flowing through a duct under a draft of 4.0 cm H₂O. The barometer indicates that the atmospheric pressure is 730 mm Hg. What is the absolute pressure of the gas in inches of mercury? See Fig. E1.24.

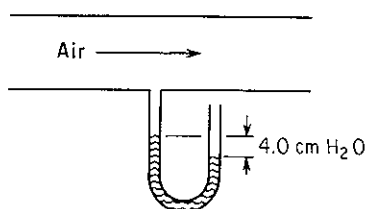


Figure E1.24

Solution

We can ignore the gas density above the manometer fluid. In the calculations we have to employ consistent units, and it appears in this case that the most convenient units are those of inches of mercury.

Basis: 730 mm Hg

$$\text{atmospheric pressure} = \frac{730 \text{ mm Hg}}{760 \text{ mm Hg}} \left| \frac{29.92 \text{ in. Hg}}{1} \right| = 28.7 \text{ in. Hg}$$

Basis: 4.0 cm H₂O draft (under atmospheric)

$$\frac{4.0 \text{ cm H}_2\text{O}}{2.54 \text{ cm}} \left| \frac{1 \text{ in}}{12 \text{ in.}} \right| \left| \frac{29.92 \text{ in. Hg}}{33.91 \text{ ft H}_2\text{O}} \right| = 0.12 \text{ in. Hg}$$

What is another way to make the conversion?

Since the reading is 4.0 cm H₂O draft (under atmospheric), the absolute reading in uniform units is

$$28.7 - 0.12 = 28.6 \text{ in. Hg absolute}$$

EXAMPLE 1.25 Vacuum Pressure Reading

Small animals such as mice can live at reduced air pressures down to 20 kPa (although not comfortably). In a test a mercury manometer attached to a tank as shown in Fig. E1.25 reads 64.5 cm Hg and the barometer reads 100 kPa. Will the mice survive?

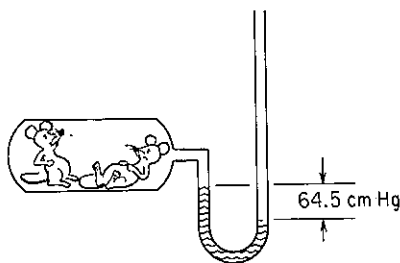


Figure E1.25

Solution

Basis: 64.5 cm Hg below atmospheric

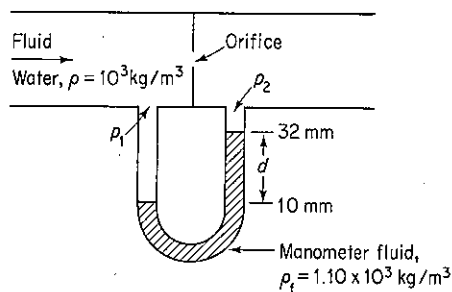
We ignore any temperature corrections to convert the mercury density and also ignore the gas density above the manometer fluid. Then, since the vacuum reading on the tank is 64.5 cm Hg below atmospheric, the absolute pressure in the tank is

$$100 \text{ kPa} - \frac{64.5 \text{ cm Hg}}{76.0 \text{ cm Hg}} \left| \frac{101.3 \text{ kPa}}{1} \right| = 100 - 86 = 14 \text{ kPa absolute}$$

The mice probably will not survive.

EXAMPLE 1.26 Calculation of Pressure Difference

In measuring the flow of fluids in a pipeline, a differential manometer, as shown in Fig. E1.26, can be used to determine the pressure difference across an orifice plate. The flow rate can be calibrated with the observed pressure drop. Calculate the pressure drop $p_1 - p_2$ in pascal.

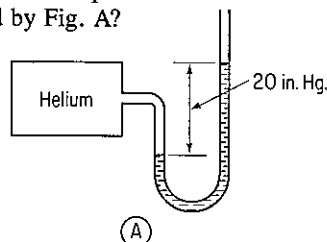

Figure E1.26
Solution

In this problem we cannot ignore the water density above the manometer fluid. Apply Eq. (1.26b), as the densities of the fluids above the manometer fluid are the same.

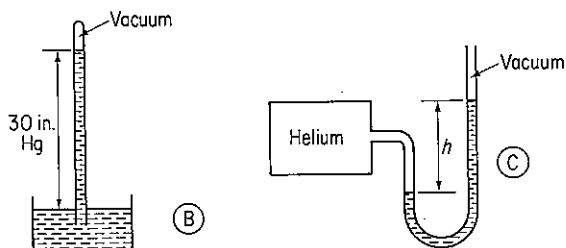
$$\begin{aligned}
 p_1 - p_2 &= (\rho_f - \rho)gd \\
 &= \frac{(1.10 - 1.00)10^3 \text{ kg}}{\text{m}^3} \left| \frac{9.807 \text{ m}}{\text{s}^2} \right| \left| \frac{(22)(10^{-3})\text{m}}{1} \right| \left| \frac{1 \text{ (N)(s}^2\text{)}}{(\text{kg})(\text{m})} \right| \left| \frac{1 \text{ (Pa)(m}^2\text{)}}{1 \text{ (N)}} \right| \\
 &= 21.6 \text{ Pa}
 \end{aligned}$$

Self-Assessment Test

- Write down the equation to convert gauge pressure to absolute pressure.
- List the values and units of the standard atmosphere for six different methods of expressing pressure.
- List the equation to convert vacuum pressure to absolute pressure.
- Convert a pressure of 800 mm Hg to the following units:
 - psia
 - kPa
 - atm
 - ft H₂O
- Your textbook lists five types of pressures: atmospheric pressure, barometric pressure, gauge pressure, absolute pressure, and vacuum pressure.
 - What kind of pressure is measured by Fig. A?



- (b) What kind of pressure is measured by Fig. B?
 (c) What would be the reading in Fig. C assuming that the pressure and temperature inside and outside the helium tank are the same as in parts (a) and (b)?



6. An evaporator shows a reading of 40 kPa vacuum. What is the absolute pressure in the evaporator in kilopascal?
 7. Answer the following questions true (T) or false (F).
 (a) Air flows in a pipeline, and a manometer containing Hg that is set up as illustrated in Fig. E1.26 shows a differential pressure of 14.2 mm Hg. You can ignore the effect of the density of the air on the height of the columns of mercury.
 (b) Lowering the He pressure 10% in Fig. A of question 5 will not cause the length of the column of Hg to decrease by 10%.

Thought Problems

1. A magic trick is to fill a glass with water, place a piece of paper over the top of the glass to cover the glass completely, and hold the paper in place as the glass is inverted 180° . On the release of your support of the paper, no water runs out! Many books state that the glass should be completely filled with water with no air bubbles present. Then the outside air pressure is said to oppose the weight of the water in the inverted glass. However, the experiment works just as well with a half-filled glass. The trick does not work if a glass plate is substituted for the piece of paper. Can you explain why?
2. A large storage tank was half full of a flammable liquid quite soluble in water. The tank needed maintenance on the roof. Since welding was involved, the foreman attached to the vent pipe on the top of the tank (in which there was a flame arrestor) a flexible hose and inserted the end of the hose into the bottom of a drum of water sitting on the ground to pick up any exhaust vapors. When the tank was emptied, the water rose up in the hose, and the tank walls collapsed inward.

What went wrong in this incident?

1.7 PHYSICAL AND CHEMICAL PROPERTIES OF COMPOUNDS AND MIXTURES

Your objectives in studying this section are to be able to:

1. Cite five sources of physical property data.
2. Go to the library and locate the five sources of data.

3. Retrieve data from the five sources to use in your calculations, employing the indexes and tables of contents.

Accurate values of physical properties are needed in almost all phases of chemical engineering design and analysis. Two ways to obtain data for the physical properties of components are: (a) by experimental measurement and/or (b) by estimation. As far as the cost of obtaining data is concerned, a substantial difference exists between the two approaches. To produce critically evaluated estimates, the time required is seldom more than a few hours, in comparison with several months needed for the experimental work. If the data you want are based on valid estimation and correlation techniques, you can bypass the expense and the waiting time required for experimental measurements.

Many of the materials we talk about and use every day are not pure compounds, but nevertheless you can obtain information about the properties of such materials. Data on materials such as coal, coke, petroleum products, and natural gas—which are the main sources of energy in this country—are available in reference books and handbooks. Some typical examples of analyses are shown in Tables 1.6 to 1.9. In the back of this book you will discover a number of appendices from which you can retrieve data necessary to solve most (but not all) of the problems at the end of the chapters. Scan through the appendices now. When you need to look

TABLE 1.6 Refinery Biological Treatment Unit Feed Characteristics

	Ranges reported*
Chlorides (mg/L)	200–960
COD (mg/L)	140–640
BOD ₅ (mg/L)	97–280
Suspended solids (mg/L)	80–450
Alkalinity (mg/L as CaCO ₃)	77–210
Temperature, (°F)	69–100
Ammonia, nitrogen (mg/L)	56–120
Oil (mg/L)	23–130
Phosphate (mg/L)	20–97
Phenolics (mg/L)	7.6–61
pH	7.1–9.5
Sulfides (mg/L)	1.3–38
Chromium (mg/L)	0.3–0.7

*Values are the averages of the minima and maxima reported by 12 refineries treating total effluent. Individual plants have reported data well outside many of these ranges.

SOURCE: *Manual on Disposal of Refinery Wastes*, American Petroleum Institute, New York, 1969, pp. 2–4.

TABLE 1.7 Typical Dry Gas Analyses

Type	Analysis (vol. %—excluding water vapor)									
	CO ₂	O ₂	N ₂	CO	H ₂	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀	C ₅ H ₁₂ [†]
Natural gas	6.5					77.5	16.0			
Natural gas, dry*	0.2		0.6			99.2				
Natural gas, wet*	1.1					87.0	4.1	2.6	2.0	3.4
Natural gas, sour†	(H ₂ S 6.4)					58.7	16.5	9.9	5.0	3.5
Butane							2.0	3.5	75.4 <i>n</i> -butane 18.1 isobutane	
Reformed refinery oil	2.3	0.7	4.9	20.8	49.8	12.3	5.5	5.5	3.7	
Coal gas, by-product	2.1	0.4	4.4	13.5	51.9	24.3				
Producer gas	4.5	0.6	50.9	27.0	14.0	3.0			3.4	
Blast furnace gas	5.4	0.7	8.3	37.0	47.3	1.3				
Sewage gas	22.0		6.0		2.0	68.0				

*Dry gas contains much less propane (C₃H₈) and higher hydrocarbons than wet gas does.

†Sour implies that the gas contains significant amounts of hydrogen sulfide.

SOURCE: *Fuel Flue Gases*, American Gas Association, New York, 1941, p. 20.

TABLE 1.8 Ultimate Analysis of Petroleum Crude

Type	Sp Gr	At °C	Weight %				
			C	H	N	O	S
Pennsylvania	0.862	15	85.5	14.2			
Humbolt, Kan.	0.921		85.6	12.4			0.37
Beaumont, Tex.	0.91		85.7	11.0	2.61		0.70
Mexico	0.97	15	83.0	11.0	1.7		
Baku USSR	0.897		86.5	12.0		1.5	

SOURCE: Data from W.L. Nelson, *Petroleum Refinery Engineering*, 4th ed., McGraw-Hill, New York, 1958.

TABLE 1.9 Chemical Analyses of Various Wastes

Material	Raw paper	Charred paper	Tire rubber	Dry sewage sludge	Charred sewage sludge	Charred animal manure	Garbage composite A	Garbage composite B
Moisture	3.8	0.8	0.5	13.6	1.2	0.0	3.4	12.3
Hydrogen*	6.9	3.1	4.3	6.7	1.4	5.4	6.6	7.0
Carbon	45.8	84.9	86.5	28.7	48.6	41.2	57.3	44.4
Nitrogen	—	0.1	—	2.6	3.7	1.5	0.5	0.4
Oxygen*	46.8	8.5	4.6	26.5	—	26.0	22.1	42.1
Sulfur	0.1	0.1	1.2	0.6	—	0.4	0.4	0.2
Ash	0.4	2.5	3.4	34.9	45.7	25.5	10.2	5.9

* The hydrogen and oxygen values reflect that due to both the presence of water and that contained within the moisture-free material.

for additional data, consult some of the appropriate references listed in Tables 1.10 and 1.11.

In the next decade, we should expect two new modes of information retrieval to be implemented and available for an individual. With the widespread introduction of modems and communication software, potential users will be able to deal directly with data base vendors, both profit and nonprofit, at reasonable cost via local phone

TABLE 1.10 Sources of Physical Property Data

- American Chemical Society, *Physical Properties of Chemical Compounds*, Vol. 22, ACS, Washington, D.C., 1959.
- American Chemical Society, *Chemical Abstracts Service*, ACS, Washington, D.C. (Continuing printed, microform, and on-line electronic information service with over 12 million abstracts.)
- American Gas Association, *Fuel Flue Gases*, AGA, New York, 1941.
- American Petroleum Institute, *Technical Data Book—Petroleum Refining*, New York, 1970.
- Aylward, G. H., and T. J. V. Findlay, *SI Chemical Data*, 2nd ed., Wiley, New York, 1976.

TABLE 1.10 (continued)

-
- Chemsoft Inc., *Chemical Compounds Data Bank*, Gulf Publishing Company, Houston, 1987. (500 compounds on disc.)
- Chen, C.-C. *Scientific and Technical Information Sources*, 2nd ed., MIT Press, Cambridge, Mass. 1987. (Reference book for publications.)
- Daubert, T. E., and R. P. Danner, *Data Compilation—Tables of Properties of Pure Compounds*, American Institute of Chemical Engineers, New York, 1985. (Properties and prediction equations for 1100 compounds; available as printed book and magnetic tape.)
- Dechema, *Chemistry Data Series*, Deutsche Gesellschaft für Chemisches Apparatewesen e.v., Frankfurt, West Germany, (Continuing series on physical and thermodynamic properties.) Also an on-line data base; see National Technical Information Service document BMFT-FB-ID-80-006.
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-

TABLE 1.11 Professional Journals

Archival journals

AIChE Journal
Angewandte Chemie
Canadian Journal of Chemical Engineering
Chemical Engineering Communications
Chemical Engineering Fundamentals
Chemical Engineering Journal (Lausanne)
Chemical Engineering Research and Design
Chemical Engineering Science
Chemical Reviews
CODATA Bulletin
Journal of Chemical and Engineering Data
Journal of Chemical Engineering of Japan
Journal of Chemical Technology and Biotechnology
Journal of the American Chemical Society
Journal of the Chinese Institute of Chemical Engineers

Other journals and magazines

Chemical Engineering
Chemical Engineering Progress
Chemical Engineer (London)
Chemical Processing
Chemical Technology
Chemical Technology Review
Chemie-Ingenieur-Technik
Chemistry and Industry (London)
German Chemical Engineering
International Chemical Engineering

calls. The other mode is information retrieval from compact read-only laser disks, which can store the order of 300,000 printed pages of material. Coupled with a microcomputer, a user will be able to search for and locate reliable information distributed by specialized professional organizations on a myriad of subjects, such as refrigerants, propellants, solvents, drugs, insecticides, and so on.

Self-Assessment Test

1. What are five sources of data on physical properties from reference books? What are two data banks that provide information on physical properties?
2. In what reference book might you find data on:
 - (a) Boiling point of inorganic liquids?
 - (b) Gas compositions for refinery gases?
 - (c) Vapor pressures of organic liquids?
 - (d) Chemical formula and properties of protocatechuic acid (3-, 4-)?
3. List the page numbers on which the items described in question 2 can be found in the reference you select.

Library Problem

1. Prepare a short description (about five lines) for numbers _____ of the following terms. You may include some sketches if you want. List the name of the author, book, and publisher and also the call number of the books that you use to prepare your report.
 - (1) Furnace
 - (2) Boiler
 - (3) Heat exchanger
 - (4) Distillation column
 - (5) Absorption tower
 - (6) Adsorption
 - (7) Liquid-liquid extraction
 - (8) Leaching process
 - (9) Evaporator
 - (10) Chemical reactor

1.8 TECHNIQUES OF PROBLEM SOLVING

Your objectives in studying this section are to be able to:

1. Memorize the eight components of effective problem solving.
2. Apply the components to all types of problems.

Weiler's Law: Nothing is impossible for the person who doesn't have to do it.

Howe's Law: Every person has a scheme which will not work.

The 90/90 Law: The first 10% of the task takes 90% of the time. The remaining 90% takes the remaining 10%.

Gordon's Law: If a project is not worth doing, it's not worth doing well.

Slack's Law: The least you will settle for is the most you can expect to get.

O'Toole's Commentary: Murphy was an optimist.⁷

One of the main objectives of this book is to enhance your problem-solving skills. If you can form good habits of problem solving early in your career, you will save considerable time and avoid many frustrations in all aspects of your work, in and out of school. Being able to solve material and energy balances means that in addition to learning basic principles, formulas, laws, and so on, *you must be able to apply them*. Routine substitution of data into an appropriate equation will by no

⁷ Arthur Bloch, *Murphy's Law and Other Reasons Why Things Go Wrong*, Price/Stern/Sloan, New York, 1977.

means be adequate to solve any material and energy balances other than the most trivial ones.

Polya⁸ recommends the use of four steps for solving problems and puzzles: define, plan, carry out the plan, and look back. The key features of this strategy are the interaction among the steps and the interplay between critical and creative thinking. A number of other helpful references are listed at the end of this chapter if you need assistance. Figure 1.14 sketches the general concept involved in problem solv-

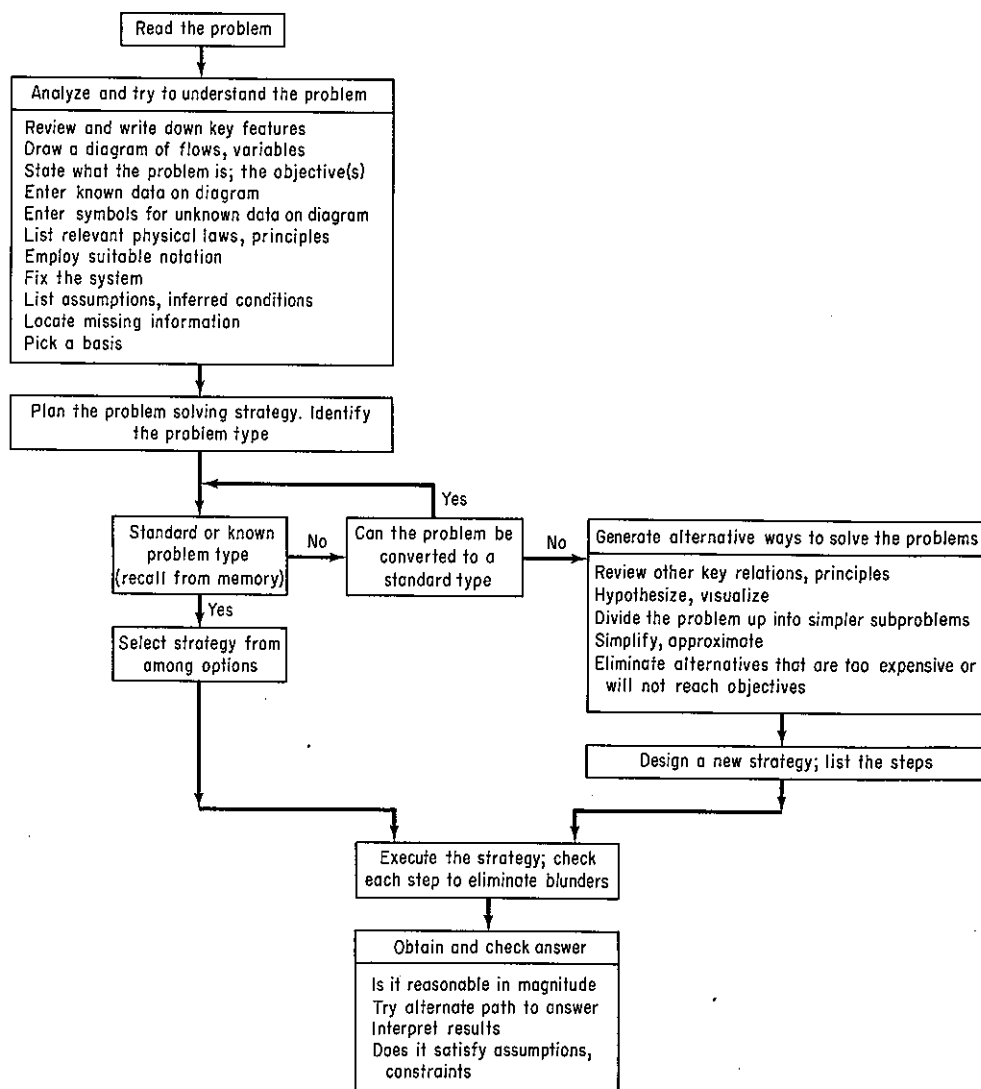


Figure 1.14 Program for problem solving.

⁸G. Polya, *How to Solve It*, 2nd ed., Doubleday, New York, 1957.

ing. First, you must identify what result you are to achieve, that is, what the problem is. Then you must define the system, perhaps with the aid of a diagram. Various physical constraints will apply as well as the time available for you to work on the solution. In almost all cases you will have to look up data and make use of general laws. Finally, the results will have to be presented properly so that you can communicate them to someone else.

A more detailed checklist for solving material and energy balance problems is as follows:

- (a) Read the available information through thoroughly and understand what is required for an answer. Sometimes, as in life, the major obstacle is to find out what the problem really is.
- (b) Determine what additional data are needed, if any, and obtain this information.
- (c) Draw a simplified picture of what is taking place and write down the available data. You may use boxes to indicate processes or equipment, and lines for the flow streams.
- (d) Pick a basis on which to start the problem, as discussed in Sec. 1.4.
- (e) If a chemical equation is involved, write it down and make sure that it is balanced.

By this time you should have firmly in mind what the problem is and a reasonably clear idea of what you are going to do about it; however, if you have not seen exactly how to proceed from what is available to what is wanted, then you should:

- (f) Decide what formulas or principles are governing in this specific case and what types of calculations and intermediate answers you will need to get the final answer. If alternative procedures are available, try to decide which are the most expedient. If an unknown cannot be found directly, give it a letter symbol, and proceed as if you knew it.
- (g) Make the necessary calculations in good form, being careful to check the arithmetic and units as you proceed.
- (h) Determine whether the answer seems reasonable in view of your experience with these types of calculations.

You can work backward as well as forward in solving problems if the forward sequence of steps to take is not initially clear. Problems that are long and involved should be divided into parts and attacked systematically piece by piece. Additional suggestions with respect to problem solving appear in Tables 2.1, 2.2, and 2.3. Glance through these tables now.

If you can assimilate the procedure discussed above and make it a part of yourself—so that you do not have to think about it step by step—you will find that you will be able to materially improve your speed, performance, and accuracy in problem solving. Novice prospective engineers usually have the most difficulty with:

- (a) Translating the physical problem into a mathematical description.
- (b) Relating the physical problem to theory
- (c) Simplifying a complex problem
- (d) Learning how to ask the right questions

Look at Table 1.12 for the differences in technique between expert and novice engineers when solving problems.

Finally, we need to remark on the quality of data; the following quotation still holds true.

The major difference between problems solved in the classroom and in the plant lies in the quality of the data available for the solution. Plant data may be of poor quality, in-

TABLE 1.12 A Comparison of the Problem-Solving Habits of a Novice and an Expert

A novice:	An expert:
Starts solving a problem before fully understanding what is wanted and/or what a good route for solution will be	Reviews the entire plan outlined in Fig. 1.14, mentally explores alternative strategies, and clearly understands what result is to be obtained
Focuses only on a known problem set that he or she has seen before and tries to match the problem with one in the set	Concentrates on similarities to and differences from known problems; uses generic principles rather than problem matching
Chooses one procedure without exploring alternatives	Examines several procedures serially or in parallel
Emphasizes speed of solution, unaware of blunders	Emphasizes care and accuracy in the solution
Does not follow an organized plan of attack such as outlined in Fig. 1.14, jumps about, and mixes problem-solving strategies	Goes through the problem-solving process step by step, checking, reevaluating, and recycling from dead ends to another valid path
Is unaware of missing data, concepts, laws	Knows what principles might be involved and where to get missing data
Exhibits bad judgment, makes unsound assumptions, poor approximations	Carefully evaluates the necessary assumptions and approximations
Gives up solving the problem because he or she does not know enough	Knows what the difficulty is and is willing to learn more that will provide the information needed
Gives up solving the problem because he or she does not have skills to branch away from a dead-end strategy	Aware that a dead end may exist for a strategy and has planned alternative strategies if a dead end is reached

conclusive, inadequate, or actually conflicting, depending on the accuracy of sampling, the type of analytical procedures employed, the skill of the technicians in the operation of analytical apparatus, and many other factors. The ability of an engineer to use the stoichiometric principles for the calculation of problems of material balance is only partly exercised in the solution of problems, even of great complexity, if solved from adequate and appropriate data. The remainder of the test lies in his ability to recognize poor data, to request and obtain usable data, and, if necessary, to make accurate estimates in lieu of incorrect or insufficient data.⁹

Self-Assessment Test

1. Prepare an information flow diagram showing the sequence (serial and parallel) of eight steps to be used in effective problem solving.
2. Take any example in Sec. 1.3, 1.5, or 1.6 and prepare an information flow diagram of your thought process in solving the example. Make a tree showing how the following classes of information are connected (put the solution as the last stage of the tree at the bottom):
 - (a) Information stated in the problem
 - (b) Information implied or inferred from the problem statement
 - (c) Information from your memory (internal data bank!)
 - (d) Information from an external data bank (reference source)
 - (e) Information determined by reasoning or calculationsLabel each class with a different-type box (circle, square, diamond, etc.) and let arrows connect the boxes to show the sequence of information flow for your procedure.
3. What should you do if you experience the following difficulties in solving problems?
 - (a) No interest in the material and no clear reason to remember
 - (b) Cannot understand after reading the material
 - (c) Read to learn "later"
 - (d) Rapidly forget what you have read
 - (e) Form of study is inappropriate

1.9 THE CHEMICAL EQUATION AND STOICHIOMETRY

Your objectives in studying this section are to be able to:

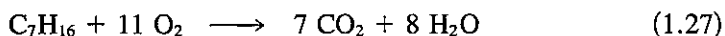
1. Write and balance chemical reaction equations.
2. Know the products of common reactions given the reactants.
3. Calculate the stoichiometric quantities of reactants and products given the chemical equation.
4. Define excess reactant, limiting reactant, conversion, degree of completion, and yield in a reaction.
5. Identify the limiting and excess reactants and calculate the percent excess reactant(s), the percent conversion, the percent completion, and

⁹B. E. Lauer, *The Material Balance*, Work Book Edition, Department of Chemical Engineering, University of Colorado, Boulder, Colo., 1954, p. 89.

yield for a chemical reaction with the reactants being in nonstoichiometric proportions.

6. Calculate the amount of products for incomplete reactions.

As you already know, the chemical equation provides a variety of qualitative and quantitative information essential for the calculation of the combining weights (mass) of materials involved in a chemical process. Take, for example, the combustion of heptane as shown below. What can we learn from this equation?



It tells us about **stoichiometric ratios**. **First, make sure that the equation is balanced!** Then you can see that 1 mole (*not* lb_m or kg) of heptane will react with 11 moles of oxygen to give 7 moles of carbon dioxide plus 8 moles of water. These may be lb mol, g mol, kg mol, or any other type of mole, as shown in Fig. 1.15. One mole of CO₂ is formed from each $\frac{1}{7}$ mole of C₇H₁₆. Also, 1 mole of H₂O is formed

C_7H_{16}	+	11O_2	$\longrightarrow$	7CO_2	+	$8\text{H}_2\text{O}$
<i>Qualitative information</i>						
heptane	reacts with	oxygen	to give	carbon dioxide	and	water
<i>Quantitative information</i>						
1 molecule of heptane	reacts with	11 molecules of oxygen	to give	7 molecules of carbon dioxide	and	8 molecules of water
6.023×10^{23} molecules of C ₇ H ₁₆	+	$11(6.023 \times 10^{23})$ molecules of O ₂	$\longrightarrow$	$7(6.023 \times 10^{23})$ molecules of CO ₂	+	$8(6.023 \times 10^{23})$ molecules of H ₂ O
1 g mole of C ₇ H ₁₆	+	11 g moles of O ₂	$\longrightarrow$	7 g moles of CO ₂	+	8 g moles of H ₂ O
1 kg mole of C ₇ H ₁₆	+	11 kg moles of O ₂	$\longrightarrow$	7 kg moles of CO ₂	+	8 kg moles of H ₂ O
1 lb mole of C ₇ H ₁₆	+	11 lb moles of O ₂	$\longrightarrow$	7 lb moles of CO ₂	+	8 lb moles of H ₂ O
1 ton mole of C ₇ H ₁₆	+	11 ton moles of O ₂	$\longrightarrow$	7 ton moles of CO ₂	+	8 ton moles of H ₂ O
1(100) g of C ₇ H ₁₆	+	11(32) g of O ₂	=	7(44) g of CO ₂	+	8(18) g of H ₂ O
100 g		352 g		308 g		144 g
		452 g	=			452 g
		452 kg	=			452 kg
		452 ton	=			452 ton
		452 lb	=			452 lb

Figure 1.15 Application of the chemical equation.

with each $\frac{7}{8}$ mole of CO_2 . Thus the equation tells us in terms of moles (*not* mass) the ratios among reactants and products. The coefficients of the compounds in the equation are known as **stoichiometric coefficients**: 1 for C_7H_{16} , 11 for O_2 , and so on.

Stoichiometry (stoi-ki-om-e-tri)¹⁰ deals with the combining weights of elements and compounds. The ratios obtained from the numerical coefficients in the chemical equation are the **stoichiometric ratios** that permit you to calculate the moles of one substance as related to the moles of another substance in the chemical equation. If the basis selected is to be mass (lb_m , kg) rather than moles, you should use the following method in solving problems involving the use of chemical equations: (1) Use the molecular weight to calculate the number of moles of the substance equivalent to the basis; (2) change this number of moles into the corresponding number of moles of the desired product or reactant by multiplying by the proper stoichiometric ratio, as determined by the chemical equation; and (3) then change the moles of product or reactant to a mass. These steps are indicated in Fig. 1.16 for the reaction in Eq. (1.27). You can combine these steps in a single dimensional equation, as shown in the examples below, for ease of calculations.

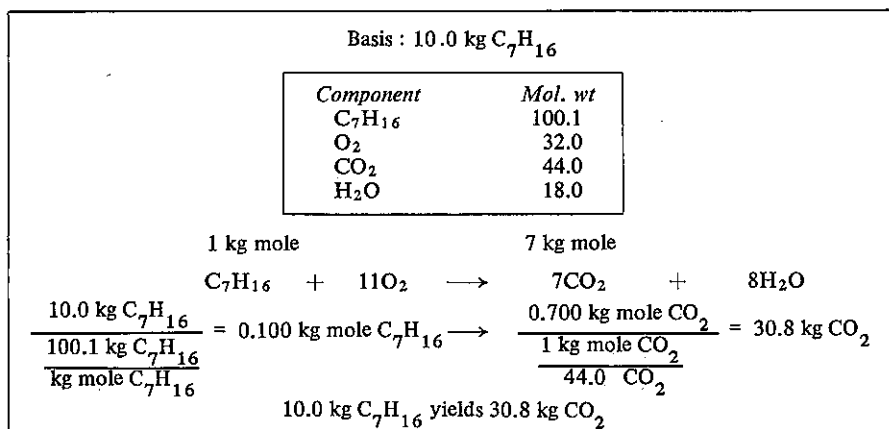


Figure 1.16 Stoichiometry.

EXAMPLE 1.27 Use of the Chemical Equation

In the combustion of heptane, CO_2 is produced. Assume that you want to produce 500 kg of dry ice per hour and that 50% of the CO_2 can be converted into dry ice, as shown in Fig. E1.27. How many kilograms of heptane must be burned per hour?

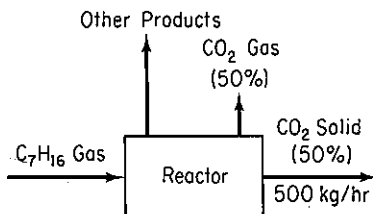


Figure E1.27

¹⁰ From the Greek *stoicheion*, basic constituent, and *metrein*, to measure.

Solution

Basis: 500 kg of dry ice (or 1 hr)

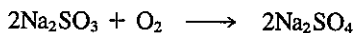
Mol. wt. heptane = 100.1. Chemical equation as in Fig. 1.15.

$$\begin{array}{c}
 \frac{500 \text{ kg dry ice}}{1 \text{ kg CO}_2} \times \frac{1 \text{ kg mol CO}_2}{44 \text{ kg CO}_2} \times \frac{1 \text{ kg mol C}_7\text{H}_{16}}{7 \text{ kg mol CO}_2} \\
 \times \frac{100.1 \text{ kg C}_7\text{H}_{16}}{1 \text{ kg mol C}_7\text{H}_{16}} = 325 \text{ kg C}_7\text{H}_{16}
 \end{array}$$

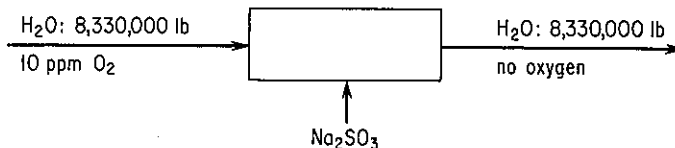
Since the basis of 500 kg of dry ice is identical to 1 hr, 325 kg of C_7H_{16} must be burned per hour. Note that kilograms are first converted to moles, then the chemical equation is applied, and finally moles are converted to kilograms again for the final answer.

EXAMPLE 1.28 Stoichiometry

Corrosion of pipes in boilers by oxygen can be alleviated through the use of sodium sulfite. Sodium sulfite removes oxygen from boiler feedwater by the following reaction:



How many pounds of sodium sulfite are theoretically required (for complete reaction) to remove the oxygen from 8,330,000 lb of water (10^6 gal) containing 10.0 parts per million (ppm) of dissolved oxygen and at the same time maintain a 35% excess of sodium sulfite? See Fig. E1.28.

**Figure E1.28****Solution**

Additional data: mol. wt. of Na_2SO_3 is 126. Chemical equation: $2\text{Na}_2\text{SO}_3 + \text{O}_2 \rightarrow 2\text{Na}_2\text{SO}_4$.

Basis: 8,330,000 lb of H_2O with 10 ppm O_2 or 83.3 lb of O_2

$$\frac{8,330,000 \text{ lb H}_2\text{O}}{(1,000,000 - 10 \text{ lb O}_2) \text{ lb H}_2\text{O}} \times 10 \text{ lb O}_2 = 83.3 \text{ lb O}_2$$

effectively same as 1,000,000

$$\frac{8,330,000 \text{ lb H}_2\text{O}}{10^6 \text{ lb H}_2\text{O}} \times \frac{10 \text{ lb O}_2}{32 \text{ lb O}_2} \times \frac{1 \text{ lb mol O}_2}{1 \text{ lb mol O}_2} \times \frac{2 \text{ lb mol Na}_2\text{SO}_3}{1 \text{ lb mol O}_2}$$

$$\frac{126 \text{ lb Na}_2\text{SO}_3}{1 \text{ lb mol Na}_2\text{SO}_3} \times \frac{1.35}{1} = 886 \text{ lb Na}_2\text{SO}_3$$

EXAMPLE 1.29 Stoichiometry

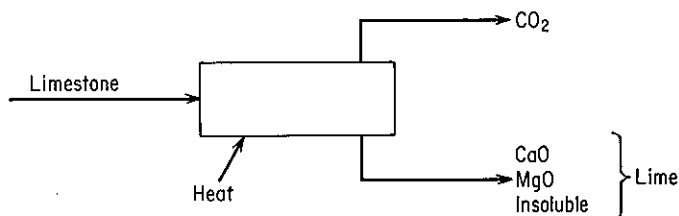
A limestone analyzes

CaCO_3	92.89%
MgCO_3	5.41%
Insoluble	1.70%

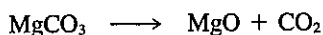
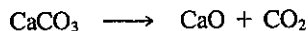
- How many pounds of calcium oxide can be made from 5 tons of this limestone?
- How many pounds of CO_2 can be recovered per pound of limestone?
- How many pounds of limestone are needed to make 1 ton of lime?

Solution

Read the problem carefully to fix in mind exactly what is required. Lime will include all the impurities present in the limestone which remain after the CO_2 has been driven off. Next, draw a picture of what is going on in this process. See Fig. E1.29.

**Figure E1.29**

To complete the preliminary analysis you need the following chemical equations:



Additional data:

	CaCO_3	MgCO_3	CaO	MgO	CO_2
Mol. wt.:	100.1	84.32	56.08	40.32	44

Basis: 100 lb of limestone

This basis was selected because pounds = percent.

Component	lb = percent	lb mol	Lime	lb	$\text{CO}_2(\text{lb})$
CaCO_3	92.89	0.9280	CaO	52.2	40.8
MgCO_3	5.41	0.0642	MgO	2.59	2.82
Insoluble	1.70		Insoluble	1.70	
Total	100.00	0.9920	Total	56.4	43.6

Note that the total pounds of products equal the 100 lb of entering limestone. Now to calculate the quantities originally asked for:

$$(a) \quad \text{CaO produced} = \frac{52.2 \text{ lb CaO}}{100 \text{ lb stone}} \left| \frac{2000 \text{ lb}}{1 \text{ ton}} \right| \frac{5 \text{ ton}}{1} = 5220 \text{ lb CaO}$$

$$(b) \quad \text{CO}_2 \text{ recovered} = \frac{43.6 \text{ lb CO}_2}{100 \text{ lb stone}} = 0.436 \text{ lb}$$

or

$$(c) \quad \text{Limestone required} = \frac{100 \text{ lb stone}}{56.4 \text{ lb lime}} \left| \frac{2000 \text{ lb}}{1 \text{ ton}} \right| = 3546 \text{ lb stone}$$

An assumption implicit in the calculations above is that the reaction takes place exactly as written in the equation and proceeds to 100% completion. When reactants, products, or degree of completion of the actual reaction differ from the assumptions of the equation, additional data must be made available to predict the outcome of reactions.

In industrial reactors you will rarely find exact stoichiometric amounts of materials used. To make a desired reaction take place or to use up a costly reactant, excess reactants are nearly always used. This excess material comes out together with, or perhaps separately from, the product—and sometimes can be used again. Even if stoichiometric quantities of reactants are used, but if the reaction is not complete or there are side reactions, the products will be accompanied by unused reactants as well as side products. In these circumstances some new definitions¹¹ must be understood:

(a) **Limiting reactant is the reactant that is present in the smallest stoichiometric amount.** In other words, if two or more reactants are mixed and if the reaction were to proceed according to the chemical equation to completion, **whether it does or not**, the reactant that would first disappear is termed the limiting reactant. For example, using Eq. (1.27), if 1 g mol of C_7H_{16} and 12 g mol of O_2 are mixed, C_7H_{16} would be the limiting reactant even if the reaction does not take place.

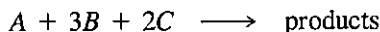
As a shortcut to determining the limiting reactant, all you have to do is to calculate the mole ratio(s) of the reactants and compare each ratio with the corresponding ratio of the coefficients of the reactants in the chemical equation thus:

	<u>Ratio in feed</u>		<u>Ratio in chemical equation</u>
$\frac{\text{O}_2}{\text{C}_7\text{H}_{16}}$:	$\frac{12}{1} = 12$	$>$	$\frac{11}{1} = 11$

If more than two reactants are present, you have to use one reactant as the reference substance, calculate the mole ratios of the other reactants in the feed relative to the reference, make pairwise comparisons versus the analogous ratios in the chemical

¹¹ You need to be aware that other definitions of the terms exist in the literature.

equation, and rank each compound. For example, given the reaction



and that 1.1 moles of A , 3.2 moles of B , and 2.4 moles of C are fed as reactants into the reactor, we choose A as the reference substance and calculate

	<u>Ratio in feed</u>		<u>Ratio in chemical equation</u>
$\frac{B}{A}$:	$\frac{3.2}{1.1} = 2.91$	$<$	$\frac{3}{1} = 3$
$\frac{C}{A}$:	$\frac{2.4}{1.1} = 2.18$	$>$	$\frac{2}{1} = 2$

We conclude that B is the limiting reactant relative to A , and that A is the limiting reactant relative to C , hence B is the limiting reactant among the set of three reactants. In symbols we have $B < A$, $C > A$ (i.e., $A < C$), so that $B < A < C$.

(b) **Excess reactant is a reactant present in excess of the limiting reactant.**

The **percent excess** of a reactant is based on the amount of any excess reactant above the amount required to react with the limiting reactant according to the chemical equation, or

$$\text{excess} = \frac{\text{excess moles}}{\text{reacted moles}}. \quad \% \text{ excess} = \frac{\text{moles in excess}}{\text{moles required to react with limiting reactant}} (100)$$

where the moles in excess frequently can be calculated as the total available moles of a reactant less the moles required to react with the limiting reactant. A common term, **excess air**, is used in combustion reactions; it means the amount of air available to react that is in excess of the air theoretically required to *completely* burn the combustible material. The *required* amount of a reactant is established by the limiting reactant and is for all other reactants the corresponding stoichiometric amount. **Even if only part of the limiting reactant actually reacts, the required and excess quantities are based on the entire amount of the limiting reactant as if it had reacted completely.**

Air requirements for combustion vary with the need to ensure full utilization of the fuel's heating value but not generate excessive air pollutants. The excess air required in practice depends on the type of fuel, the furnace, and the burner. Fuel oil, for instance, requires 5 to 20% excess air depending on burner design. Excess air is recognized as a routine measure of heater performance.

Three other terms that are used in connection with chemical reactions have less clear-cut definitions: conversion, selectivity, and yield. No universally agreed upon definitions exist for these terms—in fact, quite the contrary. Rather than cite all the possible usages of these terms, many of which conflict, we shall define them as follows:

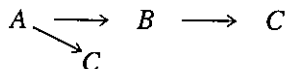
(c) **Conversion is the fraction of the feed or some material in the feed that is converted into products.** Thus, percent conversion is

$$100 \frac{\text{moles of feed (or a compound in the feed) that react}}{\text{moles of feed (or a compound in the feed) introduced}}$$

What the basis in the feed is for the calculations and into what products the basis is being converted must be clearly specified or endless confusion results. Conversion is related to the **degree of completion** of a reaction, which is usually the percentage or fraction of the limiting reactant converted into products.

(d) **Selectivity is the ratio of the moles of a particular (usually the desired) product produced to the moles of another (usually undesired) product produced in a set of reactions.**

(e) **Yield**, for a single reactant and product, is the weight (mass) or moles of final product divided by the weight (mass) or moles of initial reactant (P lb of product A per R lb of reactant B) either fed or consumed. If more than one product and more than one reactant are involved, the reactant upon which the yield is to be based must be clearly stated. Suppose that we have a reaction sequence as follows:



With B the desired product and C the undesired one. The yield of B is the moles (or mass) of B produced divided by the moles (or mass) of A fed or consumed. The selectivity of B is the moles of B divided by the moles of C produced.

The terms "yield" and "selectivity" are terms that measure the degree to which a desired reaction proceeds relative to competing alternative (undesirable) reactions. As a designer of equipment you want to maximize production of the desired product and minimize production of the unwanted products. Do you want high or low selectivity? Yield?

The employment of these concepts can best be illustrated by examples.

EXAMPLE 1.30 Limiting Reactant and Incomplete Reaction

Antimony is obtained by heating pulverized stibnite (Sb_2S_3) with scrap iron and drawing off the molten antimony from the bottom of the reaction vessel:



Suppose that 0.600 kg of stibnite and 0.250 kg of iron turnings are heated together to give 0.200 kg of Sb metal. Calculate:

- The limiting reactant
- The percentage of excess reactant
- The degree of completion (fraction)
- The percent conversion
- The yield

Solution

The molecular weights needed to solve the problem and the gram moles forming the basis are:

Component	kg	Mol. wt.	g mol
Sb_2S_3	0.600	339.7	1.77
Fe	0.250	55.85	4.48
Sb	0.200	121.8	1.64
FeS		87.91	

The process is illustrated in Fig. E1.30.

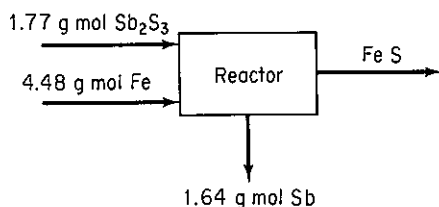


Figure E1.30

- (a) To find the limiting reactant, we examine the chemical reaction equation and note that if 1.77 g mol of Sb_2S_3 reacts, it requires $3(1.77) = 5.31$ g mol of Fe, whereas if 4.48 g mol of Fe reacts, it requires $(4.48/3) = 1.49$ g mol of Sb_2S_3 to be available. Thus Fe is present in the smallest stoichiometric amount and is the limiting reactant; Sb_2S_3 is the excess reactant.
- (b) The percentage of excess reactant is

$$\% \text{ excess} = \frac{1.77 - 1.49}{1.49}(100) = 18.8\% \text{ excess } \text{Sb}_2\text{S}_3$$

- (c) Although Fe is the limiting reactant, not all the limiting reactant reacts. We can compute from the 1.64 g mol of Sb how much Fe actually does react:

$$\frac{1.64 \text{ g mol Sb}}{2 \text{ g mol Sb}} \left| \frac{3 \text{ g mol Fe}}{2 \text{ g mol Sb}} \right| = 2.46 \text{ g mol Fe}$$

If by the fractional degree of completion is meant the fraction conversion of Fe to FeS, then

$$\text{fractional degree of completion} = \frac{2.46}{4.48} = 0.55$$

- (d) Let us assume that the percent conversion refers to the Sb_2S_3 since the reference compound is not specified in the question posed.

$$\frac{1.64 \text{ g mol Sb}}{2 \text{ g mol Sb}} \left| \frac{1 \text{ g mol } \text{Sb}_2\text{S}_3}{2 \text{ g mol Sb}} \right| = 0.82 \text{ g mol } \text{Sb}_2\text{S}_3$$

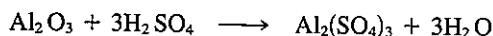
$$\% \text{ conversion of } \text{Sb}_2\text{S}_3 \text{ to Sb} = \frac{0.82}{1.77}(100) = 46.3\%$$

- (e) The yield will be stated as kilograms of Sb formed per kilogram of Sb_2S_3 that was fed to the reaction:

$$\text{yield} = \frac{0.200 \text{ kg Sb}}{0.600 \text{ kg } \text{Sb}_2\text{S}_3} = \frac{1 \text{ kg Sb}}{3 \text{ kg } \text{Sb}_2\text{S}_3} = \frac{0.33 \text{ kg Sb}}{1 \text{ kg } \text{Sb}_2\text{S}_3}$$

EXAMPLE 1.31 Limiting Reactant and Incomplete Reactions

Aluminum sulfate can be made by reacting crushed bauxite ore with sulfuric acid, according to the following equation:



The bauxite ore contains 55.4% by weight of aluminum oxide, the remainder being impurities. The sulfuric acid solution contains 77.7% H_2SO_4 , the rest being water.

To produce crude aluminum sulfate containing 1798 lb of pure aluminum sulfate, 1080 lb of bauxite ore and 2510 lb of sulfuric acid solution are used.

- Identify the excess reactant.
- What percentage of the excess reactant was consumed?
- What was the degree of completion of the reaction?

Solution

We will omit the figure for this problem. You need to look up or calculate the molecular weights of the compounds involved. The pound moles of substances forming the basis of the problem can be computed as follows:

$$\begin{array}{l} \frac{1798 \text{ lb Al}_2(\text{SO}_4)_3}{342.2 \text{ lb Al}_2(\text{SO}_4)_3} = 5.25 \text{ lb mol} \\ \frac{1080 \text{ lb bauxite}}{1 \text{ lb bauxite}} \times \frac{0.554 \text{ lb Al}_2\text{O}_3}{101.96 \text{ lb Al}_2\text{O}_3} = 5.87 \text{ lb mol} \\ \frac{2510 \text{ lb acid}}{98.1 \text{ lb H}_2\text{SO}_4} \times \frac{0.777 \text{ lb H}_2\text{SO}_4}{1 \text{ lb acid}} = 19.88 \text{ lb mol} \end{array}$$

- (a) The excess reactant can be determined as follows:

$$\begin{array}{ll} \text{Ratio in feed} & \text{Ratio in chemical equation} \\ \text{H}_2\text{SO}_4 : \frac{19.88}{5.87} = 3.39 & \frac{3}{1} = 3 \end{array}$$

Hence H_2SO_4 is the excess reactant.

- (b) The $\text{Al}_2(\text{SO}_4)_3$ actually formed indicates that

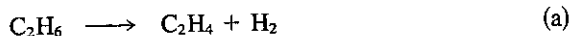
$$\begin{array}{l} \frac{5.25 \text{ lb mol Al}_2(\text{SO}_4)_3}{1 \text{ lb mol Al}_2(\text{SO}_4)_3} \times \frac{3 \text{ lb mol H}_2\text{SO}_4}{1 \text{ lb mol Al}_2(\text{SO}_4)_3} = 15.75 \text{ lb mol H}_2\text{SO}_4 \text{ was consumed} \\ \frac{15.75}{19.88}(100) = 79.2\% \end{array}$$

- (c) The fractional degree of completion refers to the limiting reactant. For each mole of $\text{Al}_2(\text{SO}_4)_3$, 1 mole of Al_2O_3 was used:

$$\frac{5.25}{5.87} = 0.89$$

EXAMPLE 1.32 The Meaning of Selectivity and Yield

Two well-known reactions take place in the dehydrogenation of ethane:



Given the following product distribution (in the gas-phase reaction of C_2H_6 in the presence of H_2) from the reaction of C_2H_6

Component	Percent
C_2H_6	35
C_2H_4	30
H_2	28
CH_4	7
Total	100

what is (a) the selectivity of C_2H_4 relative to CH_4 and (b) the yield of C_2H_4 in kilogram moles of C_2H_4 per kilogram mole of C_2H_6 ?

Solution

Basis: 100 kg mol of products

(a) The selectivity (as defined) is

$$\frac{30 \text{ kg mol C}_2\text{H}_4}{7 \text{ kg mol CH}_4} = 4.29 \frac{\text{mol C}_2\text{H}_4}{\text{mol CH}_4}$$

(b) The moles of C_2H_6 entering into the reaction can be determined from the C_2H_4 and the CH_4 formed.

$$\begin{array}{l} \frac{30 \text{ kg mol C}_2\text{H}_4}{1 \text{ kg mol C}_2\text{H}_4} = 30 \text{ kg mol C}_2\text{H}_6 \\ \frac{7 \text{ kg mol CH}_4}{2 \text{ kg mol CH}_4} = 3.5 \text{ kg mol C}_2\text{H}_6 \\ \hline 33.5 \text{ kg mol C}_2\text{H}_6 \end{array}$$

$$\text{Total C}_2\text{H}_6 = 33.5 + 35 = 68.5 \text{ kg mol.}$$

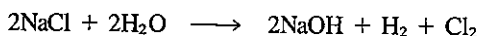
$$\frac{30 \text{ kg mol C}_2\text{H}_4}{68.5 \text{ kg mol C}_2\text{H}_6} = 0.44 \frac{\text{kg mol C}_2\text{H}_4}{\text{kg mol C}_2\text{H}_6}$$

You should remember that the chemical equation does not indicate the true mechanism of the reaction or how fast or to what extent the reaction will take place. For example, a lump of coal in air will sit unaffected at room temperature, but at higher temperatures it will readily burn. All the chemical equation indicates is the stoichiometric amounts required for the reaction and obtained from the reaction if it

proceeds in the manner in which it is written. Also, remember to make sure that the chemical equation is balanced before using it.

Self-Assessment Test

1. Write balanced reaction equations for the following reactions:
 - (a) C_9H_{18} and oxygen to form carbon dioxide and water
 - (b) FeS_2 and oxygen to form Fe_2O_3 and sulfur dioxide
2. If 1 kg of benzene (C_6H_6) is oxidized with oxygen, how many kilograms of O_2 are needed to convert all the benzene to CO_2 and H_2O ?
3. The electrolytic manufacture of chlorine gas from a sodium chloride solution is carried out by the following reaction:



How many kilograms of Cl_2 can one produce from 10 m^3 of a brine solution containing 5% by weight of sodium chloride? The specific gravity of the solution relative to water at 4°C is 1.07.

4. Calcium oxide (CaO) is formed by decomposing limestone (pure $CaCO_3$). In one kiln the reaction goes to 70% completion.
 - (a) What is the composition of the solid product withdrawn from the kiln?
 - (b) What is the yield in terms of pounds of CO_2 produced per pound of limestone charged?
5. In problem 3, suppose that 50.0 kg of $NaCl$ reacts with 10.0 kg of H_2O .
 - (a) What is the limiting reactant?
 - (b) What is the excess reactant?
 - (c) What components will the product solution contain if the reaction is 60% complete?

Thought Problems

1. An accident occurred in which one worker lost his life. A large steel evaporator in magnesium chloride service, containing internal heating tubes, was to be cleaned. It was shut down, drained, and washed. The next day two employees who were involved in the maintenance of the evaporator entered the vessel to repair the tubes. They were overcome, apparently from lack of oxygen. Subsequently, one employee recovered and escaped, but the other never regained consciousness, and died several days later.

What in your opinion might have caused the accident (the lack of oxygen)?

2. A leaky pen will make itself known in Technicolor. Leaked information appears in the national news. But a leaking underground gasoline storage tank is extremely difficult to detect.

Suggest some ways which can be used to detect leaky tanks. How effective are inventory control methods?

3. OSHA requires the use of breathing apparatus when working in or around tanks containing traces of solvents. While demolishing an old tank, a contractor purchased several cylinders of compressed air, painted gray. After two days he found that he needed more cylinders, and sent a truck for another cylinder. The driver returned with a black cylinder. None of the workers, including the man in charge of the breathing apparatus, noticed the change or, if they did, attached any importance to it. When the new cylinder was brought into use, a welder's facepiece caught fire. Fortunately, he pulled it off at once and was not injured.

What would be the most likely cause of this incident?

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PROBLEMS

An asterisk designates problems appropriate for solution using a computer. Refer also to the problems that require writing computer programs at the end of the chapter.

Section 1.1

- 1.1. An engineer went to the liquor supermarket—which advertised the lowest prices in town—and bought what he thought was the usual $\frac{1}{2}$ -gal jug for \$7.77, but the jug turned out to be in reality a 1.75-L bottle. Was this a bargain in comparison with two "fifths" (of a gallon) at \$7.99?
- 1.2. Convert the following quantities to the units shown in parentheses.
 - (a) The speed of light in a vacuum is 2.998×10^8 m/s (mi/hr).
 - (b) The speed of sound in air at standard temperature and pressure is 340.294 m/s (ft/s).
 - (c) The speed of a car traveling 60 mi/hr (km/hr).
 - (d) The standard football field is 100 yards long (m).

- (e) A fifth (of a gallon) of rum is cm^3 .
 (f) A passenger car gets 25 mi/gal (km/L).
- 1.3. The following test will measure your SIQ. List the correct answer.
- (a) Which is the correct symbol?
 (1) nm (2) $^{\circ}\text{K}$ (3) sec (4) N/mm
- (b) Which is the wrong symbol?
 (1) MN/m^2 (2) GHz/s (3) $\text{kJ}/(\text{s} \cdot \text{m}^3)$ (4) $^{\circ}\text{C/M/s}$
- (c) Atmospheric pressure is about:
 (1) 100 Pa (2) 100 kPa (3) 10 Mpa (4) 1 GPa
- (d) The temperature 0°C is defined as:
 (1) 273.15 $^{\circ}\text{K}$ (2) Absolute zero
 (3) 273.15 K (4) The freezing point of water
- (e) Which height and mass are those of a petite woman?
 (1) 1.50 m, 45 kg (2) 2.00 m, 95 kg
 (3) 1.50 m, 75 kg (4) 1.80 m, 60 kg
- (f) Which is a recommended room temperature in winter?
 (1) 15°C (2) 20°C (3) 28°C (4) 45°C
- (g) The watt is:
 (1) One joule per second (2) Equal to $1 \text{ kg} \cdot \text{m}^2/\text{s}^3$
 (3) The unit for all types of power (4) All of the above
- (h) What force may be needed to lift a heavy suitcase?
 (1) 24 N (2) 250 N (3) 25 kN (4) 250 kN
- 1.4. Convert $9 \times 10^8 \text{ (lb)/(ft}^3\text{)(hr)}$ to $\text{g}/(\text{s})(\text{cm}^3)$.
- 1.5. In the United States a diet soft drink would be labeled "low cal" cola. In Australia, such drinks are sold as "low _____" cola.
- 1.6. Your boss announced that the speed of the company Boeing 727 is to be cut from 525 mi/hr to 475 mi/hr to "conserve fuel," thus cutting consumption from 2200 gal/hr to 2000 gal/hr. How many gallons are saved in a 1000-mi trip?
- 1.7. It can be determined that the rate of energy loss from a person's body will normally vary by approximately 20 Btu/hr for each 1°F change in skin temperature. Therefore, in going from a comfortably cool skin temperature of 90°F to a comfortably warm skin temperature of 93°F , the heat rejection rate is increased by only 60 Btu/hr. How much are the two rates, respectively, in watts?
- 1.8. Change the underlined quantities in the following popular phrases to the units given in parentheses.
- (a) An ounce (g) of prevention is worth a pound (kg) of cure.
 (b) Give her an inch (cm) and she'll take a mile (m).
 (c) He demanded his pound (g) of flesh.
 (d) First down and 10 yards (m) to go.
 (e) A miss is as good as a mile (cm).
 (f) More bounce to the ounce (kg).
 (g) I love you a bushel (L) and a peck (cm³).
 (h) Walk a mile (ft) in his shoes.
- 1.9. Match the following quantities in list A with the most closely associated terms in list B.

LIST A

- | | |
|----------------------|---------------------|
| a. 1 million phones | c. 10 millipedes |
| b. 2000 mockingbirds | d. 1/1000 of a tent |

- | | |
|-----------------------------|----------------------|
| e. 2×10^{18} mines | j. 10^{21} picolos |
| f. 10^{12} tories | k. 0.000001 phone |
| g. two tenths of a fur | l. 100 oars |
| h. 10 cards | m. $1/10^6$ graph |
| i. 10^{-6} scope | |

LIST B

- | | |
|------------------------|------------------|
| 1. 1 deckacards | 8. to examine |
| 2. 1 micrograph | 9. 1 gigolo |
| 3. 1 militant | 10. 1 microphone |
| 4. 1 centipede | 11. 1 microscope |
| 5. to decipher | 12. 1 megaphone |
| 6. 1 territory | 13. 1 hector |
| 7. 2 kilo-mockingbirds | |

- 1.10. A technical publication describes a new model 20-hp Stirling (air cycle) engine that drives a 68-kW generator. Is this possible?
- 1.11. A freeze-dried coffee is advertised as "97% caffeine-free." Is it possible for this coffee to contain only 0.14% caffeine? Explain.
- 1.12. The following is a synopsis of an article to appear in an engineering journal. Convert the quantities mentioned to the nearest whole unit shown in parentheses.

The ocean thermal energy conversion (OTEC) to electrical power utilizes the temperature difference between the warm surface water and the cold deep water. A proposed (a) 37,912 Btu/s (MW) OTEC power plant will require a (b) 9.15 m (ft)-diameter cold water supply pipe reaching a depth of (c) 900 m (ft). A test system was designed to measure the environmental loading and dynamic response of the barge and pipe. This system consisted of a (d) 2.4 m (ft) diameter fiberglass/foam pipe (e) 122 m (ft) long suspended vertically by a gimbal from a barge anchored (f) 3.7 km (mi) south of Honolulu, Hawaii, in (g) 374.9 m (ft) of water. The pipe wall was a composite of alternating (h) 9.7 mm (in.) fiberglass-reinforced plastic (FRP) and insulating foam of (i) 0.254 cm (in.), for an overall thickness of (j) 33 mm (in.).

Strain gauges and pressure transducers were attached at (k) 15.25 m (ft) intervals along the pipe and an accelerometer at (l) 29.3 m (ft) below the surface. The barge and pipe system, weighing a total of (m) 81,648 kg (ton) was anchored by (n) 22,680 kg (lb) concrete blocks on either side. A cable system held the barge in place.

The cable, rising from the bottom, consisted of a (o) 4.5 cm (in.) wire rope (p) 45.7 m (yd) long attached to a (q) 4,536 kg (ton) sinker followed by a second wire rope of (r) 579 m (ft). This was connected to a more flexible nylon rope (s) 365 m (ft) long and (t) 13 cm (in.) in diameter. The nylon ropes were attached to mooring buoys which were chained to the barge.

- 1.13. An elevator that weighs 10,000 lb is pulled up 10 ft between the first and second floors of a building 100 ft high. The greatest velocity the elevator attains is 3 ft/s. How much kinetic energy does the elevator have in (ft)(lb) at this velocity?
- 1.14. Find the kinetic energy of a ton of water moving at 60 mi/hr expressed as (a) (ft)(lb_f)

- (b) joules
- (c) (hp)(s)
- (d) (watt)(s)
- (e) (liter)(atm)

- 1.15. Calculate the kinetic and potential energy of a missile moving at 12,000 mi/hr above the earth where the acceleration due to gravity is 30 ft/s².
- 1.16. Although gold is itself biologically inert, bacteria can degrade gold sulfide to open up the matrix of gold ore, making the gold more amenable to attack by cyanide. With this biotechnical step added in processing, yields may then be increased from, say, the original 60% up to as much as 90% recovery of gold in the ore, depending on the mineralogy.

Typically, this additional biological step is carried out in a pachuca, a cone-bottomed column familiar to the mining industry. Ground or milled ore, mixed with the aqueous bacterial solution, is introduced into the top of the column, and air is injected at the base. The injected air serves a number of functions: it maintains the solid in suspension, it mixes the solid with liquid—giving a three-phase gas/liquid/solid system—and it provides the oxygen and carbon dioxide required by the bacteria. The bacteria also require a feed of nitrogen and phosphorous, which can be added to the column if they are not indigenous to the ore.

The theory in simple; in practice it is a matter of economics as to whether the added steps in gold recovery are worthwhile. If a company sends in a sample containing on the average 2.5 g of gold per metric ton of ore, what could be the maximum additional capital and operating costs per ton of ore if the additional biological recovery step is to be employed?

- 1.17. Explain in detail whether or not the following equation for flow through a rectangular weir is dimensionally consistent. (This is the modified Francis formula.)

$$q = 0.415(L - 0.2h_0)h_0^{1.5}\sqrt{2g}$$

where q = volumetric flow rate, ft³/s

L = crest height, ft

h_0 = weir head, ft

g = acceleration of gravity, 32.2 ft/(s)²

- 1.18. Consider the following equation for the rate of heat transfer across a boundary layer:

$$Q = hA \Delta T$$

where Q = rate of heat transfer

h = heat transfer coefficient

A = area perpendicular to the transfer

ΔT = temperature difference

What are the SI units of the heat transfer coefficient?

- 1.19. Densities are sometimes expressed as a linear function of temperature, such as

$$\rho = \rho_0 + aT$$

where ρ is in g/cm³ and T is in degrees K. What must the units of ρ_0 and a be?

- 1.20. An experimental investigation of the rate of mass transfer of SO₂ from an airstream into water indicated that the mass transfer coefficient could be correlated by an equation of the form

$$k_x = Ku^{0.487}$$

where k_x is the mass transfer coefficient in $\text{mol}/(\text{cm}^2)(\text{s})$ and u is the velocity in cm/s . Does the constant K have dimensions? What are they? If the velocity is to be expressed in ft/s , and we want to retain the same form of the relationship, what would the units of K' have to be if k_x is still $\text{mol}/(\text{cm}^2)(\text{s})$, where K' is the new coefficient in the formula.

- 1.21. The equation for the flow of water through a nozzle is as follows:

$$q = C \sqrt{\frac{2g}{1 - (d_1/d_2)^4}} \left(A \sqrt{\frac{\Delta p}{\rho}} \right)$$

where q = volume flowing per unit time

C = dimensionless constant

g = local gravitational acceleration

d_1 = smaller nozzle diameter

d_2 = larger nozzle diameter

A = area of nozzle outlet

Δp = pressure drop across nozzle

ρ = density of fluid flowing

State whether this equation is dimensionally consistent. Show how you arrived at your conclusion.

- 1.22. A useful dimensionless number called the *Reynolds number* is

$$\frac{DU\rho}{\mu}$$

where D = diameter or length

U = some characteristic velocity

ρ = fluid density

μ = fluid viscosity

Calculate the Reynolds number for the following cases:

	1	2	3	4
D	2 in.	20 ft	1 ft	2 mm
U	10 ft/s	10 mi/hr	1 m/s	3 cm/s
ρ	62.4 lb/ft ³	1 lb/ft ³	12.5 kg/m ³	25 lb/ft ³
μ	0.3 lb _m /(hr)(ft)	0.14×10^{-4} lb _m /(s)(ft)	2×10^{-6} centipoise (cP)	1×10^{-6} centipoise

- 1.23. Computers are used extensively in automatic plant process control systems. The computers must convert signals from devices monitoring the process, evaluate the data using the programmed engineering equations, and then feed back the appropriate control adjustments. The equations must be dimensionally consistent. Therefore, a conversion factor must be part of the equation to change the measured field variable into the proper units.

Crude oil pumped from a storage unit to a tanker is to be expressed in tons/hr, but the field variables of density and the volumetric flow rate are measured in lb/ft³ and gal/min, respectively. Determine the units and the numerical value of the factor necessary to convert the field variables to the desired output.

- 1.24. The pressure drop for flow through a circular pipe may be calculated from the Darcy formula:

$$\Delta p = \frac{KfLpv^2}{D}$$

where $\Delta p = \text{lb}_f/\text{in.}^2$

$K = \text{a constant}$

$f = \text{friction factor, dimensionless}$

$L = \text{ft}$

$p = \text{lb}_m/\text{ft}^3$

$v = \text{ft/s}$

$D = \text{in.}$

What are the units of the constant, K ?

- 1.25. The Francis weir formula is used in distillation column design to find the height of the liquid crest over the weir that controls liquid flow from the plate in the column

$$h = 0.48F\left(\frac{Q}{L}\right)^{0.67}$$

where $h = \text{liquid crest height, in.}$

$F = \text{dimensionless correction factor}$

$Q = \text{liquid flow rate, gal/min}$

$L = \text{weir length, in.}$

- (a) What are the units associated with 0.48?
 (b) Convert the equation above to the SI system so that Q can be introduced into the equation in the units of m^3/s and L and h in meters.

Section 1.2

- 1.26. Write the formula and calculate the molecular weight in grams of:
 (a) Silver sulfate
 (b) Anhydrous barium chloride
 (c) Propyl benzoate
 (d) Thyroxin (thyro-oxyindol)
- 1.27. Write the formula and calculate the molecular weight in pounds of (a) silver sulfate, (b) anhydrous barium chloride, (c) propyl benzoate, and (d) thyroxin (thyro-oxyindol).
- 1.28. How many pounds of compound are contained in each of the following?
 (a) 64 g of BaMnO_4
 (b) 40 lb mol of dimethyl amine
 (c) 11 kg mol of nicotine
- 1.29. (a) How many g mol are represented by 100 g of CO_2 ?
 (b) Calculate the weight in pounds of 3.5 g mol of nitrogen.
- 1.30. Convert the following:
 (a) 105 g mol of potassium oxide to g
 (b) 2 lb mol of aluminum acetate to lb
 (c) 114 g mol of oxalic acid to lb
 (d) 3 lb of quinine to g mol
 (e) 40 g of trichlorobenzene to lb mol

- 1.31.** How many pounds of compound are contained in each of the following?
- 130 g mol of sodium hydroxide (anhydrous).
 - 300 g mol of sodium oleate.
 - 165 lb mol of sodium perchromate.
 - 62 lb mol of pure nitric acid.
 - 36 g mol of cupric tartrate.
 - 36 lb mol of cuprous phosphide (Cu_3P_2).
 - 72 lb mol of nickel bromide.
 - 12 g mol of nitrogen sulfide (N_4S_4).
 - 120 lb mol of potassium nitrate.
 - 11 g mol of stannic fluoride.
- 1.32.** The structural formulas in Fig. P1.32 are for vitamins:
- How many pounds of compound are contained in each of the following (do for each vitamin):
 - 2.00 g mol
 - 16 g
 - How many grams of compound are contained in each of the following (do for each vitamin):
 - 1.00 lb mol
 - 12 lb

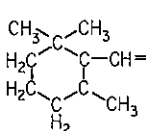
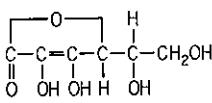
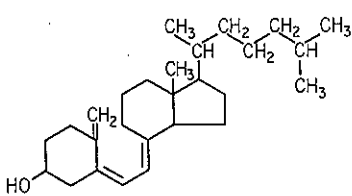
Vitamin	Structural formula	Dietary sources	Deficiency symptoms
Vitamin A	 Retinol	Fish liver oils, liver, eggs, fish, butter, cheese, milk; a precursor, β -carotene, is present in green vegetables, carrots, tomatoes, squash	Night blindness, eye inflammation
Ascorbic acid (vitamin C)		Citrus fruit, tomatoes, green peppers, strawberries, potatoes	Scurvy
Vitamin D	 Vitamin D₃	Fish liver oils, butter, vitamin-fortified milk, sardines, salmon; the body also obtains this compound when ultraviolet light converts 7-dehydrocholesterol in the skin to vitamin D	Rickets, osteomalacia, hypoparathyroidism

Figure P1.32

- 1.33.** The recent correspondence from Dr. Morris and Professor Mullin concerning the mole seems to have left its dimensions unresolved. Morris asserts that the "mole has the dimensions of mass"; Mullin counters that "the mole is certainly related to mass, but this does not confer dimensions of mass on it."
- State which authority is correct, Mullin or Morris, and explain in no more than two sentences why.

- 1.34. With the proper catalyst and temperature, ethane (C_2H_6) can be completely decomposed to acetylene (C_2H_2) and hydrogen (H_2). If the flow rate into the reactor is 100 lb/hr, what is the exit flow rate of H_2 in lb/hr?

Section 1.3

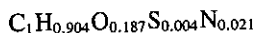
- 1.35. A proposed New Mexico coal gasification project is predicted to produce about 573,000 lb/hr of wet ash from one unit. It would form a pile 231 ft at the base, 100 ft high, and almost 7 mi long in 1 year! Based on these estimates, what was the density used for the ash in pounds per cubic foot?
- 1.36. The density of a certain solution is 8.80 lb/gal at 80°F. How many cubic feet will be occupied by 10,010 lb of this solution at 80°F?
- 1.37. The density of benzene at 60°F is 0.879 g/cm³. What is the specific gravity of benzene at 60°F/60°F?
- 1.38. A solution of sulfuric acid at 60°F is found to have a specific gravity of 1.22. If this is equivalent to 30% H_2SO_4 , what is the concentration of H_2SO_4 in lb mol/gal?
- 1.39. A liquid has a specific gravity of 0.90 at 25°C. What is its
 (a) Density at 25°C in kg/m³?
 (b) Specific volume at 25°C in ft³/lb_m?
 (c) If the liquid is placed in a 1.5-L bottle that has a mass of 232 g, how much will the full bottle weigh?
- 1.40. Given a water solution that contains 1.704 kg of HNO_3 /kg H_2O and has a specific gravity of 1.382 at 20°C, express the composition in the following ways:
 (a) Weight percent HNO_3
 (b) Pounds HNO_3 per cubic foot of solution at 20°C
 (c) Molarity (gram moles of HNO_3 per liter of solution at 20°C)
- 1.41. Two immiscible liquids are allowed to separate in a vessel. One liquid has a specific gravity of 0.936; the second liquid weighs 9.63 lb/gal. A block that is 9 in. by 9 in. by 9 in. and weighs 25.8 lb is dropped into the vessel. Will the block float on the top, stop at the interface where the two liquids are separated, or sink? What fraction of the volume of the block is in one or both of the liquids?
- 1.42. Five thousand barrels of 28°API gas oil are blended with 20,000 bbl of 15°API fuel oil. What is the density of the mixture in lb/gal and lb/ft³? Assume that the volumes are additive. 1 bbl = 42 gal.

$$\text{Specific gravity} \frac{60^\circ F}{60^\circ F} = \frac{141.5}{^\circ\text{API} + 131.5}$$

Density of water at 60°F = 0.999 g/cm³.

- 1.43. In a handbook you find that the conversion between °API and density is 0.800 density = 45.28°API. Is this a misprint?
- 1.44. One barrel each of gasoline (55°API), kerosene (40°API), gas oil (31°API), and isopentane (96°API) are mixed. What is the composition of the mixture expressed in weight percent and volume percent? What is the API gravity and the density of the mixture in g/cm³ and lb/gal?
- 1.45. A tank for deep-sea diving consists of 60% He (atomic weight = 4.00), 20% N_2 , and 20% O_2 . The weight of the gas in the tank, excluding the tank weight, is 1.437 kg. What is the mass fraction of each of the components in the tank?

- 1.46. In the analysis of coal, a pseudo coal molecule can be postulated as represented by the following formula:



- (a) What is the analysis of the coal by weight (mass) percent?
 (b) What is the molecular weight of the pseudo molecule?
- 1.47. Solid fuel is used in many commercial areas from rocket boosters to rock concert light shows. Components of solid fuel fall into four main categories: fuel, oxidizer, other special additives, and a binder. For example, various metal compounds and metal powders are used in fireworks to produce the brilliant colors that you see. An analysis shows that the powder from a gold cone fountain firework contained the following:

Percent by weight	Component	Purpose
2	C	Fuel
8	S	Fuel
53	KNO ₃	Oxidizer
32	Fe	Color
4	Al	Color
1	Stearic acid — CH ₃ (CH ₂) ₁₆ CO ₂ H	Organic binder
Total	100	

Calculate the mol % of each component.

- 1.48. Harbor sediments in the New Bedford, Massachusetts, area contain PCBs at levels up to 190,000 ppm according to a report prepared by Grant Weaver of the Massachusetts Coastal Zone Management Office (*Environ. Sci. Technol.*, v. 16, no. 9, p. 491A, 1982). What is the concentration in percent?
- 1.49. NIOSH sets standards for CCl₄ in air at 12.6 mg/m³ of air (a time weighted average over 40 hr). The CCl₄ found in a sample is 4800 ppb (parts per billion; billion = 10⁹). Does the sample exceed the NIOSH standard? Be careful!
- 1.50. The following table shows the annual inputs of phosphorus to Lake Erie.

	Short tons/yr
Source	
Lake Huron	2,240
Land drainage	6,740
Municipal waste	19,090
Industrial waste	2,030
	<u>30,100</u>
Outflow	<u>4,500</u>
Retained	25,600

SOURCE: *23rd Report to Committee on Government Operations*, U.S. Government Printing Office, Washington, D.C., 1970.

- (a) Convert the retained phosphorus to concentration in micrograms per liter as-

suming that Lake Erie contains 1.2×10^{14} gal of water and that average phosphorus retention time is 2.60 yr.

- (b) What percentage of the input comes from municipal water?
 - (c) What percentage of the input comes from detergents, assuming they represent 70% of the municipal waste?
 - (d) If 10 ppb of phosphorus triggers nuisance algal blooms, as has been reported in some documents, would removing 30% of the phosphorus in the municipal waste and all the phosphorus in the industrial waste be effective in reducing the eutrophication (i.e., the unwanted algal blooms) in Lake Erie?
 - (e) Would removing all the phosphate in detergents help?
- 1.51. A single piston reciprocating pump has a 15-cm-diameter piston with a 15-cm length of stroke. If the pump makes 16 strokes per minute, what is the pumping rate in (a) L/min? (b) gal/min?
- 1.52. A 15-in.-diameter sewer pipe is flowing full and carrying a flow of 2.7 million gallons per day. What is the average velocity of the water in the pipe?
- 1.53.* Solubility of formaldehyde in a solvent is measured by a spectrophotometer operating at 570 nm. The data collected are optical density versus concentration expressed in grams per liter.

Concentration, c	Optical density, d
100	0.086
300	0.269
500	0.445
600	0.538
700	0.626

Based on Beer's law, obtain the best estimates of the coefficients for a linear relation

$$c = b_0 + b_1 d$$

and plot the equation so obtained together with the data. Refer to Appendix M.

Section 1.4

- 1.54. Twenty-five pounds of gas consisting of 30% CO_2 , 40% N_2 , and 30% O_2 are placed in a cylinder 200 ft^3 in volume. What is the
- (a) Average molecular weight of the gas?
 - (b) Density of the gas (in lb/ft^3)?
- 1.55. A mixture of gases contains 10.5% CO_2 , 13.0% Cl_2 , 12.7% N_2 , and the balance hydrogen (all are mole percentages). The molecular weights are: $\text{CO}_2 = 44.01$, $\text{Cl}_2 = 70.90$, $\text{N}_2 = 28.01$, and $\text{H}_2 = 2.02$.
- (a) What is the average molecular weight of the gas?
 - (b) Calculate the gas composition as weight fractions.
- 1.56. You are given 200 kg of a gas mixture containing
- | | |
|---------------|-----|
| SO_2 | 20% |
| NO_2 | 20% |
| N_2 | 60% |

What is

- (a) The lb of N_2 in the gas?
(b) The average molecular weight of the gas?

- 1.57. You have 100 lb of gas of the following composition:

CH_4	30%
H_2	10%
N_2	60%

What is the average molecular weight of this gas?

- 1.58. A fuel gas is reported to analyze, on a mole basis, 20% methane, 5% ethane, and the remainder CO_2 . Calculate the analysis of the fuel gas on a mass percentage basis.
- 1.59. Recent developments in ion chromatography are filling the analytical gap between the atomic adsorption spectroscopy and inductively coupled plasma metal spectroscopy. Ion chromatography can now not only determine what trace metals are present, but also their oxidation state, the degree of complexation, and the stability of the complex. For example, a nickel electroplating solution was analyzed by diluting it with a water eluant solution. The analysis revealed the following ion concentrations.

Metal cation	Concentration (mg/L)
Pb^{2+}	4
Cu^{2+}	0.5
Cd^{2+}	4
Co^{2+}	1
Zn^{2+}	2
Ni^{2+}	4

The other material in the solution was water.

- (a) Calculate the concentration in mass % of each cation present.
(b) Calculate the composition in mol % of each cation present.

- 1.60. A gas mixture consists of three components: argon, *B*, and *C*. The following analysis of this mixture is given:

40.0 mol % argon

18.75 mass % *B*

20.0 mol % *C*

The molecular weight of argon is 40 and the molecular weight of *C* is 50. Find:

- (a) The molecular weight of *B*
(b) The average molecular weight of the mixture

- 1.61. Two engineers are calculating the average molecular weight of a gas mixture containing oxygen and other gases. One of them uses the correct molecular weight of 32 for oxygen and determines the average molecular weight as 39.2. The other uses an incorrect value of 16 and determines the average molecular weight as 32.8. This

is the only error in his calculations. What is the percentage of oxygen in the mixture expressed as mol %?

- 1.62. A liquid mixture of compounds A, B, and C containing 10 kg of A analyzes 25% B and contains 1.5 moles of C per mole of B. The respective molecular weights of A, B, and C are 56, 58, and 72; and the specific gravities are 0.58, 0.60, and 0.67. Calculate the analysis of the mixture in mole percent, the molecular weight of the mixture, the volume percent A on a B-free basis (ignore volume change on mixing), the density of the mixture, and the total number of moles of the mixture.

Section 1.5

- 1.63. In a report on the record low temperatures in Antarctica, *Chemical and Engineering News* said at one point that "the mercury dropped to -76°C ." In what sense is that possible? Mercury freezes at -39°C .
- 1.64. Two engineers are arguing about the physical properties of a certain product. One claims that the melting point is -40°F , while the other claims that it is -40°C .
 (a) Who is correct?
 (b) The boiling point is 325 K. Determine the boiling point in $^{\circ}\text{C}$, $^{\circ}\text{R}$, and $^{\circ}\text{F}$.
 (c) Find $T_B - T_m$ (T_B = boiling point, T_m = melting point) for K, $^{\circ}\text{C}$, $^{\circ}\text{R}$, and $^{\circ}\text{F}$.
- 1.65. "Further, the degree Celsius is exactly the same as a kelvin. The only difference is that zero degree Celsius is 273.15 kelvin. Use of Celsius temperature gives us one less digit in most cases" from [*Eng. Educ.*, p. 678 (April 1977)]. Comment on the quotation. Is it correct? If not, in what way or sense is it wrong?
- 1.66. (a) Convert 70°F to $^{\circ}\text{C}$, K, $^{\circ}\text{R}$.
 (b) Convert 210 K to $^{\circ}\text{C}$, $^{\circ}\text{F}$, $^{\circ}\text{R}$.
- 1.67. From what we know now, it is not going to be easy to make practical room-temperature superconductors. For one thing, scientists believe that getting a superconductor to carry more than a trickle of electrical current at room temperature would be very difficult. Even if we had room-temperature superconductivity, we would probably still work at the liquid nitrogen temperature, at which the current-carrying capacity of the superconductors is much greater.

Liquid nitrogen boils at 77 K and is required as a refrigerant for the highest-temperature superconductors proven to date. Its use represents a big improvement over the more costly and formerly required liquid helium, which boils at about -452°F . But it is far short of the room-temperature goal, which if attained, could lead rapidly to a revolution in cars, personal computers, and other yet-to-be-imagined consumer devices.

Convert the boiling point of liquid nitrogen to degrees Fahrenheit and degrees Rankine, and the boiling point of liquid helium to degrees Celsius and kelvin.

- 1.68. Given $T_1 = 25^{\circ}\text{F}$, $T_2 = 60^{\circ}\text{C}$, and $\Delta T = T_1 - T_2$, express T_1 , T_2 , and ΔT in all of the following units: $^{\circ}\text{F}$, $^{\circ}\text{C}$, $^{\circ}\text{R}$, K.
- 1.69. Calculate all temperatures from the one value given:

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
$^{\circ}\text{F}$	140				1000			
$^{\circ}\text{R}$			500			1000		
K		298					1000	
$^{\circ}\text{C}$				-40				1000

- 1.70. The emissive power of a blackbody depends on the fourth power of the temperature and is given by

$$W = AT^4$$

where W = emissive power, Btu/(ft²)(hr)

A = Stefan-Boltzmann constant, 0.171×10^{-8} Btu/(ft²)(hr)(°R)⁴

T = temperature, °R

What is the value of A in the units J/(m²)(s)(K⁴)?

Section 1.6

- 1.71. The amplitude of sound at any given point is expressed as sound-pressure level (SPL). Its physical unit is the decibel, which is given as

$$\text{SPL} = 20 \ln (p/p_0) \quad \text{in dB}$$

where p is the sound pressure being measured and p_0 is a reference pressure, usually 20 micronewtons per square meter ($\mu\text{N/m}^2$). The reference pressure of $20 \mu\text{N/m}^2$ is approximately equal to the lowest pressure which a young person with normal hearing can barely detect at a frequency of 1000 Hz. Other measures of sound pressure may be encountered in the literature, such as dynes per square centimeter, microbars, and pounds per square inch.

Common examples of representative SPL (in dB) include:

Business office	50
Speech at 3 ft	65
Subway at 20 ft	95
Jet aircraft at 35 ft	130
On gantry during Saturn V launch	172

Convert the dB values of 0, 20, and those in the list above to:

- (a) N/m²
- (b) μbar
- (c) psi

- 1.72. Fill in the values in the blanks in the following:

- (a) Consider a tank that is to take a pressure of 10 psi. This is the equivalent of (approximately) _____ newtons/square meter (N/m²), or _____ kilonewtons/square meter (kN/m²), or _____ kilopascals (kPa).
- (b) It may help to remember that 14.7 psi (_____ atm) = _____ kN/m² or approximately _____ MN/m².
- (c) If your automobile tires take 24 psi of air, you will have to set the air-pump pressure to _____ kN/m²; if you need 26 psi, the pressure will be _____ kN/m². If you are used to asking just for 30 lb of air in your car tires, you would in the future have to ask for _____ kilograms.
- (d) An absolute pressure of 10 micrometers of mercury is almost exactly equivalent to _____ N/m², or _____ Pa.
- (e) If you are designing a corrosion-resistant tank of rolled (annealed) Hastelloy alloy C, the tensile strength needed for your calculations is 896 MN/m², which is

the equivalent of _____ psi. Perhaps it would be better to round it off to _____ GN/m^2 .

- 1.73. "Holiday drivers stranded, Icicles blossom in Florida; 50 cities set record lows" was the headline in the December 26 newspaper. After citing all of the cities with new low temperatures, the paper went on to say: "As the full force of the cold wave descended on the East it was 5 degrees Sunday morning in New York City. As one example of the strength of the cold air mass flowing from northwest Canada, the highest pressure ever recorded in the United States—31.42 inches of mercury—was measured on Christmas Eve in Miles City, Mont." Can this last statement be true?
- 1.74. The pressure gauge on a tank reads 50 psig on a day when the barometric pressure is 28.0 in. Hg. Find the absolute pressure in the tank in psia.
- 1.75. The growing use of ceramic fibers in the electronics, aerospace, and chemical process industries has prompted the development of a much faster manufacturing process, one several hundred times faster than current chemical vapor deposition (CVD) methods. The fibers are stronger and more flexible because they have a more uniform composition (i.e., no carbon core required). The reactor conditions may be varied from 1000 to 2500 K and the pressure from 100 to 700 kPa to accommodate a wide range of material processes. For example, silicon carbide fibers could be produced by mixing silicon chloride with methane and then heating the mix with an yttrium/aluminum laser, thus allowing strict control of the process temperature in the reactor. Convert the ranges for the reactor temperature and pressure to their corresponding American Engineering units ($^{\circ}\text{R}$ and psi).
- 1.76. Suppose that a submarine inadvertently sinks to the bottom of the ocean at a depth of 1000 m. It is proposed to lower a diving bell to the submarine and attempt to enter the conning tower. What must the minimum air pressure be in the diving bell at the level of the submarine to prevent water from entering into the bell when the opening valve at the bottom is cracked open slightly? Give your answer in absolute kilopascal. Assume that seawater has a constant density of 1.024 g/cm^3 .
- 1.77. Air in a scuba diver's tank shows a pressure of 300 kPa absolute. What is the pressure in:
- | | | | |
|------------|-----------------------------|----------------------|---------|
| (a) psi | (b) kPa | (c) atm | (d) bar |
| (e) in. Hg | (f) ft H_2O | (g) kg/cm^2 | |
- 1.78. Flat-roof buildings are a popular architectural style in dry climates because of the economy of materials of construction. However, during the rainy season water may pool up on the roof decks so that structural considerations for the added weight must be taken into account. If 25 cm of water accumulates on a 10-m by 10-m area during a heavy rain storm, determine:
- (a) The total added weight the building must support
- (b) The force of the water on the roof in psi
- 1.79. A problem with wastewater treatment tanks set below ground was realized when the water table rose and an empty tank floated out of the ground. This buoyancy problem was overcome by installing a check valve in the wall of the tank so that if the water table rose high enough to float the tank, it would fill with water. If the density of concrete is 2080 kg/m^3 , determine the maximum height at which the valve should be installed to prevent a buoyant force from raising a rectangular tank with inside dimensions of 30 m by 27 m and 5 m deep. The walls and floor have a uniform thickness of 200 mm.
- 1.80. A centrifugal pump is to be used to pump water from a lake to a storage tank that is

148 ft above the surface of the lake. The pumping rate is to be 25.0 gal/min, and the water temperature is 60°F. The pump on hand can develop a pressure of 50.0 psig when it is pumping at a rate of 25.0 gal/min. (Neglect pipe friction kinetic energy effects, or factors involving pump efficiency.)

- (a) How high (in feet) can the pump raise the water at this flow rate and temperature?
- (b) Is this pump suitable for the intended service?

- 1.81. You are asked to recommend a liquid for an open-top vertical manometer. As much accuracy as possible is needed, but the ceiling in the room restricts the column height to 3.35 m. Pressures will range up to 50 kPa gauge. Choose one of the substances from the following table, and explain the reasons for your choice. Give any calculations needed.

Substance	Density (g/cm ³)
Mercury	13.5939
Water	0.99823
Carbon tetrachloride	1.5940

- 1.82. A manometer uses kerosene, sp gr 0.82, as the fluid. A reading of 5 in. on the manometer is equivalent to how many millimeters of mercury?
- 1.83. The pressure gauge on the steam condenser for a turbine indicates 26.2 in. Hg of vacuum. The barometer reading is 30.4 in. Hg. What is the pressure in the condenser in psia?
- 1.84. A pressure gauge on a process tower indicates a vacuum of 3.53 in. Hg. The barometer reads 29.31 in. Hg. What is the absolute pressure in the tower in millimeters of mercury?
- 1.85. John Long says he calculated from a formula that the pressure at the top of Pikes Peak is 9.75 psia. John Green says that it is 504 mm Hg because he looked it up in a table. Which John is right?
- 1.86. The floor of a cylindrical water tank was distorted into 7-in. bulges due to the settling of improperly stabilized soil under the tank floor. However, several consulting engineers restored the damaged tank to use by placing plastic skirts around the bottom of the tank wall and devising an air floatation system to move it to an adjacent location. The tank was 30.5 m in diameter and 13.1 m deep. The top, bottom, and sides of the tank were made of 9.35-mm-thick welded steel sheets. The density of the steel is 7.86 g/cm³.
- (a) What is the gauge pressure in kPa of the water at the bottom of the tank when it is completely full of water?
- (b) What must the air pressure be in kPa beneath the empty tank in order to just raise it up for movement?
- 1.87. A pressure instrument has failed on a process line that requires constant monitoring. A bell-type gauge as shown in Fig. P1.87, is available which has oil (density of 0.800 g/cm³) as a sealant liquid. Construction of the gauge limits the sealant liquid's travel to 12.7 cm before blowout of the oil occurs. What maximum pressures can this gauge measure in kPa?
- 1.88. One end of a mercury manometer at 70°F is attached to an evaporator at low pressure; the other is open to the atmosphere. The height of the mercury column is

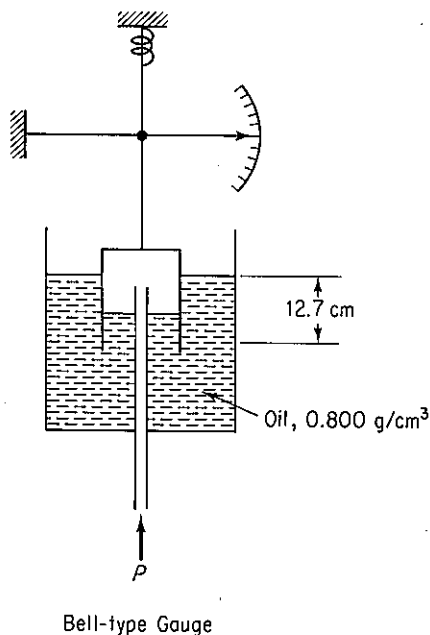


Figure P1.87

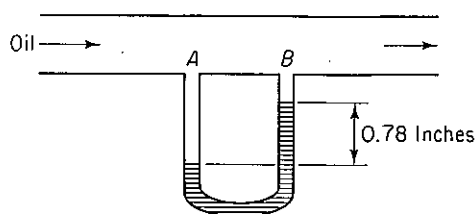


Figure P1.89

- 8.1 mm. How many kPa absolute is this? What is the equivalent reading in inches of Hg vacuum?
- 1.89. Examine Fig. P1.89. Oil (density = 0.91 g/cm^3) flows in a pipe, and the flow rate is measured via a mercury (density = 13.546 g/cm^3) manometer. If the difference in height of the two legs of the manometer is 0.78 in., what is the corresponding pressure difference between points A and B in mm Hg? At which point, A or B, is the pressure higher? The temperature is 60°F .
 - 1.90. Water at 60°F is flowing through a pipeline, and the line contains an orifice plate for flow measurement. The pressure difference between the upstream and the downstream sides of the orifice plate is to be measured using an available U-tube manometer which is 48 in. high. If the maximum expected pressure drop across the orifice plate is 3.55 psi, what is the minimum specific gravity required of a manometer liquid that can be used to measure this differential pressure? Density of water at $60^\circ\text{F} = 0.999 \text{ g/cm}^3$.
 - 1.91. An inclined manometer is often used to measure small pressure differences. See Fig. P1.91.
 - (a) Find the equation for h in terms of L and θ .
 - (b) The process fluid is a gas at low pressure, the angle of inclination is 20° , $L = 10 \text{ cm}$, and the manometer fluid is oil with a specific gravity of 0.8. What is the pressure differential (pascal) $p_1 - p_2$?
 - (c) If the process fluid is oil (sp gr = 0.8), the manometer fluid is water, and $L = 12 \text{ in.}$, what is the pressure drop in psi?
 - 1.92. Pressure in a gas cell is measured with an inverted manometer, as shown in Fig. P1.92. The scale on the far right-hand side of the figure shows the distances in mm (not to scale) of the interfaces of the liquids in the manometer. What is the pressure

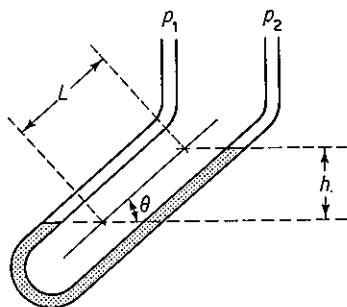


Figure P1.91

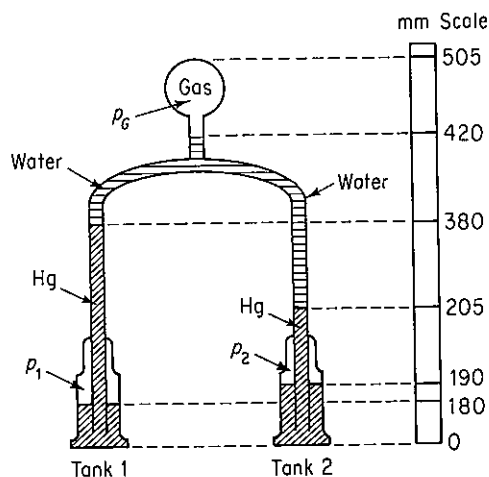


Figure P1.92

of the gas in psia if the pressures in the closed tanks are $p_1 = 12.50$ psia and $p_2 = 9.17$ psia? The density of Hg is 13.6 g/cm^3 .

- 1.93. The pressure reading on the gauge shown in Fig. P1.93 is 2.4 psi. If the liquid level in the glass pipe connecting the two tanks is 16 ft below the bottom of the water tank, is the pressure gauge working properly? Show all calculations. What should the gauge read in psig?

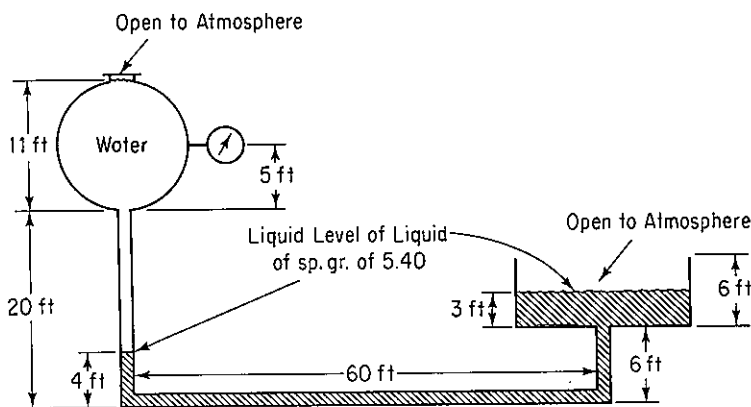


Figure P1.93

- 1.94. The indicating liquid in the manometer shown in Fig. P1.94 is water, and the other liquid is benzene. These two liquids are essentially insoluble in each other. If the manometer reading is $\Delta Z = 36.3 \text{ cm}$ water, what is the pressure difference in kPa? The temperature is 25°C .
- 1.95. Examine Fig. P1.95. The barometric pressure is 720 mm Hg. The density of the oil is 0.80 g/cm^3 . The Bourdon gauge reads 33.1 psig. What is the pressure in kPa of the gas?

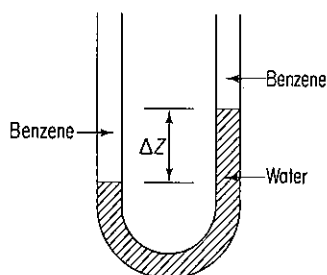
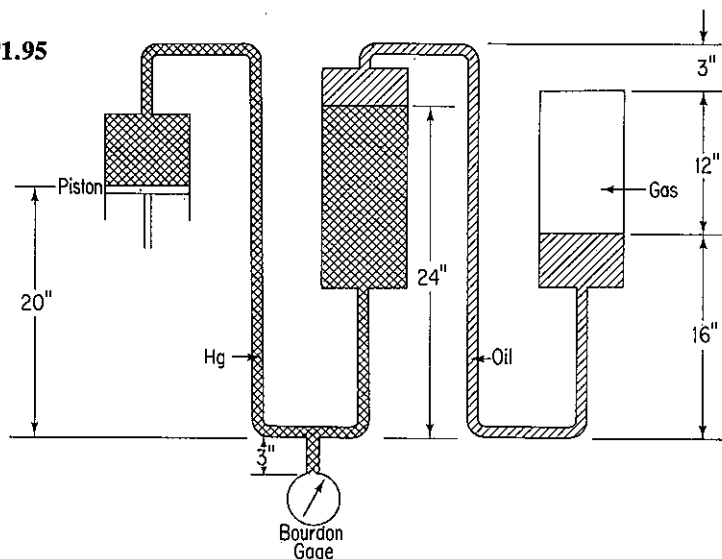


Figure P1.94



Section 1.7

- 1.96. Visit your library and carry out data searches for the following information. Report the values or equations found and their respective sources in complete detail: names of authors, article name, book or journal name, volume (if applicable), page number, and date of publication.
 - (a) Density of lead thiocyanate $[\text{Pb}(\text{CNS})_2]$ at 20°C
 - (b) Boiling point in $^\circ\text{C}$ of glyceryl tributyrate $[(\text{C}_2\text{H}_5\text{CH}_2\text{CO}_2)_3 \cdot \text{C}_3\text{H}_5]$ at atmospheric pressure
 - (c) Solubility of ammonium oxalate $[(\text{NH}_4)_2 \cdot \text{C}_2\text{O}_4]$ at 40°C in water
 - (d) Volume of methyl chloride gas at 200°F and 6 psia
- 1.97. Does Perry's *Chemical Engineers' Handbook* contain information on densities of alcohol-water mixtures? osmotic pressure of sodium chloride solutions? corrosion properties of metals?
- 1.98. List four journals that publish physical property data and the libraries in which they are located.

Section 1.8

- 1.99.** How would you estimate the rate at which a spherical water-filled capsule in outer space cools. List the types of information needed to solve such a problem, the as-

sumptions that must be made, and what physical principles that you have studied in physics might be used to help solve the problem. Draw a picture of the process and indicate what the independent and dependent variables involved in the process might be.

- 1.100.** Read the introductory portions of the book by Moshe F. Rubenstein, *Problem Solving* (Prentice-Hall, Englewood Cliffs, N.J., 1975), and compare the techniques outlined there with those listed in Sec. 1.8.

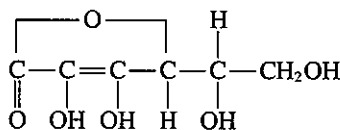
Section 1.9

- 1.101.** $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + 2\text{NaCl}$
- How many grams of barium chloride will be required to react with 5.00 g of sodium sulfate?
 - How many grams of barium chloride are required for the precipitation of 5.00 g of barium sulfate?
 - How many grams of barium chloride are needed to produce 5.00 g of sodium chloride?
 - How many grams of sodium sulfate are necessary for the precipitation of the barium of 5.00 g of barium chloride?
 - How many grams of sodium sulfate have been added to barium chloride if 5.00 g of barium sulfate is precipitated?
 - How many pounds of sodium sulfate are equivalent to 5.00 lb of sodium chloride?
 - How many pounds of barium sulfate are precipitated by 5.00 lb of barium chloride?
 - How many pounds of barium sulfate are precipitated by 5.00 lb of sodium sulfate?
 - How many pounds of barium sulfate are equivalent to 5.00 lb of sodium chloride?
- 1.102.** $\text{AgNO}_3 + \text{NaCl} \longrightarrow \text{AgCl} + \text{NaNO}_3$
- How many grams of silver nitrate will be required to react with 5.00 g of sodium chloride?
 - How many grams of silver nitrate are required for the precipitation of 5.00 g of silver chloride?
 - How many grams of silver nitrate are equivalent to 5.00 g of sodium nitrate?
 - How many grams of sodium chloride are necessary for the precipitation of the silver of 5.00 g of silver nitrate?
 - How many grams of sodium chloride have been added to silver nitrate if 5.00 g of silver chloride is precipitated?
 - How many pounds of sodium chloride are equivalent to 5.00 lb of sodium nitrate?
 - How many pounds of silver chloride are precipitated by 5.00 lb of silver nitrate?
 - How many pounds of silver chloride are precipitated by 5.00 lb of sodium chloride?
 - How many pounds of silver chloride are equivalent to 5.00 lb of silver nitrate?
- 1.103.** The chemical composition of the material formed by an organism that grows in an incubator by degrading hexadecane ($\text{C}_{16}\text{H}_{34}$) can be written as $\text{C}_{4.4}\text{H}_{7.3}\text{N}_{0.86}\text{O}_{1.2}$. How many moles of oxygen are used to degrade 1 mole of hexadecane according to the following reaction?



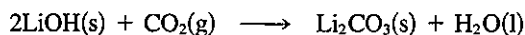
where a_1 , a_2 , b_1 , b_2 are constants to be determined.

- 1.104.** When benzene (C_6H_6) is burned with oxygen, how many pounds of oxygen are required to burn 10 lb of benzene to carbon dioxide and water vapor?
- 1.105.** A plant makes liquid CO_2 by treating dolomitic limestone with commercial sulfuric acid. The dolomite analyzes 68.0% CaCO_3 , 30.0% MgCO_3 , and 2.0% SiO_2 ; the acid is 94% H_2SO_4 and 6% H_2O . Calculate:
- Pounds of CO_2 produced per ton dolomite treated
 - Pounds of commercial acid required per ton of dolomite treated Mol. wt. data: CaCO_3 , 100; MgCO_3 , 84.3; SiO_2 , 60; H_2SO_4 , 98.
- 1.106.** The formula for vitamin C is as follows:

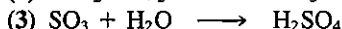
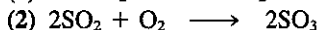
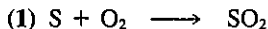


How many pounds of this compound are contained in 2 g mol?

- 1.107.** Removal of CO_2 from a manned spacecraft has been accomplished by absorption with lithium hydroxide according to the following reaction:



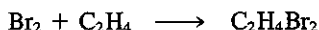
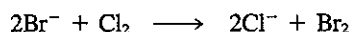
- If 1.00 kg of CO_2 is released per day per person, how many kilograms of LiOH are required per day per person?
 - What is the penalty (i.e., the percentage increase) in weight if the cheaper NaOH is substituted for LiOH ?
- 1.108.** Some states and cities are placing a ban on laundry detergents containing more than 20% phosphates. In reading the detailed provisions of one law, it prohibits the sale of any detergent with a phosphorus pentoxide content of over 20% by weight. How much is this when expressed as weight percent sodium triphosphate, the compound found in the detergent?
- 1.109.** Sulfuric acid can be manufactured by the contact process according to the following reactions:



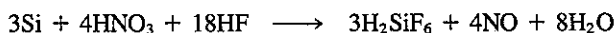
You are asked as part of the preliminary design of a sulfuric acid plant with a design capacity of 2000 tons/day of 66° Be (Baumé) (93.2% H_2SO_4 by weight) to calculate the following:

- How many tons of pure sulfur are required per day to run this plant?
 - How many tons of oxygen are required per day?
 - How many tons of water are required per day for reaction (3)?
- 1.110.** Seawater contains 65 ppm of bromine in the form of bromides. In the Ethyl-Dow recovery process, 0.27 lb of 98% sulfuric acid is added per ton of water, together with the theoretical Cl_2 for oxidation; finally, ethylene (C_2H_4) is united with the

bromine to form $\text{C}_2\text{H}_4\text{Br}_2$. Assuming complete recovery and using a basis of 1 lb of bromine, find the weights of acid, chlorine, seawater, and dibromide involved.



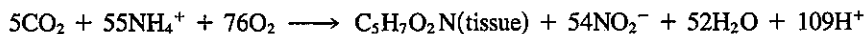
- 1.111. The burning of limestone, $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, goes only 70% to completion in a certain kiln.
 (a) What is the composition (mass %) of the solid withdrawn from the kiln?
 (b) How many kilograms of CO_2 are produced per kilogram of limestone fed? Assume that the limestone is pure CaCO_3 .
- 1.112. In the semiconductor industry, integrated circuit (IC) production begins with the mechanical slicing of silicon rod into wafers. Once the wafers are sliced, the surfaces are lapped and polished to uniform flat surfaces. Contaminants and microscopic defects (work damage) are then removed chemically by etching. A traditional etching solution consists of a 4:1:3 volumetric ratio of 49% hydrofluoric, 70% nitric, and 100% acetic acids, respectively. Although work damage is usually only 10 μm deep, overetching to 20 μm per side is common. The reaction for dissolving the silicon surface is



Calculate the flow rate of the etching solution in kilograms per hour if 20 μm per side is to be etched for 6000 wafers per hour of 150 mm diameter. What is the limiting reagent?

Data	Mol. wt.
Si—2.33 g/cm ³	28.09
Sp gr 49% HF is 1.198	20.01
Sp gr 70% HNO_3 is 1.4134	63.01
Sp gr 100% $\text{CH}_3\text{CO}_2\text{H}$ is 1.0492	60.05

- 1.113. One method of synthesizing the aspirin substitute, acetaminophen, involves a three-step procedure as outlined in Fig. P1.113. First, *p*-nitrophenol is catalytically hydrogenated in the presence of aqueous hydrochloric acid to the acid chloride salt of *p*-aminophenol with a 86.9% degree of completion. Next the salt is neutralized to obtain *p*-aminophenol with a 0.95 fractional conversion. Finally, the *p*-aminophenol is acetalated by reacting with acetic anhydride, resulting in a yield of 3 kg mol of acetaminophen per 4 kg mol. What is the overall conversion fraction of *p*-nitrophenol to acetaminophen?
- 1.114. The most economic method of sewage wastewater treatment is bacterial digestion. As an intermediate step in the conversion of organic nitrogen to nitrates, it is reported that the *Nitrosomonas* bacteria cells metabolize ammonium compounds into cell tissue and expel nitrite as a by-product by the following overall reaction:



If 20,000 kg of wastewater containing 5% ammonium ions by weight flows

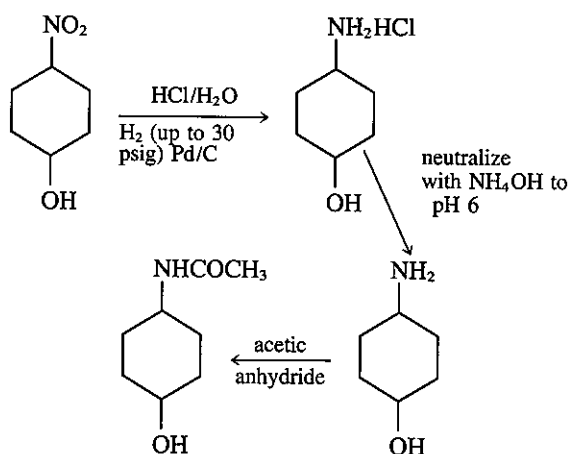
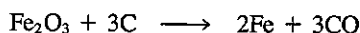


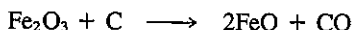
Figure P1.113

through a septic tank inoculated with the bacteria, how many kilograms of cell tissue are produced provided that 95% of the NH_4^+ is consumed?

- 1.115.** One can view the blast furnace from a simple viewpoint as a process in which the principal reaction is

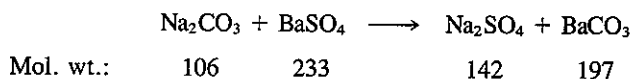


but some other undesired side reactions occur, mainly



After mixing 600.0 lb of carbon (coke) with 1.00 ton of pure iron oxide, Fe_2O_3 , the process produces 1200.0 lb of pure iron, 183 lb of FeO, and 85.0 lb of Fe_2O_3 . Calculate the following items:

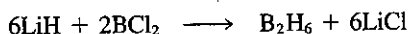
- The percentage of excess carbon furnished, based on the principal reaction
 - The percentage conversion of Fe_2O_3 to Fe
 - The pounds of carbon used up and the pounds of CO produced per ton of Fe_2O_3 charged
 - What is the selectivity in this process?
- 1.116.** Barite mineral (BaSO_4) is processed by heat fusion with soda ash (Na_2CO_3) followed by leaching with water to produce BaCO_3 , which has several uses, including that of a rat poison. The overall reaction is



The analysis shows the weight percents as: solid residue, 33.6% BaSO_4 and 66.4% BaCO_3 ; and soluble salts, 58.1% Na_2SO_4 and 41.9% Na_2CO_3 . For the process, calculate:

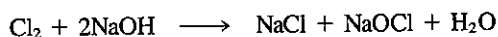
- The composition of the mix before fusion
- The percent excess reactant
- The degree of completion

- 1.117. Diborane, B_2H_6 , a possible rocket propellant, can be made by using lithium hydride (LiH):



If you mix 200 lb of LiH with 1000 lb of BCl_3 , you recover 45.0 lb of B_2H_6 . Determine:

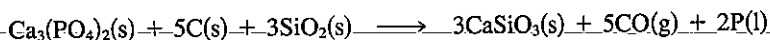
- The limiting reactant
 - The excess reactant
 - The percent excess reactant
 - The percent conversion of LiH to B_2H_6
 - The degree of completion of the reaction
 - The yield of B_2H_6 based on the LiH charged (in lb/lb)
 - The lb of LiCl produced
- 1.118. A common method used in manufacturing sodium hypochlorite bleach is by the reaction



Chlorine gas is bubbled through an aqueous solution of sodium hydroxide, after which the desired product is separated from the sodium chloride (a by-product of the reaction).

A water-NaOH solution that contains 1145 lb of pure NaOH is reacted with 851 lb of gaseous chlorine. The NaOCl formed weighs 618 lb:

- What is the limiting reactant?
 - What was the percentage excess of the excess reactant used?
 - What is the degree of completion of the reaction, expressed as the moles of NaOCl formed to the moles of NaOCl that would have formed if the reaction had gone to completion?
 - What is the yield of NaOCl per amount of chlorine used (on a weight basis)?
- 1.119. An electric furnace produces phosphorus by the reaction



The specific compounds in the slag (*solid product*) are $Ca_3(PO_4)_2$, $CaSiO_3$, and SiO_2 . You analyze the slag from the reaction and find 2.0% P, 1% C, 36.0% CaO, and the rest silicon and oxygen compounds. Determine:

- The limiting reactant
 - The excess reactants
 - The kg of P actually produced per kg of $Ca_3(PO_4)_2$ charged to the process
 - The kg mol of CO produced per kg of $Ca_3(PO_4)_2$ charged to the process
 - The degree of completion of the reaction
- 1.120. An electric furnace produces iron (Fe, mol. wt. 55.85) from Fe_2O_3 by reaction with powdered coal (carbon). The reaction products are Fe, FeO, and CO. Twenty-five kilograms of Fe_2O_3 and 6 kg of C are charged to the furnace and after drawing off the Fe, the slag (FeO solid product remaining) is 3 kg. Determine:
- The limiting reactant
 - The excess reactant
 - The percent excess reactant
 - The fractional degree of completion of the reaction

PROBLEMS THAT REQUIRE WRITING COMPUTER PROGRAMS

- 1.1. Write a program to convert °C into K, °F, and °R at 1°C intervals. Have the program arranged so that you can do the following:
 - (a) Read in a starting value of °C.
 - (b) Read in a stopping value of °C.
 - (c) Use a loop to convert °C to the other temperature scales at 1°C intervals between the starting and stopping values, inclusive.
 - (d) Print out the four temperatures for each conversion with appropriate column headings.
 - (e) Use comment lines where appropriate to indicate your procedure in conversion.
- 1.2. Write a computer program to convert millimeters of mercury into inches of mercury, feet of H₂O, and psia at 2 mm Hg (even-numbered) increments below 760 mm Hg. Have the computer arranged so that you can do the following:
 - (a) Read in a starting value for the millimeters of mercury.
 - (b) Stop at 760 mm Hg (the last value to be converted).
 - (c) Read in the necessary conversion factors, such as CPSIA in

$$\text{PSIA} = \text{CPSIA} * \text{HGMM}$$
 - (d) Use one-dimensional arrays to store values.
 - (e) Use a DO loop to compute all values. You must compute the number of iterations required with your program.
 - (f) Print out the results after all the conversions have been made.
 - (1) Print your name at the top of the first page.
 - (2) Print appropriate column headings at the top of each new page.
 - (3) Print only 50 lines of conversion results per page.
- 1.3. Prepare a computer program to give the specific gravity of a fluid as a function of °Be (Baumé) and °API over the interval 0°–30° in 1° increments. Equation (1.10) gives the relation between specific gravity and °API (for liquids less dense than water), while the relationships for °Be are

liquids more dense than water

$$\text{sp gr} = \frac{145.0}{145.0 - ^\circ\text{Be}}$$

liquids less dense than water

$$\text{sp gr} = \frac{140.0}{130.0 + ^\circ\text{Be}}$$

Read the °API or °Be into the program as data.

MATERIAL BALANCES

2

2.1	The Material Balance	104
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Conservation laws occupy a special place in science and engineering. Common statements of these laws take the form of "mass (energy) is neither created nor destroyed," "the mass (energy) of the universe is constant," "the mass (energy) of any isolated system is constant," or equivalent statements. To refute a conservation law, it would be sufficient to find just one example of a violation.

But what degree of accuracy is needed to prove a violation? In an industrial environment, in spite of considerable effort, it is not possible to make a 99.9% closure of a mass balance. The results of errors in measurement in determining the amount of material in tanks, bins, drums, bags, and bottles; in obtaining representative samples; in analyzing the samples; and in determining the fate of raw materials fed into the system and products removed from it preclude such accuracy.

Our belief in the validity of the conservation laws rests on the experiences of Lavoisier and many of the scientists following in his path who studied chemical changes quantitatively and found invariably that the sum of the weights of the substances entering into a reaction equaled the sum of the weights of the products of the reaction. Thus our collective experience has been summed up and generalized as the law of the conservation of matter. Of course, we must exclude processes involving nuclear transformations, or else extend our law to include the conservation of both energy and matter.

Why study material balances as a separate topic? You will find that material balance calculations are almost invariably a prerequisite to all other calculations in the solution of both simple and complex chemical engineering problems. Furthermore, skills that you develop in analyzing material balances are easily transferred to other types of balances and other types of problems.

In this chapter we discuss the principle of the conservation of matter and how it can be applied to engineering calculations, making use of the background information discussed in Chap. 1. Figure 2.0 shows the relations between the topics discussed in this chapter and the general objective of making material and energy balances. In approaching the solution of material balance problems, we first consider how to analyze them in order to clarify the method and the procedure of solution. The aim will be to help you acquire a generalized approach to problem solving so that you may avoid looking upon each new problem, unit operation, or process as entirely new and unrelated to anything you have seen before. As you scrutinize the examples used to illustrate the principles involved in each section, explore the method of analysis, but avoid memorizing each example by rote, because, after all, they are only samples of the myriad of problems that exist or could be devised on the subject of material balances. Most of the principles we consider are of about the same degree of complexity as the law of compensation devised by some unknown, self-made philosopher who said: "Things are generally made even somewhere or some place. Rain always is followed by a dry spell, and dry weather follows rain. I have found it an invariable rule that when a man has one short leg, the other is always longer!"

In working these problems you will find it necessary to employ some engineering judgment. You think of mathematics as an exact science. For instance, suppose that it takes 1 man 10 days to build a brick wall; then 10 men can finish it in 1 day. Therefore, 240 men can finish the wall in 1 hr, 14,400 can do the job in 1 min. and with 864,000 men the wall will be up before a single brick is in place! Your password to success is the famous IBM motto: THINK.

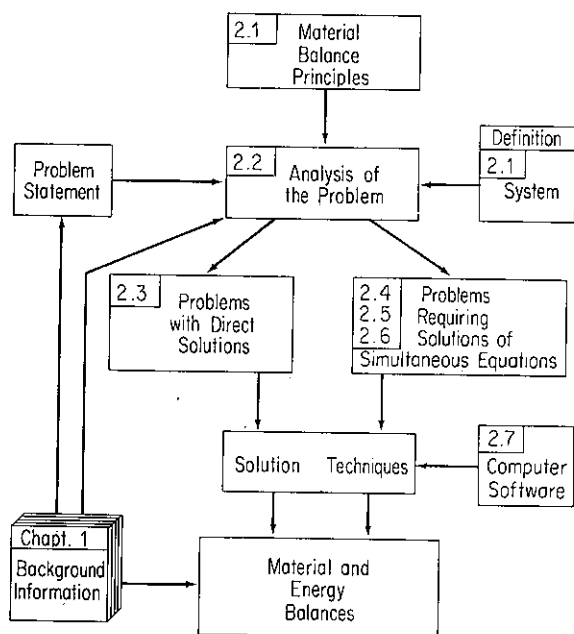


Figure 2.0 Hierarchy of topics to be studied in this chapter (section numbers are in the upper left-hand corner of the boxes).

2.1 THE MATERIAL BALANCE

Your objectives in studying this section are to be able to:

1. Define the system and draw the system boundaries for which the material balance is to be made.
2. Explain the difference between an open and a closed system.
3. Write the general material balance in words including all terms. Be able to apply the balance to simple problems.
4. Cite examples of processes in which no accumulation takes place; no generation or consumption takes place; no mass flow in and out takes place.
5. Apply the material balance equation for the simplified case of input = output to the total mass of material and to an individual species.
6. Explain the circumstances in which the mass of a compound entering the system equals the mass of the compound leaving the system. Repeat for moles.

To make a material balance (or an energy balance as discussed in Chap. 4) for a process, you need to specify what the system is and outline its boundaries. According to the dictionary, a **process** is one or a series of actions or operations or treatments that result in an end [product]. Chemical engineering focuses on operations that cause physical and chemical change in materials. Innumerable textbooks¹ and reference books² give examples of processes such as

Chemical manufacture
Fluid transport
Handling of bulk solids
Size reduction and enlargement
Heat generation and transport
Distillation
Gas absorption
Bioreactions

¹McCabe, W. L., J. C. Smith, and P. Harriott, *Unit Operations of Chemical Engineering*, 4th ed., McGraw-Hill, NY, 1986.

²Perry, R. H. and D. W. Green, eds., *Chemical Engineers' Handbook*, 6th ed., McGraw-Hill, NY, 1984; Shreve, R. N. and J. A. Brink, *Chemical Process Industries*, 4th ed., McGraw-Hill, NY, 1977.

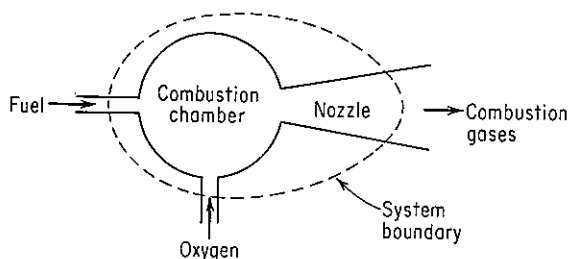


Figure 2.1 Flow (open) system defined for combustion.

and so on. The examples we use in this book often refer to abstractions of these processes, because we do not have the space here to describe the details of any of them. By **system** we mean any arbitrary portion or whole of a process set out specifically for analysis. Figure 2.1 shows a system in which flow and reaction take place; note particularly that the **system boundary** is formally circumscribed about the process itself to call attention to the importance of carefully delineating the system in each problem you work. An **open (or flow) system** is one in which material is transferred across the system boundary, that is, enters the system, leaves the system, or both. A **closed (or batch) system** is one in which there is no such transfer *during the time interval of interest*. Obviously, if you charge a reactor with reactants and take out the products, and the reactor is designated as the system, material is transferred across the system boundary. But you might ignore the transfer, and focus attention solely on the process of reaction that takes place only after charging is completed and before the products are withdrawn. Such a process would occur within a closed system.

A system boundary may be fixed with respect to the process equipment as in Fig. 2.1, or the boundary may be an imaginary surface that grows or shrinks as the process goes on. Think of a tube of toothpaste that is squeezed. A fixed boundary might be the tube itself, in which case mass crosses the boundary as you squeeze the tube. Or, you can imagine a flexible boundary surrounding the toothpaste itself that follows the extruded toothpaste, in which case no mass crosses the boundary.

A **material balance** is nothing more than an accounting for material flows and changes in inventory of material for a system. Examine Fig. 2.2. Equation (2.1) de-

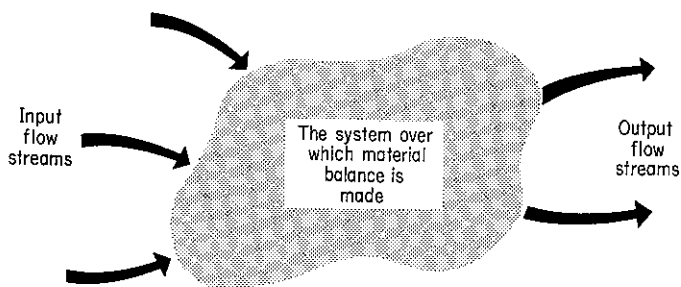


Figure 2.2 Any enclosed volume or system over which material balances are to be made. We are not concerned with the internal details, only with the passage of material across the volume boundaries and the change of material inside the system.

scribes in words the principle of the material balance applicable to processes both with and without chemical reaction:

$$\left\{ \begin{array}{c} \text{accumulation} \\ \text{within} \\ \text{the} \\ \text{system} \end{array} \right\} = \left\{ \begin{array}{c} \text{input} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} - \left\{ \begin{array}{c} \text{output} \\ \text{through} \\ \text{system} \\ \text{boundaries} \end{array} \right\} + \left\{ \begin{array}{c} \text{generation} \\ \text{within} \\ \text{the} \\ \text{system} \end{array} \right\} - \left\{ \begin{array}{c} \text{consumption} \\ \text{within} \\ \text{the} \\ \text{system} \end{array} \right\} \quad (2.1)$$

As a generic term, material balance can refer to a balance on a system for the

1. Total mass
2. Total moles
3. Mass of a chemical compound
4. Mass of an atomic species
5. Moles of a chemical compound
6. Moles of an atomic species
7. Volume (possibly)

With respect to a total mass balance, in this book the generation and consumption terms are zero whether a chemical reaction occurs in the system or not (we neglect the transfer between mass and energy in ordinary chemical processing)³; hence

$$\text{accumulation} = \text{input} - \text{output} \quad (2.2)$$

With respect to a balance on the total moles, if a chemical reaction does occur, you most likely will have to take into account the generation or consumption terms. In the absence of chemical reaction, the generation and consumption terms do not apply to a single chemical compound such as water or acetone; with a chemical reaction present in the system, the terms do apply.

From the viewpoint of both a mass balance or a mole balance for elements themselves, such as C, H, or O, the generation and consumption terms are not involved in a material balance. Finally, Eq. (2.1) should not be applied to a balance on a volume of material unless ideal mixing occurs (see Sec. 3.1) and the densities of the streams are the same. In this chapter, information about the generation and consumption terms for a chemical compound will be given *a priori* or can be inferred from the stoichiometric equations involved in the problem. Texts treating chemical reaction engineering describe how to calculate from basic principles gains and losses of chemical compounds.

³For information concerning the conservation of mass and Einstein's principle in circumstances in which ordinary chemical reactions occur, read R. S. Treptow, *J. Chem. Educ.*, v. 63, pp. 103, 1052 (1986), or J. Maddox, *Nature*, v. 328, p. 755 (1987).

In Eq. (2.1) the accumulation term refers to a change in mass or moles (plus or minus) within the system with respect to time, whereas the transfers through the system boundaries refer to inputs to and outputs of the system. If Eq. (2.1) is written in symbols so that the variables are functions of time, the equation so formulated would be a differential equation. As an example, the differential equation for the O_2 material balance for the system illustrated in Fig. 2.1 might be written as

$$\frac{dn_{O_2 \text{ within system}}}{dt} = \dot{n}_{O_2 \text{ in}} - \dot{n}_{O_2 \text{ out}} - \dot{n}_{O_2 \text{ reacted}} \quad (2.1a)$$

where $n_{O_2 \text{ within system}}$ denotes the moles of oxygen within the system boundary, and $\dot{n}_{O_2}$ denotes the rate at which oxygen enters, leaves or reacts, respectively, as indicated by the subscript. Each term in the differential equation represents a *rate* with the units of, say, moles per unit time. Problems formulated as differential equations with respect to time are called **unsteady-state** (or transient) problems and are discussed in Chap. 6. In contrast, in **steady-state** problems the values of the variables in the system do not change with time, hence the accumulation term in Eq. (2.1) is zero by definition.

In this Chapter for convenience in treatment we use an **integral balance** form of Eq. (2.1). What we do is to take as a basis a time period such as one hour or minute, and integrate Eq. (2.1a) with respect to time. The derivative (the left hand side) in the differential equation becomes

$$\int_{t_1}^{t_2} \frac{dn_{O_2 \text{ within system}}}{dt} dt = \int_{t_1}^{t_2} dn_{O_2 \text{ within system}} = n_{O_2} |_{t_2} - n_{O_2} |_{t_1} = \Delta n$$

where Δn is the difference in the n_{O_2} within the system at t_2 less that at t_1 . A term on the right hand side of the differential equation becomes, as for example the first term,

$$\int_{t_1}^{t_2} \dot{n}_{O_2 \text{ in}} dt = n_{O_2 \text{ in}}$$

where $n_{O_2 \text{ in}}$ represents the entire net quantity of oxygen introduced into the system between t_1 and t_2 . If the flow rate of O_2 into the system shown in Fig. 2.1 is constant at the rate of 1200 moles/hr, by choosing a basis of one hour

$$\int_0^1 \frac{1200 \text{ mol}}{\text{hr}} \left| \frac{dt}{\text{hr}} \right| = 1200 \text{ mol} (1 - 0) = 1200 \text{ mol}$$

Most, but not all, of the problems discussed in this chapter are steady-state problems treated as integral balances for fixed time periods. If no accumulation occurs in a problem, and the generation and consumption terms can be omitted from consideration, the material balances reduce to the very simple relation

$$\left\{ \begin{array}{c} \text{mass/mole input} \\ \text{through the} \\ \text{system boundaries} \end{array} \right\} = \left\{ \begin{array}{c} \text{mass/mole output} \\ \text{through the} \\ \text{system boundaries} \end{array} \right\} \quad (2.3)$$

or briefly: "What comes in must go out."

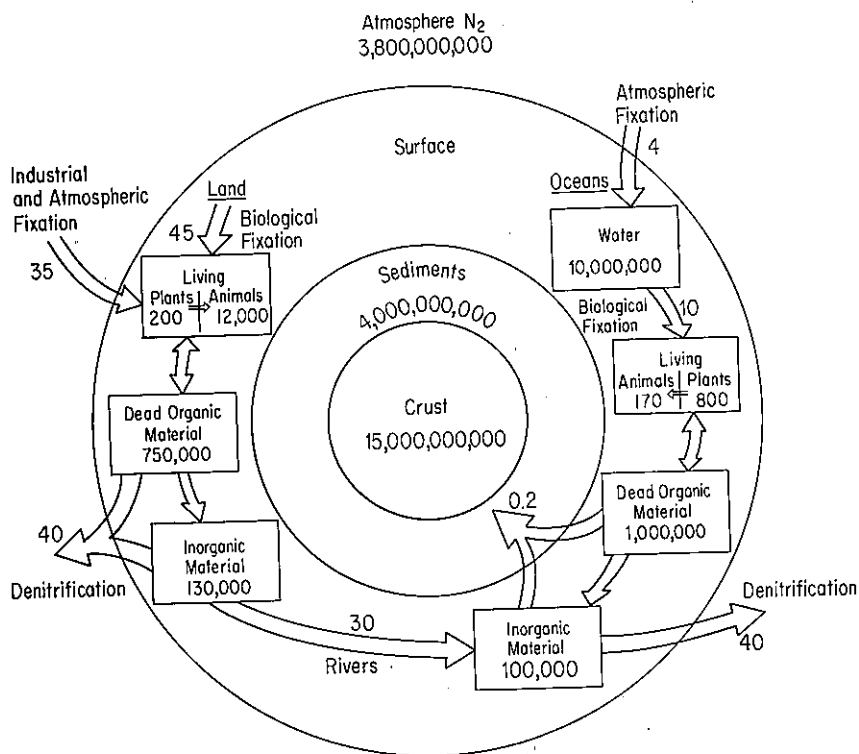


Figure 2.3 Distribution and annual rates of transfer of nitrogen in the biosphere (in millions of metric tons).

Material balances can be made for a wide variety of materials, at many scales of size for the system and in various degrees of complication. To obtain a perspective as to the scope of material balances, examine Figs. 2.3 and 2.4. Figure 2.3 illustrates a world-scale set of mass balances for nitrogen for different systems, while Fig. 2.4 shows a **flowsheet** for a chemical plant that includes both mass and energy flows.

In the process industries, material balances assist in the planning for process design, in the economic evaluation of proposed and existing processes, in process control, and in process optimization. For example, in the extraction of soybean oil from soybeans, you could calculate the amount of solvent required per ton of soybeans or the time needed to fill up the filter press, and use this information in the design of equipment or in the evaluation of the economics of the process. All sorts of raw materials can be used to produce the same end product, and quite a few different types of processing can achieve the same end result, so that case studies (simulations) of the processes can assist materially in the financial decisions that must be made.

Material balances are also used in the hourly and daily operating decisions of plant managers. If there are one or more points in a process where it is impossible or

We should also note in passing that balances using Eq. (2.1) can be made on many other quantities in addition to mass and moles. Balances on dollars are common (your bank statement, for example) as are balances on the number of entities, as in traffic counts, population balances, and social services.

We now look at some simple examples of the application of Eq. (2.1)

EXAMPLE 2.1 Total Mass Balance

A thickener in a waste disposal unit of a plant removes water from wet sewage sludge as shown in Fig. E2.1. How many kilograms of water leave the thickener per 100 kg of wet sludge that enter the thickener? The process is in the steady state.

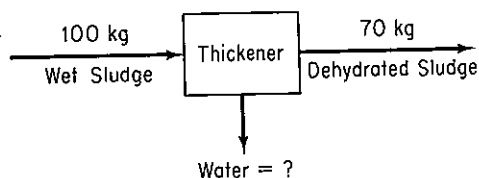


Figure E2.1

Solution

The system is the thickener (an open system). No accumulation, generation, or consumption occur. Use Eq. (2.3). The total mass balance is

$$\begin{array}{ccc} \text{In} & = & \text{Out} \\ \hline 100 \text{ kg} & = & 70 \text{ kg} + \text{kg of water} \end{array}$$

Consequently, the water amounts to 30 kg.

EXAMPLE 2.2 Mass Balances for a Fluidized Bed

Hydrogenation of coal to give hydrocarbon gases is one method of obtaining gaseous fuels with sufficient energy content for the future. Figure E2.2 shows how a free-fall fluidized-bed reactor can be set up to give a product gas of high methane content.

Suppose, first, that the gasification unit is operated without steam at room temperature (25°C) to check the gas flow rate monitoring instruments.

- If 1200 kg of coal per hour (assume that the coal is 80% C, 10% H, and 10% inert material) is dropped through the top of the reactor without the air flowing, how many kg of coal leave the reactor per hour?
- If, in addition to the coal supplied, 15,000 kg of air per hour is blown into the reactor, at 25°C, how many kg of air per hour leave the reactor?
- Finally, suppose that the reactor operates at the temperatures shown in Fig. E2.2, and that 2000 kg of steam (H₂O vapor) per hour are blown into the reactor along with 15,000 kg/hr of air and the 1200 kg of coal. How many kg of gases exit the reactor per hour assuming complete combustion of the coal?

Solution

Basis: 1 hr

The system is the fluidized bed.

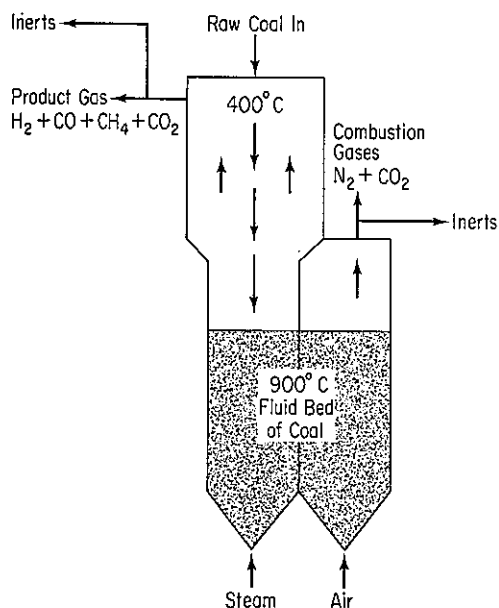


Figure E2.2

- (a) If coal is dropped into the vessel without airflow or reaction, as would be the case at 25°C, 1200 kg of coal must remain in the reactor representing the accumulation:

$$\frac{\text{Accumulation}}{1200} = \frac{\text{Input}}{1200} - \frac{\text{Output}}{0}$$

Hence 0 kg of coal leave the reactor per hour.

- (b) Because the accumulation is zero for air in the reactor, and no reaction occurs

$$\frac{\text{Output}}{15,000} = \frac{\text{Input}}{15,000} - \frac{\text{Accumulation}}{0}$$

The output is 15,000 kg/hr.

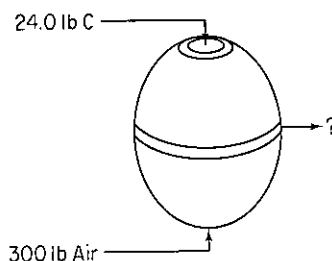
- (c) All the material except the inert portion of the coal leaves as a gas. Consequently, we can add up the total mass of material entering the unit, subtract the inert material, and obtain the mass of combustion gases by difference:

$$\frac{1200 \text{ kg coal}}{100 \text{ kg coal}} \left| \frac{10 \text{ kg inert}}{100 \text{ kg coal}} \right. = 120 \text{ kg inert}$$

Entering material	kg
Coal	1,200
Air	15,000
Steam	2,000
Total	18,200
	18,200 - 120 = 18,080 kg/hr of gases

EXAMPLE 2.3 Material Balances

- (a) If 300 lb of air and 24.0 lb of carbon are fed to a reactor (see Fig. E2.3) at 600°F and after complete combustion no material remains in the reactor, how many pounds of carbon will have been removed? How many pounds of oxygen? How many pounds total?
- (b) How many moles of carbon and oxygen enter? How many leave the reactor?
- (c) How many total moles enter the reactor and how many leave the reactor?

**Figure E2.3****Solution**

This is a problem without any accumulation. The system is the reactor and will be treated as an open system as shown in Fig. E2.3. We want to make a total mass balance and CO_2 and O_2 mole balances.

Basis: 300 lb air

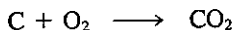
First we calculate the lb mol of oxygen, nitrogen, and carbon entering:

$$\frac{300 \text{ lb air}}{29.0 \text{ lb air}} \times \frac{1 \text{ lb mol air}}{29.0 \text{ lb air}} \times \frac{21.0 \text{ lb mol O}_2}{100 \text{ lb mol air}} = 2.17 \text{ lb mol O}_2$$

$$\frac{2.17 \text{ lb mol O}_2}{21.0 \text{ lb mol O}_2} \times \frac{79.0 \text{ lb mol N}_2}{100 \text{ lb mol air}} = 8.17 \text{ lb mol N}_2$$

$$\frac{24.0 \text{ lb C}}{12.0 \text{ lb C}} \times \frac{1 \text{ lb mol C}}{12.0 \text{ lb C}} = 2.00 \text{ lb mol C}$$

The chemical reaction is



From the stoichiometry, 2.00 lb mol C requires 2.00 lb mol O_2 for complete combustion, so that a mole balance for O_2 leads to the conclusion that oxygen is the excess reactant, and that

$$\frac{\text{out}}{\text{O}_2 \text{ out}} = \frac{\text{in}}{2.17} - \frac{\text{consumption}}{2.00} = 0.17 \text{ lb mol}$$

The carbon as C that exits is zero because all the C as such is consumed; from the stoichiometry we conclude for the CO_2 that

$$\frac{\text{out}}{\text{CO}_2} = \frac{\text{in}}{0} + \frac{\text{generation}}{2.00} = 2.00 \text{ lb mol}$$

Summing up all the calculations in the form of a table, we have (mol. wt. atmospheric $N_2 = 28.2$)

	In		Out	
	lb	lb mol	lb	lb mol
C	24.0	2.00	0	0
O ₂	69.5	2.17	5.5	0.17
N ₂	230.5	8.17	230.5	8.17
CO ₂	0	0	88.0	2.00
Total	324	12.34	324	10.34

We can now answer the questions posed in the problem.

- No carbon will be removed as the element C, but 88 lb of CO₂ will be removed, which contains 24 lb of C. Only 0.17 lb mol of O₂ will be removed as O₂; the remainder of the O₂ is in the CO₂. The total pounds removed will be that placed initially in the reactor, namely 324 lb.
- 2.00 lb mol C and 2.17 lb mol of O₂ enter the reactor, and 0 lb mol of C and 0.17 lb mol of O₂ as elemental species leave the reactor.
- 12.34 lb mol enters the reactor and 10.34 lb mol leaves the reactor.

Note that instead of making mole balances on CO₂ and O₂ as in Example 2.3, we could have made mole, or mass, balances on the elements C and O (or oxygen expressed as O₂) that would lead to the same final information. **You can avoid using the generation and consumption terms in Eq. (2.1) if you make element balances.**

Mole Balance

	In			Out		
	C	CO ₂	O ₂	C	CO ₂	O ₂
C:	2	0	0	= 0	2	0
O:	0	0	2.17(2)	= 0	2(2)	0.17(2)
or O ₂ :	0	0	2.17	= 0	2	0.17

Mass Balance

C:	24	0	0	= 0	24	0
O:	0	0	69.5	= 0	64	5.5
or O ₂ :	0	0	69.5	= 0	64	5.5

Of course, the element balances make use of the same stoichiometric information, as do the component balances.

What can we conclude from Example 2.3? You can observe that although the total mass put into a process and the total mass recovered from a process have been

shown to be equal, there is no such equality on the part of the *total* moles in and out, *if* a chemical reaction takes place. What is true is that the number of atoms of an element (such as C, O, or even oxygen expressed as O_2) put into a process must equal the atoms of the same element leaving the process. In Example 2.3 the total moles in and out are shown to be unequal, but the moles of O in (or moles expressed as O_2) equal the moles of O (or moles expressed as O_2) leaving.

Keeping in mind the remarks above for processes involving chemical reactions, we can summarize the circumstances under which the input equals the output for *steady-state processes* (no accumulation) as follows:

Type of balance	Equality required for input and output of steady-state process	
	Without chemical reaction	With chemical reaction
<i>Total balances</i>		
Total mass	Yes	Yes
Total moles	Yes	No
<i>Component balances</i>		
Mass of a pure compound	Yes	No
Moles of a pure compound	Yes	No
Mass of an atomic species	Yes	Yes
Moles of an atomic species	Yes	Yes

We now turn to more detailed consideration of material balance problems. Your basic task is to turn the problem, expressed in words, into a quantitative form, expressed in mathematical symbols and numbers, and then solve the resulting mathematical equations.

Self-Assessment Test

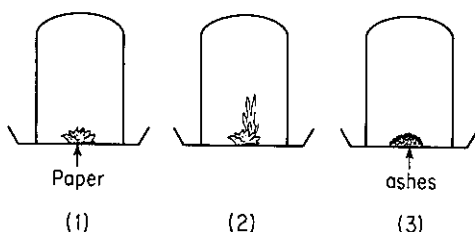
- Draw a sketch of the following processes and place a dashed line around the system:
 - Tea kettle
 - Fireplace
 - Swimming pool
- Show the materials entering and leaving the systems in problem 1. Designate the time interval of reference and classify the system as open or closed.
- Write down the general material balance in words. Simplify it for each process in problem 1 above, stating the assumptions made in each simplification.
- Classify the following processes as (1) batch, (2) flow, (3) neither, or (4) both on a time scale of one day:
 - Oil storage tank at a refinery
 - Flush tank on a toilet
 - Catalytic converter on an automobile
 - Gas furnace in a home

5. What is a steady-state process?
6. Do the inputs and outputs of the chemicals in Fig. 2.3 agree? Why not? Repeat for Fig. 2.4 for the chemicals.
7. Define a material balance. A mass balance.
8. Answer the following true (T) or false (F).
 - (a) If a chemical reaction occurs, the total masses entering and leaving the system for a steady-state process are equal.
 - (b) In combustion, all of the moles of C that enter a steady-state process exit from the process.
 - (c) The number of moles of a chemical compound entering a steady-state process in which a reaction occurs with that compound can never equal the number of moles of that same compound leaving the process.
9. List the circumstances for a steady-state process in which the number of moles entering the system equals the number of moles leaving the system.

Thought Problem

1. Examine the Figure. A piece of paper is put into the bell in (1). In picture (2) we set fire to the paper. Ashes are left in (3). If everything has been weighed (the bell, the dish and the substances) in each case, we would observe that:
 - (a) Case 1 would have the larger weight.
 - (b) Case 2 would have the larger weight.
 - (c) Case 3 would have the larger weight.
 - (d) None of the above.

Explain your answer.



2.2 PROGRAM OF ANALYSIS OF MATERIAL BALANCE PROBLEMS

Your objectives in studying this section are to be able to:

1. Define what the term "solution of a material balance problem" means.
2. Ascertain that a unique solution exists for a problem using the given data, and/or ascertain the number of degrees of freedom in a problem so that additional information can be obtained (and get it).
3. Decide which equations to use if you have redundant equations.

4. Solve a set of n independent equations containing n variables whose values are unknown.
5. Retain in memory and recall as needed the implicit constraints in a problem.
6. Prepare material flow diagrams from word problems.
7. Translate word problems and the associated diagrams into material balances with properly defined symbols for the unknown variables and consistent units for steady-state processes with and without chemical reaction.
8. State the maximum number of independent equations that can be generated in a specific problem.
9. Recite the 10 steps used to analyze material balance problems so that you have an organized strategy for solving material balance problems.

One of the main objectives you should have in studying this chapter is to develop a logical methodology of your own to solve material balance problems. Descartes summed up the matter more than three centuries ago, when he wrote in his "Discours de la Methode": "Ce n'est pas assez d'avoir l'esprit bon, mais le principal est de l'appliquer bien." In English: "It is not enough to have a good intelligence—the principal thing is to apply it well."

Much of the remaining portion of this chapter demonstrates the techniques of analyzing and solving problems involving material balances. Later portions of the text consider the case of combined material and energy balances. By **solving** we mean obtaining a **unique** solution. Refer to Appendix L for more information about the concept of a unique solution if the concept is not clear to you. Because material balance problems all involve the same principle, although the details of the applications of the principle may differ slightly, we shall examine in this section a generalized method of analyzing such problems which can be applied to the solution of any type of material balance problem.

We are going to discuss a strategy of analysis of material balance problems that will enable you to understand, first, how similar these problems are, and second, how to solve them in the most expeditious manner. For some types of problems the method of approach is relatively simple and for others it is more complicated, but the important point is to regard problems in distillation, crystallization, evaporation, combustion, mixing, gas absorption, or drying not as being different from each other but as being related from the viewpoint of how to proceed to solve them.

An orderly method of analyzing problems and presenting their solutions represents training in logical thinking that is of considerably greater value than mere knowledge of how to solve a particular type of problem. Understanding how to approach these problems from a logical viewpoint will help you to develop those fundamentals of thinking that will assist you in your work as an engineer long after you have read this material. But keep in mind the old Chinese proverb:

None of the secrets of success will work unless you do.

First, let us examine how many material balances you can write in any given problem using Eq. (2.1). Whether a chemical reaction occurs or not, you can write for the defined system a balance on

- (a) Total mass
- (b) Mass (or moles) of each atomic species (H, C, O, etc.)

without using the generation or consumption terms in Eq. (2.1). If a chemical reaction does not occur, you can also write a mass (or mole) balance for each component present in the system without involving the generation and consumption terms. How many balances are required to solve a problem? **You have to have the same number of independent balances as the number of variables whose values are unknown** (and are to be calculated). Refer to Appendix L for instances in which too few or too many equations are evolved from a problem. Thus, there is no point in beginning to solve a set of material balances unless you can be certain that the equations have a unique solution.

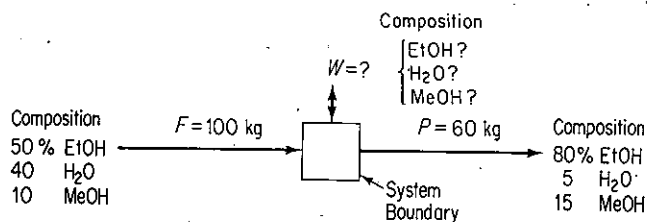


Figure 2.5 A process with three components without chemical reaction.

You can gain some experience by making mass balances on one unit for a simple problem comprised of just three components, as illustrated in Fig. 2.5. In Secs. 2.5 and 2.6 we will take up material balance problems involving more than one system. In Fig. 2.5 the system is the box, and we assume that the process is in the steady state without reaction, so that Eq. (2.3) applies. An independent mass balance equation based on Eq. (2.3) can be written for each compound involved in the process defined by the system boundary. We will use the symbol ω with an appropriate subscript to denote the mass fraction of a component in the streams F , W , and P , respectively. Each mass balance will have the form

$$\omega_{i,F} F = \omega_{i,P} P + \omega_{i,W} W \quad (2.4)$$

but we will, of course, insert the known values of the stream flow rates and mass fractions instead of the symbols where possible. Here are the three balances:

	In	Out	
EtOH:	$(0.50)(100)$	$= (0.80)(60) + \omega_{\text{EtOH},W}(W)$	(2.4a)
H ₂ O:	$(0.40)(100)$	$= (0.05)(60) + \omega_{\text{H}_2\text{O},W}(W)$	(2.4b)
MeOH:	$(0.10)(100)$	$= (0.15)(60) + \omega_{\text{MeOH},W}(W)$	(2.4c)

In addition, we can write a balance for the total material in and out:

$$\begin{array}{rcc} & \text{In} & \text{Out} \\ \text{Total:} & (1.00) 100 = & (1.00)(60) + (\omega_{\text{EtOH},W} \\ & & + \omega_{\text{H}_2\text{O},W} + \omega_{\text{MeOH},W})(W) \end{array} \quad (2.4d)$$

Remember that implicit constraints (equations) exist in the problem formulation because of the definition of mass fraction: namely, that the sum of the mass fractions in each stream must be unity:

$$\omega_{\text{EtOH}} + \omega_{\text{H}_2\text{O}} + \omega_{\text{MeOH}} = 1 \quad (2.5)$$

We could substitute Eq. (2.5) into Eq. (2.4d) to simplify the latter equation, or retain it as a separate equation.

First, let us count the number of independent equations. You should recognize that not all of the four mass balances [Eqs. (2.4a)–(2.4d)], are independent equations. Do you see how the sum of the three component balances (2.4a), (2.4b), and (2.4c) equals the total mass balance? In any problem you can substitute the total material balance for any one of the component material balances if you plan to solve two or more equations simultaneously or if the substitution makes the solution procedure simpler.

Let us just consider Eq. (2.4a)–(2.4c) and (2.5). Are they a set of independent equations? If the answer (yes) is not obvious to you from the way in which the equations were formulated, you should review Appendix L to ascertain how to determine in a formal way whether the equations in a set of equations are independent. In general, **but not always**, in the absence of reactions, the number of independent equations equals the number of chemical compounds present, and with reactions occurring, the number of independent equations equals the number of atomic species present. But you may say the equations as posed are not linear because a mass fraction, an unknown, multiplies W , also unknown. However, Eq. (2.5) can be substituted into Eq. (2.4d), the latter solved directly for W , and W substituted into each of Eq. (2.4a)–(2.4c). The result is three linear equations. Can you show that they are independent?

In the first formulation of the problem, we had four independent equations to solve, Eqs. (2.4a)–(2.4c) plus Eq. (2.5). How many variables can have unknown values if a unique solution is to be obtained? Four! Let us count the number of variables whose values are unknown: W , $\omega_{\text{EtOH},W}$, $\omega_{\text{H}_2\text{O},W}$, and $\omega_{\text{MeOH},W}$. The problem specifications worked out quite well—we have the same number of independent equations as unknown values of variables, and can solve the four equations to get a unique solution. To solve the equations as simply as possible, in this particular case you should note by inspection that the equations are not badly *coupled together* by the unknowns and can be reduced to three uncoupled independent linear equations by substituting Eq. (2.5) into Eq. (2.4), solving the latter for W , and substituting the value of W into Eq. (2.4a)–(2.4c). How many variables can have unknown values with three independent equations? Just three. You can solve each of the Eqs. (2.4a)–(2.4c) as simplified separately.

What should you do if the count of independent equations and unknown variables does not match up? The best procedure is to review your analysis of the problem to make sure that you have not ignored some equation(s) or variable(s), double counted, forgotten to look up some missing data, or made some error in your assumptions. For example, let us suppose that the value of P in Fig. 2.5 is not known. The mass balances and summation of mole fraction equations remain the same except that the symbol P representing a variable whose value is unknown is substituted for 60 kg. As a result, the problem contains four independent equations and five unknown variables, and thus has no unique solution unless one more independent equation or piece of information can be assembled. In some problems, no basis is cited, and the number of unknowns is one greater than the number of independent equations. In such a case, you can select an arbitrary basis of 1 or 100 kg, lb, mol, and so on, to provide the essential extra piece of information needed to obtain a unique solution.

Finally, examine Fig. 2.6. How many component mass balances can you make for the process in the figure? The answer is three!

$$50 = 0.80P + 0.05W$$

$$40 = 0.05P + 0.925W$$

$$10 = 0.15P + 0.025W$$

How many unknown variables are there? Two is the answer. Clearly, the problem is overspecified and does not have a unique solution as posed unless one of the equations is redundant. Can you show by the method described in Appendix L, or by substitution, that only two of the component balances (any two) are independent and the third is dependent (redundant)?

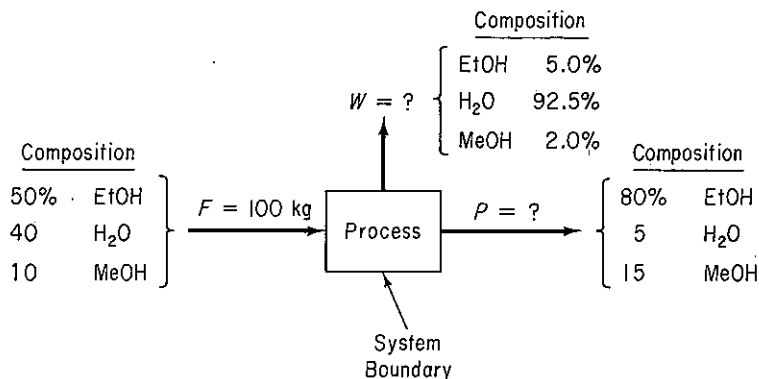


Figure 2.6 A steady state process with one input and two outputs.

If you use the Gauss-Jordan method outlined in Sec. L.1, you will find that the transformed augmented matrix for the set of equations above in the format of Eq. (L.4) will have only zeros in one row, indicating that one equation is not independent and hence redundant. (As explained in Appendix L, if the determinant of the augmented matrix were not equal to zero, the rank of the augmented matrix

would be 3 and the rank of the coefficient matrix would be equal to 2, and consequently the equations would not have a unique solution.)

Could we have made mole balances instead of mass balances for the problem illustrated in Fig. 2.5? Certainly, because no chemical reaction occurs, but it would be inefficient to write such balances. However, if the concentrations had been given in mole percent (which they were not because the flows were of liquids) and F and P were stated in moles, mole balances would be more convenient to write than mass balances. In such cases, Eq. (2.6), the summation of mole fractions

$$x_{\text{EtOH}} + x_{\text{H}_2\text{O}} + x_{\text{MeOH}} = 1 \quad (2.6)$$

would be an implied independent equation analogous to Eq. (2.5).

Now let us look at material balances in which chemical reactions are involved. How does the procedure for analysis differ from the previous analysis for the case without chemical reaction? As mentioned in Sec. 2.1, because moles of a species and total moles are not conserved when a chemical reaction takes place, you will find it convenient to make the balances on the total mass and the mass or moles of each atomic species or multiple thereof (i.e., hydrogen expressed as H_2). For example, look at Fig. 2.7. Note that the flows in Fig. 2.7 are expressed as mass flows, but the compositions are in mole percent because the compounds are gases. We have no information about the extent of reaction or fraction conversion, but, you can write a total mass (not mole) balance, and a carbon, a hydrogen, a nitrogen, and an oxygen balance on the atomic species. Only four of the balances would be independent. The carbon balance might be in terms of C, and the hydrogen, nitrogen, and oxygen balances in terms of H_2 , N_2 , and O_2 , respectively, or, if you prefer, in terms of H, N, and O. In making the balances on the atomic species, keep in mind that the units for each species balance can be either mass or moles. Do you understand why?

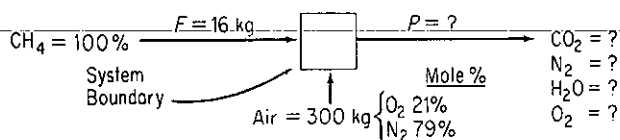


Figure 2.7 A steady state flow process with chemical reactions occurring and complete combustion.

Because you can make four independent balances for the process shown in Fig. 2.7, you can solve for four variables whose values are unknown in the process, such as n_{CO_2} , n_{N_2} , n_{O_2} , and $n_{\text{H}_2\text{O}}$, where n is the number of moles of each species in P ; P would be the sum of the four n 's:

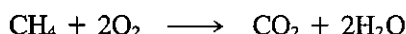
$$P = n_{\text{CO}_2} + n_{\text{H}_2\text{O}} + n_{\text{O}_2} + n_{\text{N}_2} \quad (2.7a)$$

The element balances in moles (on the basis of $16 \text{ kg of CH}_4 \equiv 1 \text{ kg mol CH}_4$) are

$$\frac{300 \text{ kg air}}{29 \text{ kg air}} \left| \frac{1 \text{ kg mol air}}{29 \text{ kg air}} \right| = 10.35 \text{ kg mol air}$$

<i>Balance</i>	<i>CH₄ in</i>	<i>Air in</i>	<i>P out</i>	
C:	1		n_{CO_2}	(2.7b)
H ₂ :	2		$n_{\text{H}_2\text{O}}$	(2.7c)
O ₂ :		$10.35(.21) = 2.17$	$= 0.5n_{\text{H}_2\text{O}} + n_{\text{O}_2} + n_{\text{CO}_2}$	(2.7d)
N ₂ :		$10.35(.79) = 8.17$	$= n_{\text{N}_2}$	(2.7e)

Use of the stoichiometric equation



and the concepts discussed in Sec. 1.9 (plus the assumption of complete combustion of CH₄) also leads to the conclusion that $n_{\text{CO}_2} = 1$ and $n_{\text{H}_2\text{O}} = 2$. The oxygen exiting would have to be calculated from the oxygen balance.

Instead of letting the unknown variables exiting the process in Fig. 2.7 be n , they could have been x_i , the mole fractions of each component. Then five variables whose values were unknown would exist: x_{CO_2} , $x_{\text{H}_2\text{O}}$, x_{O_2} , x_{N_2} , and P . Only four balances are independent. Have we made it impossible to solve this problem by just changing the variables? Of course not. *Hint*: Recall Eq. (2.6). Does an analogous independent equation apply if you make the unknowns mole fractions of the exiting components? The equations would be

$$\text{C : } 1(1.0) = P(x_{\text{CO}_2}) \quad (2.8a)$$

$$\text{H}_2: 1(2.0) = P(x_{\text{H}_2\text{O}}) \quad (2.8b)$$

$$\text{O}_2: 2.17 = P(0.5x_{\text{H}_2\text{O}} + x_{\text{O}_2} + x_{\text{CO}_2}) \quad (2.8c)$$

$$\text{N}_2: 8.17 = P(x_{\text{N}_2}) \quad (2.8d)$$

$$x_{\text{CO}_2} + x_{\text{H}_2\text{O}} + x_{\text{O}_2} + x_{\text{N}_2} = 1 \quad (2.8e)$$

You will find by experience that the second formulation of the problem is as straightforward as the first, but is a bit more complex to solve. Ask yourself the question: How would the formulation in Eq. (2.8e) compare with that in Eq. (2.7a) if you multiplied Eq. (2.8e) by P ? Is $Px_i = n_i$? What are the units of the product of P and x ?

EXAMPLE 2.4 Material Balances for a Distillation Column

A continuous still is to be used to separate acetic acid, water, and benzene from each other. On a trial run, the calculated data were as shown in Fig. E2.4. Data recording the benzene composition of the feed were not taken because of an instrument defect. The problem is to calculate the benzene flow in the feed per hour. How many independent material balance equations can be formulated for this problem? How many variables whose values are unknown exist in the problem?

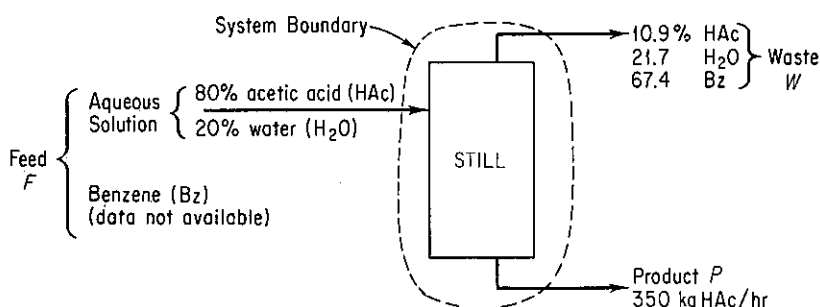


Figure E2.4

Solution

Examine Fig. E2.4. No reaction takes place, and the process is in the steady state. Values of two streams, W and F , are not known if 1 hr is taken as a basis, nor is the concentration of the benzene in F , $\omega_{Bz,F}$. (If you know the concentration of benzene in F , you know all the concentrations in the aqueous feed.) Three components exist in the problem, hence three mass balances can be written down (the units are kg):

Balance	F in	W out	P out	
HAc	$0.80(1 - \omega_{Bz,F})F$	$0.109W$	350	(a)
H ₂ O	$0.20(1 - \omega_{Bz,F})F$	$0.217W$	0	(b)
Benzene	$\omega_{Bz,F}F$	$0.674W$	0	(c)

The total balance would be: $F = W + 350$ in kg. Are the three component balances independent? Because of the zero terms in the right-hand sides of Eqs. (b) and (c), no variation or combination of Eqs. (b) and (c) will lead to Eq. (a). Are Eqs. (b) and (c) redundant equations? No constant exists that when multiplied into Eq. (b) gives Eq. (c), hence the three mass balances are independent.

A more formal way of establishing independence is to form the coefficient matrix of the equations as explained in Sec. L.1:

$$\begin{array}{ccc}
 F & \omega_{Bz,F}F & W \\
 \hline
 \begin{bmatrix} 0.80 & -0.80 & -0.109 \\ 0.20 & -0.20 & -0.217 \\ 0 & 1 & -0.674 \end{bmatrix}
 \end{array}$$

Can you show by elementary operations that the matrix is of full rank, hence the three component mass balances are independent?

EXAMPLE 2.5 Determination of the Number of Independent Balances

The organism *Lymomonos mobilis* is used to convert carbohydrates to ethanol. Glucose (formula $C_6H_{12}O_6$, mol. wt. 180 g/gmol) in a 100 g/L feed solution is converted to ethanol and carbon dioxide in a fermentation tank. The problem is to determine the final molar concentration of ethanol in the product stream and the number of liters of carbon dioxide gas

produced at atmospheric pressure and temperature per liter of feed. Assume that carbon dioxide is insoluble in the liquid. Note that the water in the solution is not involved in the overall metabolism, and that for our purposes the mass flow rates of the materials out of and into the fermenter are constant.

How many independent material balances can be made for this problem, and how many values of the variables are unknown?

Solution

Figure E2.5 illustrates the steady-state process. All of the compositions are known. Three variables have values that are unknown if F is the basis: B , P , and G . Because the concentrations are in moles, we will make mole balances on the elements present. For convenience in presentation, we will take as a basis:

$$\frac{100\text{g}}{180\text{g}} \left| \frac{1\text{ g mol}}{180\text{g}} \right| = 0.556\text{ g mol of glucose that enters in } F$$

<u>Balance</u>	<u>F in</u>	<u>B out</u>	<u>P out</u>	<u>G out</u>
C:	$0.556(6)$	$= B(1) + P(2) + G(6)$		
H:	$0.556(12)$	$= B(0) + P(6) + G(12)$		
O:	$0.556(6)$	$= B(2) + P(0) + G(6)$		

The solvent water can be ignored.

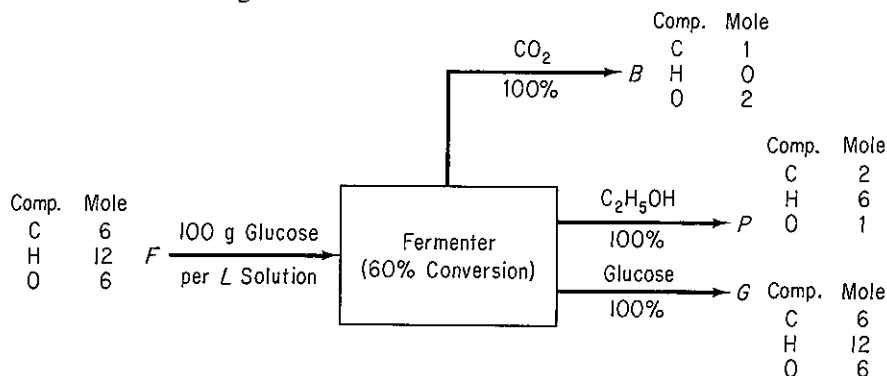


Figure E2.5

The question is: Are those equations independent? Perhaps you can tell by inspection. But, if not, form the coefficient matrix and determine its rank as described in Sec. L.1:

$$\begin{bmatrix} 1 & 2 & 6 \\ 0 & 6 & 12 \\ 2 & 1 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 6 \\ 0 & 6 & 12 \\ 0 & -3 & -6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 6 \\ 0 & 6 & 12 \\ 0 & 6 & 12 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 6 \\ 0 & 6 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

You can see that the three material balances are not independent: The rank of the coefficient matrix is only 2, hence we need one more piece of independent information to solve this problem.

For the extra piece of information needed, assume that 60% of the glucose reacts to form products in passing through the reactor.

	<u>In</u>	<u>Out</u>	<u>Consumed</u>
Glucose balance:	0.556	$= G$	$+ 0.556(0.6)$

Then the glucose exiting is the unreacted glucose, namely $G = 0.556(0.4) = 0.222$ gmol, and the material balances become:

<u>Balance</u>	<u>F in</u>	<u>B out</u>	<u>P out</u>	<u>G out</u>
C:	$0.556(6)$	$= B(1)$	$+ P(2)$	$+ 0.222(6)$
H:	$0.556(12)$	$= B(0)$	$+ P(6)$	$+ 0.222(12)$
O:	$0.556(6)$	$= B(2)$	$+ P(1)$	$+ 0.222(6)$

Now we have two unknowns B and P , and seemingly three equations! But only two of the equations are independent; the rank of the coefficient matrix

$$\begin{bmatrix} 1 & 2 \\ 0 & 6 \\ 2 & 1 \end{bmatrix}$$

can at most be two.

We can conclude from this example that **usually the number of independent material balances equals the number of atomic species in cases in which a reaction occurs, but not always**. If you assume that the balances are independent, but encounter difficulty in solving the equations you formulate, check to see if the equations are independent while checking for other errors.

To sum up the results of the discussion so far, associated with any stream entering or leaving a process are one or more of the so-called **stream variables** composed of the amounts of each of the n species in the stream. Keep in mind that the stream-variables may be comprised of the total flow plus $(n-1)$ components or just the flows of the n species themselves. You want to formulate m independent material balances that can be solved for m variables whose values are unknown (not specified or chosen as a basis). (Additional stream variables such as pressure and temperature will be considered in subsequent chapters as well as other types of balances, such as energy balances.) If more variables whose values are unknown exist than independent equations, an infinite number of solutions exists for a material balance problem—not a satisfactory outcome. Such problems are deemed **underspecified**. Either values of additional variables must be found to make up the deficit or the problem must be posed as an optimization problem such as minimize some cost or revenue function subject to the constraints comprised of the material balances. On the other hand, if fewer values of the variables whose values are unknown exist than independent equations, the problem is **overspecified**, and no solution exists to the problem, as the equations are inconsistent. Again, the problem might be posed as an optimization problem, namely to minimize the sum of the squares of the deviations of the equations from zero (or their right-hand constants).

The difference between the number of variables whose values are unknown and the number of independent equations is known as the **number of degrees of**

freedom. If the degrees of freedom are positive, such as 2, you *must* seek out two additional independent equations or specifications of variables to get a unique solution to your material balance problem. If the degrees of freedom are negative, such as -1 , you have too many equations or not enough variables in the problem. Perhaps you forgot to include one variable in setting up the information diagram for the problem. Perhaps some of the information you used was not correct. **Zero degrees of freedom means that the material balances problem is properly specified**, and you can then proceed to solve the equations for the variables whose values are unknown. (If the independent equations are nonlinear, possibly more than one solution exists, as, for example, in solving a quadratic equations $ax^2 + bx + c = 0$, the solutions for which are $x = (-b \pm \sqrt{b^2 - 4ac})/2a$; refer to Sec. L.2.)

EXAMPLE 2.6

Examine Fig. E2.6a, which represents a simple flow sheet for a single unit. Only the value of D is known. What is the minimum number of other measurements that must be made to determine all the other stream and composition values?

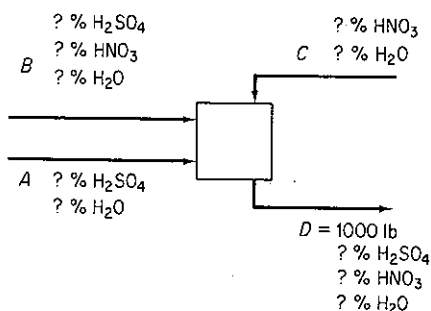


Figure E2.6a

Solution

What you are asked to find is the number of degrees of freedom for the problem in Fig. E2.6. Here is the count of the number of unknown values taking into account that in any stream, one of the composition values can be determined by difference from 100%. Do you remember why?

Stream	Net unknown variables
A	2
B	3
C	2
D	2
Total	9

As example of the count, in stream A specification of A plus one composition makes it possible to calculate the other composition and thus the mass flow of both the H_2SO_4 and the H_2O . In total only three independent material balance equations can be written (do you remember why?), leaving $9 - 3 = 6$ compositions and stream values that have to be specified, and three values to be solved for from the material balances.

Will any six values of the variables in Fig. E2.6a do? No. Only those values can be specified that will leave a number of independent material balances equal to the number of unknown variables. As an example of a satisfactory set of measurements, choose one composition in stream *A*, two in *B*, one in *C*, and two in *D* leaving the flows *A*, *B*, *C* as unknowns. What do you think of the selection of the following set of the measurements: *A*, *B*, *C*, two compositions in *D*, and one composition in *B*? Draw a diagram of the information, as in Fig. E2.6b (* = known quantity). Write down the three material balance equations. Are the three equations independent? You will find that they are not independent. Remember that the sum of the mass fractions is unity for each stream.

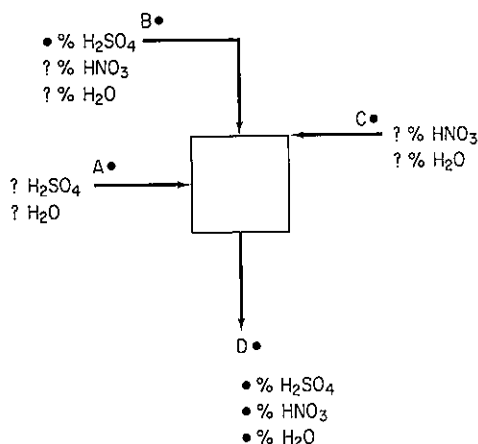


Figure E2.6b

So far we have focused attention on how to write down the material balances and the requirements that have to be met for the equations to have a solution. Now it is time to examine other important aspects of developing skills for successful problem solving. **Problem solving** is an interactive process involving a number of skills, some requiring more practice and experience than others. Figure 2.8 identifies many of the skills you need to develop to be successful in solving problems in this book, while Table 2.1 lists many of the common pitfalls you will encounter in problem solving. Table 2.2 offers some general suggestions to help you solve problems in addition to the obvious suggestion of avoiding the traps listed in Table 2.1. Table 2.3 is a checklist for self-assessment of your problem solving traits. How many of the items in the table pertain to your problem-solving techniques? Practice visual thinking, stress management, and awareness of the process whereby you solve problems. As formulated by Woods,⁴ developing your awareness of your problem-solving skills is an important factor in improving them because

1. You can identify where you are when solving a problem.
2. You can develop a methodical approach.
3. You can, whenever you are stuck, identify the obstacle.

⁴D. R. Woods, *Unit 1, Developing Awareness*, the McMaster Problem Solving Program, McMaster University, Hamilton, Ont., Canada, 1985.

4. You can describe to others what you have done and any difficulties that you are encountering.
5. You become aware of what skills need improvement.
6. You increase your level of confidence.
7. You develop traits of carefulness.

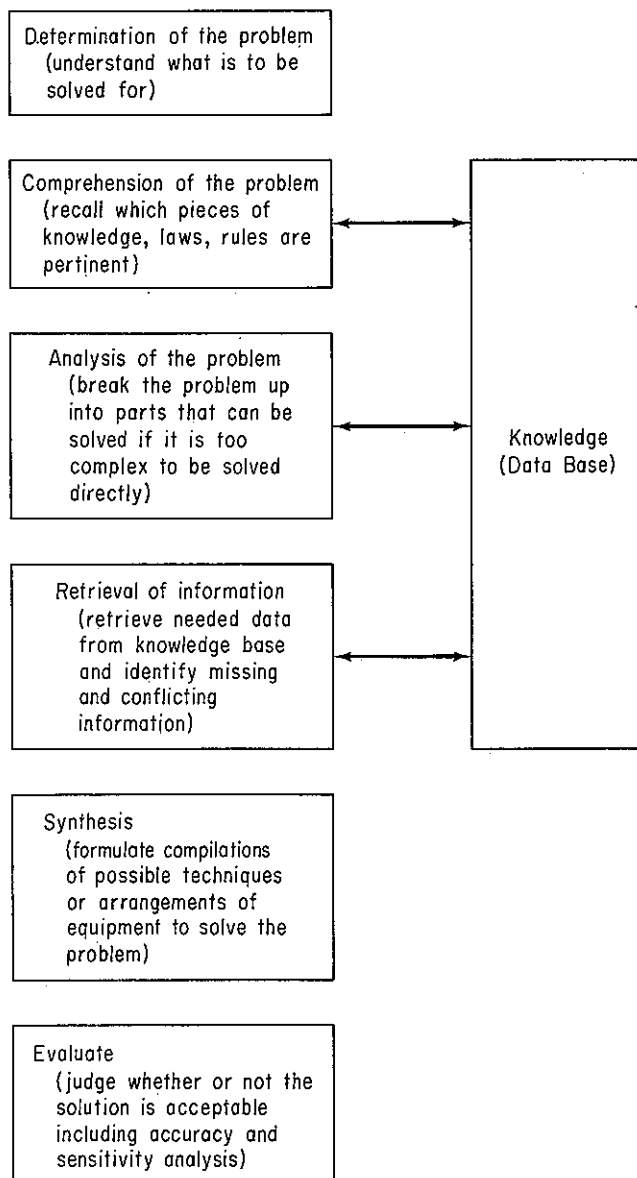


Figure 2.8 Skills needed to solve problems efficiently.

TABLE 2.1 Diagnosis of Reasons for Failing to Solve Problems
(*"Experience is the name everyone gives to their mistakes."* Oscar Wilde)

Failure to work on a problem in a systematic rather than a scatterbrained way (start too soon; skip essential steps)
Failure to read/understand the problem thoroughly
Failure to draw a diagram and enter all data thereon and the symbols for the unknowns
Failure to ascertain the unknown
Fixing on the first, a poor, or an incorrect strategy of solution without considering alternative strategies
Selection of the wrong principle or equation to use (total moles instead of total mass, ideal gas instead of real gas) and solution of the wrong problem
Working with false information
Picking the wrong entry from a data base, chart, or table (wrong sign, wrong units, decimal misplaced, etc.)
Entering incorrect inputs/parameters into calculations (transpose numbers, wrong units, etc.)
Failure to include units in each step of the calculations
Sloppy execution of calculations introduce errors (add instead of subtract, invert coefficients, etc.)
Difficulty in distinguishing new features in a problem that superficially looks familiar
Incorrect algebraic manipulations
Use of unsatisfactory computer code for the problem (too much error, premature termination)

Unable to locate needed data, coefficients by not reading the problem thoroughly or looking in the wrong data base
Unable to estimate what the answer should be to use in comparison with calculated answer
Knowledge (your data base) is inadequate (you have forgotten, or never learned, some essential laws, equations, values of coefficients, conversion factors, etc.)
Only forward reasoning rather than both forward and backward reasoning is employed
Emotional stress (fear of making a mistake, looking foolish or stupid)
Lack of motivation
Inability to relax

TABLE 2.2 Techniques Used by Experts to Overcome Barriers to Problem Solving

Read the problem over several times but at different times. Be sure to understand all facets of it. Emphasize different features each time.

Restate the problem in your own words. List assumptions.

Draw a comprehensive diagram of the process and enter *all* known information on the diagram. Enter symbols for unknown variables and parameters.

Formally write down what you are going to solve for: "I want to calculate . . .".

Choose a basis.

Relate the problem to similar problems you have encountered before, but note any differences.

Plan a strategy for solution; write it down if necessary. Consider different strategies.

Write down all the equations and rules that might apply to the problem.

Formally write down everything you know about the problem and what you believe is needed to execute a solution.

Talk to yourself as you proceed to solve the problem.

Ask yourself questions as you go along concerning the data, procedures, equations involved, etc.

Talk to other people about the problem.

Break off problem solving for a few minutes and carry out some other activity.

Break up the solution of the problem into more manageable parts, and start at a familiar stage. Write down the objective for each subproblem (i.e., convert mole fraction to mass fraction, find the pressure in tank 2, etc.).

Repeat the calculations but in a different order.

Work both forward and backward in the solution scheme.

Consider if the results you obtained are reasonable. Check both units and order of magnitude of the calculations. Are the boundary conditions satisfied?

Use alternative paths to verify your solution.

Maintain a positive attitude—you know the problem can be solved—just how is the question.

TABLE 2.3 A Checklist of Personal Traits to Avoid In Problem Solving

1. When I fail to solve a problem, I do not examine how I went wrong.
2. When confronted with a complex problem, I do not develop a strategy of finding out exactly what the problem is.
3. When my first efforts to solve a problem fail, I become uneasy about my ability to solve the problem (or I panic!).
4. I am unable to think of effective alternatives to solve a problem.
5. When I become confused about a problem, I do not try to formalize vague ideas or feelings into concrete terms.
6. When confronted with a problem, I tend to do the first thing I can think of to solve it.
7. Often I do not stop and take time to deal with a problem, but just muddle ahead.
8. I do not try to predict the overall result of carrying out a particular course of action.
9. When I try to think of possible techniques of solving a problem, I do not come up with very many alternatives.
10. When faced with a novel problem, I do not have the confidence that I can resolve it.
11. When I work on a problem, I feel that I am grasping or wandering, and not getting a good lead on what to do.
12. I make snap judgments (and regret them later).
13. I do not think of ways to combine different ideas or rules into a whole.
14. Sometimes I get so charged up emotionally that I am unable to deal with my problem.
15. I jump into a problem so fast, I solve the wrong problem.
16. I depend entirely on the worked-out sample problems to serve as models for other problems.

SOURCE: Based on the ideas in a questionnaire in P.P. Heppner, *P.S.I.*, Department of Psychology, University of Missouri-Columbia, 1982.

If you are sure that the problem is one you have encountered before and solved successfully, you can simply follow your previous procedure. However, if the problem is or looks as if it is a new problem, you need to follow carefully the recommended checklist outlined in Table 2.4. An expert proceeds in problem solving by abbreviated steps; many are done only mentally. A beginner should go through each step explicitly until he or she becomes experienced. For guidance, perhaps you should turn to Sherlock Holmes (as cited in Arthur Conan Doyle's "The Naval Treaty"):

"Do you see any clue?"

"You have furnished me with seven, but of course I must test them before I can pronounce upon their value."

TABLE 2.4 Strategy For Analyzing Material Balance Problems
(“A problem recognized is a problem half-solved.” Ann Landers)

-
1. Draw a sketch of the process; define the system by a boundary.
 2. Label the flow of each stream and the associated compositions with symbols.
 3. Put all the known values of compositions and stream flows on the figure by each stream; calculate additional compositions from the given data as necessary. Or, at least initially identify the known parameters in some fashion.
 4. Select a basis.
 5. List by symbols each of the unknown values of the stream flows and compositions, or at least mark them distinctly in some fashion.
 6. List the number of independent balances that can be written; ascertain that a unique solution is possible. If not, look for more information or check your assumptions.
 7. Select an appropriate set of balances to solve; write the balances down with type of balance listed by each one. Do not forget the implicit balances for mass or mole fractions.
 8. State whether the problem is to be solved by direct addition or subtraction or by the solution of simultaneous equations.
 9. Solve the equations. Each calculation must be made on a consistent basis.
 10. Check your answers by introducing them, or some of them, into the material balances. Are the equations satisfied? Are the answers reasonable?
-

“You must suspect someone?”

“I suspect myself.”

“What?”

“Of coming to conclusions too rapidly.”

If you use the steps in Table 2.4 as a mental checklist each time you start to work on a problem, you will have achieved the major objective of this chapter and substantially added to your professional skills. These steps do not have to be carried out in the order prescribed, and you may repeat steps as the issues in the problem become clearer. But they are all essential.

Past experience has indicated that the two major difficulties you will experience in solving material balance problems have nothing to do with the mechanics of solving the problem (once they are properly formulated!). Instead, you will find that the two main stumbling blocks are:

- (a) After reading the problem you do not really understand what the process in the problem is all about. Help yourself achieve understanding by sketching the system—just a box with arrows as in Figs. 2.5 and 2.6 will do.

- (b) After a first reading of the problem you are confused as to which values of compositions and flows are known and unknown. Help reduce the confusion by putting the known values down on your diagram as you discover them or calculate them, and by putting question marks or alphabetic symbols for the unknown quantities. Try to record the compositions as weight, or mole, fractions (or percents) as you proceed so as to assist in clarifying the choice of basis and the material balances that can be employed. Because you will have to set up and solve a material balance for each unknown you enter on the diagram, try to keep the number of unknown labels to as few as possible.

You should also be concerned with efficient problem solving. You will discover many correct ways to solve a given problem; all will give the same correct answer if properly applied. Not all, however, will require the same amount of time and effort. For example, you may want to solve a problem initially stated in units of pounds by converting the given units to grams or gram moles, solving for the required quantities in SI units and converting the answer back to American engineering units. Such a method is valid but can be quite inefficient—it consumes your time in unnecessary steps, and it introduces unnecessary opportunities for numerical errors to occur. You should, therefore, start developing the habit of looking for efficient ways to solve a problem, not just *a* way. A good example is to substitute the total mass balance for a component mass balance to reduce the number of unknown variables. Cadman publishes computer aided analysis codes that enable you to practice efficient problem solving; refer to the references at the end of this chapter.

We conclude the discussion in this section with a few remarks about solving the material balances you formulate in a problem. Appendix L describes how to solve sets of simultaneous equations, hence the details will not be repeated here. In many problems, you will find that the equations are sufficiently simple that selection of the obvious choice for the sequence of solution of the equations permits you to avoid having to solve two or more equations simultaneously. Typical examples of such problems will be found in the next section, Sec. 2.3, entitled "Solving Material Balance Problems Without Solving Simultaneous Equations." For example, problems in which the mass (weight) of one stream and the composition of one stream are unknown can be solved without difficulty by direct addition or direct subtraction. All you need is a simple hand calculator to compute the values of the unknown variables.

On the other hand, if you cannot avoid solving a set of simultaneous coupled equations because the set you are working with is tightly coupled in its initial structure and cannot be simplified even by successive substitution of one equation into another, you should read Sec. 2.7, in which computer-aided solutions are discussed. As explained in that section, you can use readily available software in computer center libraries or the computer codes in the pocket in the back of this book to solve sets of independent material balances.

In the next section we examine examples of material balance problems that can be solved without solving simultaneous equations.

Self-Assessment Test

1. What does the concept "solution of a material balance problem" mean?
2. (a) How many values of unknown variables can you compute from one independent material balance?
(b) From three?
(c) From four material balances, three of which are independent?
3. A water solution containing 10% acetic acid is added to a water solution containing 30% acetic acid flowing at the rate of 20 kg/min. The product P of the combination leaves at the rate of 100 kg/min. What is the composition of P ? For this process,
 - (a) Determine how many independent material balances can be written.
 - (b) List the names of the balances.
 - (c) Determine how many unknown variables can be solved for.
 - (d) List their names and symbols.
 - (e) Determine the composition of P .

In your solution set out the 10 steps specifically and label them.

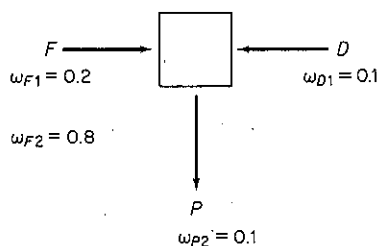
4. Can you solve these three material balances for F , D , and P ?

$$0.1F + 0.3D = 0.2P$$

$$0.9F + 0.7D = 0.8P$$

$$F + D = P$$

5. Cite two ways to solve a set of linear equations.
6. If you want to solve a set of independent equations that contain fewer unknown variables than equations (the overspecified problem), how should you proceed with the solution?
7. What is the major category of implicit constraints (equations) you encounter in material balance problems?
8. If you want to solve a set of independent equations that contain more unknown variables than equations, what must you do to proceed with the solution?
9. How many values of the concentrations and flow rates variables in the process shown in the figure are unknown? List them. The streams contain two components, 1 and 2.



10. How many material balances are needed to solve problem 9? Is the number the same as the number of unknown variables? Explain.
11. A synthesis gas analyzing CO_2 : 6.4%, O_2 : 0.2%, CO : 40.0%, and H_2 : 50.8% (the balance is N_2) is burned with excess dry air. The problem is to determine the quantitative composition of the flue gas. How many degrees of freedom exist in this problem, that is, how many additional variables have to have their values specified?

2.3 SOLVING MATERIAL BALANCE PROBLEMS THAT DO NOT ENTAIL SOLVING SIMULTANEOUS EQUATIONS

Your objectives in studying this section are to be able to:

1. Define flue gas, stack gas, Orsat analysis, dry basis, wet basis, theoretical air (oxygen), required air (oxygen), and excess air (oxygen).
2. Given two of the three factors: entering air (oxygen), excess air (oxygen), and required air (oxygen), compute the third factor.
3. Apply the 10-step strategy to solve problems (with or without chemical reaction) having a direct solution (i.e., problems in which the equations are decoupled so that simultaneous equations do not have to be solved).

Problems in which the material balances are decoupled can be solved without the solution of sets of simultaneous equations. An example would be a problem in which one mass (weight) and one composition are unknown. Such a problem can be solved by direct addition or subtraction as shown in the examples below. You may find it necessary to make some brief preliminary calculations to decide whether or not all the information about the compositions and weights that you need to have is available. Of course, in a stream containing just one component, the composition is known, because that component is 100% of the stream. Once you find out that the number of degrees of freedom is zero, you can proceed to solve the equations in sequence one at a time. Before examining some examples, we need to emphasize some terms commonly used in combustion problems.

In dealing with problems involving combustion, you should become acquainted with a few special terms:

- (a) **Flue or stack gas**—all the gases resulting from a combustion process including the water vapor, sometimes known as *wet basis*.
- (b) **Orsat analysis or dry basis**—all the gases resulting from the combustion process not including the water vapor. (Orsat analysis refers to a type of gas analysis apparatus in which the volumes of the respective gases are measured over and in equilibrium with water; hence each component is saturated with water vapor. The net result of the analysis is to eliminate water as a component being measured.)

Pictorially, we can express this classification for a given gas as in Fig 2.9. To convert from one analysis to another, you have to adjust the percentages for the components as shown in Example 2.12.

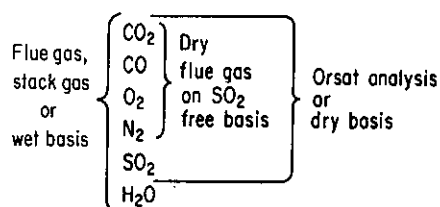


Figure 2.9 Comparison of gas analyses on different bases.

- (c) **Theoretical air (or theoretical oxygen)**—the amount of air (or oxygen) required to be brought into the process for **complete combustion**. Sometimes this quantity is called the **required air** (or oxygen).
- (d) **Excess air (or excess oxygen)**—in line with the definition of excess reactant given in Chap. 1, excess air (or oxygen) would be the amount of air (or oxygen) **in excess of that required for complete combustion** as computed in (c).

The calculated amount of excess air does not depend on how much material is actually burned but what can be burned. Even if only partial combustion takes place, as, for example, C burning to both CO and CO₂, the excess air (or oxygen) is computed as if the process of combustion produced only CO₂. The percent excess air is identical to the percent excess O₂ (a more convenient calculation):

$$\% \text{ excess air} = 100 \frac{\text{excess air}}{\text{required air}} = 100 \frac{\text{excess O}_2/0.21}{\text{required O}_2/0.21} \quad (2.9)$$

Note that the ratio 1/0.21 of air to O₂ cancels out in Eq. (2.9). Percent excess air may also be computed as

$$\% \text{ excess air} = 100 \frac{\text{O}_2 \text{ entering process} - \text{O}_2 \text{ required}}{\text{O}_2 \text{ required}} \quad (2.10)$$

or

$$\% \text{ excess air} = 100 \frac{\text{excess O}_2}{\text{O}_2 \text{ entering} - \text{excess O}_2}$$

since

$$\text{O}_2 \text{ entering process} = \text{O}_2 \text{ required for complete combustion} + \text{excess O}_2 \quad (2.11)$$

The precision of these different relations for calculating the percent excess air may not be the same. If the percent excess air and the chemical equation are given in a problem, you know how much air enters with the fuel, and hence the number of unknowns is reduced by one.

In the burning of coal, you may wonder how to treat the oxygen found in most coals in some combined form. Just assume that the oxygen is already combined with some of the hydrogen in the coal in the proper proportions to make water. Consequently, the O₂ does not enter into the combustion process, and the hydrogen equivalent to this oxygen does not enter the combustion reaction either. No corresponding

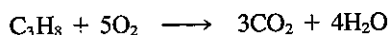
oxygen is counted toward the required oxygen needed in the air. Only the remaining or "net" hydrogen requires oxygen from the air to form water vapor on burning.

Now let us look at some examples of solving material balance problems without having to solve simultaneous equations.

EXAMPLE 2.7 Excess Air

Fuels for motor vehicles other than gasoline are being eyed because they generate lower levels of pollutants than does gasoline. Compressed propane has been suggested as a source of economic power for vehicles. Suppose that in a test 20 lb of C_3H_8 is burned with 400 lb of air to produce 44 lb of CO_2 and 12 lb of CO . What was the percent excess air?

Solution



Basis: 20 lb of C_3H_8

Since the percentage of excess air is based on the *complete combustion* of C_3H_8 to CO_2 and H_2O , the fact that combustion is not complete has no influence on the definition of "excess air." The required O_2 is

$$\frac{20 \text{ lb } C_3H_8}{44 \text{ lb } C_3H_8} \times \frac{1 \text{ lb mol } C_3H_8}{1 \text{ lb mol } C_3H_8} \times \frac{5 \text{ lb mol } O_2}{1 \text{ lb mol } C_3H_8} = 2.27 \text{ lb mol } O_2$$

(The molecular weight of propane is actually 44.09, but we use the value of 44 for convenience.)

The entering O_2 is

$$\frac{400 \text{ lb air}}{29 \text{ lb air}} \times \frac{1 \text{ lb mol air}}{100 \text{ lb mol air}} \times \frac{21 \text{ lb mol } O_2}{100 \text{ lb mol air}} = 2.90 \text{ lb mol } O_2$$

The percent excess air is

$$100 \times \frac{\text{excess } O_2}{\text{required } O_2} = 100 \times \frac{\text{entering } O_2 - \text{required } O_2}{\text{required } O_2}$$

$$\% \text{ excess air} = \frac{2.90 \text{ lb mol } O_2 - 2.27 \text{ lb mol } O_2}{2.27 \text{ lb mol } O_2} \times 100 = 28\%$$

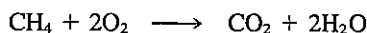
In the following problems each step cited in Table 2.4 will be identified so that you can follow the strategy of the solution.

EXAMPLE 2.8 Material Balances with Combustion

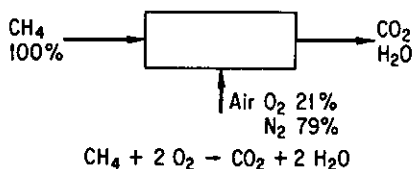
A salesperson comes to the door selling a service designed to check "chimney rot." He explains that if the CO_2 content of the gases leaving the chimney rises above 15%, it is dangerous to your health, is against the city code, and causes your chimney to rot. On checking the flue gas from the furnace, he finds it is 30% CO_2 . Suppose that you are burning natural gas which is about 100% CH_4 and that the air supply is adjusted to provide 130% excess air. Do you need his service?

Solution

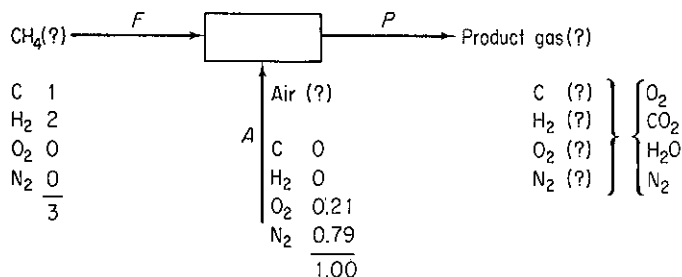
Let us calculate the actual percentage of CO_2 in the gases from the furnace assuming that complete combustion takes place and the process is in the steady state. See Fig. E2.8a. The 130% excess air means 130% of the air required for complete combustion of CH_4 . The chemical reaction is



Step 1 The system is defined in Fig. E2.8a; note that a chemical reaction occurs.

**Figure E2.8a**

Steps 2, 3, and 5 For clarity in labeling, Fig. E2.8a is repeated as Fig. E2.8b with all the species listed by each related stream, and the known (and easily calculated) compositions listed next to each elemental species.

**Figure E2.8b**

Step 4 By picking a basis of 1 mole of CH_4 , we can reduce the unknown quantities by one.

Basis: 1 mole of $\text{CH}_4 = F$

Step 6 We can only make four elemental material balances, C, H as H_2 , O as O_2 , and N as N_2 , and one summation of moles in the product stream to equal P ; hence one more piece of information is needed, as we have five unknowns: A , and the moles of C, H_2 , O_2 , and N_2 in P .

Step 3 (Repeated) The quantity of air can be ascertained because of the specification of the excess air, and is computed as follows: On the basis of 1 mole of CH_4 , 2 moles of O_2 are required for complete combustion, or

$$\frac{2 \text{ mol O}_2}{0.21 \text{ mol O}_2} \times \frac{1.00 \text{ mol air}}{1.00 \text{ mol O}_2} = 9.52 \text{ mol air required}$$

composed of 2 moles of O_2 and 7.52 moles of N_2 . The excess entering air is $9.52 (1.30) = 12.4$ moles of air, composed of 2.60 moles of O_2 and 9.80 moles of N_2 . Now we have left four quantities with unknown values. We can in principle make four independent material balances (C, H_2 , O_2 , N_2), and use one equation summing the moles in the product gas to calculate the product gas, so that the problem can be solved for the relative quantity of CO_2 .

To sum up our calculations so far, we have entering the furnace together with 1 mole of C and 2 moles of H₂ the following moles:

	Air	O ₂	N ₂
Required portion	9.52	2.00	7.52
Excess portion	12.4	2.60	9.80
Total	21.92	4.60	17.32

Steps 7, 8, and 9 We could formally write down the balances if you wish:

Balance	CH ₄ in	Air in	P out
C:	1		= n_{CO_2}
H ₂ :	2		= $n_{\text{H}_2\text{O}}$
O ₂ :		4.60	= $n_{\text{O}_2} + n_{\text{CO}_2} + \frac{1}{2}n_{\text{H}_2\text{O}}$
N ₂ :		17.32	= n_{N_2}

The n_{CO_2} , $n_{\text{H}_2\text{O}}$, and n_{N_2} can be calculated directly; the n_{O_2} can then be obtained from the oxygen balance. Or you might prefer to calculate the CO₂ and H₂O as follows using the combustion equation since all the entering C exits as CO₂ and all the H exits as H₂O:

$$\text{C balance: } \frac{1 \text{ mol C in}}{1 \text{ mol CH}_4 \text{ in}} \left| \frac{1 \text{ mol CO}_2 \text{ out}}{1 \text{ mol C out}} \right| = 1 \text{ mol CO}_2 \text{ out}$$

$$\text{H}_2 \text{ balance: } \frac{2 \text{ mole H}_2 \text{ in}}{1 \text{ mol CH}_4 \text{ in}} \left| \frac{1 \text{ mol H}_2\text{O out}}{1 \text{ mol H}_2 \text{ out}} \right| = 2 \text{ mol H}_2\text{O out}$$

and the nitrogen as follows:

$$\text{N}_2 \text{ balance: } \frac{2 \text{ mol O}_2 \text{ in}}{1 \text{ mol CH}_4 \text{ in}} \left| \frac{0.79 \text{ mol N}_2}{0.21 \text{ mol O}_2} \right| (1.30 \text{ excess} + 1.00 \text{ req}) \text{ mol O}_2$$

$$= 17.3 \text{ mol out}$$

In either case, $n_{\text{O}_2} = 4.60 - 1 - 1 = 2.60$.

Finally, we calculate the composition of the stack gas.

Component	Moles	Percent
CH ₄	0	0
CO ₂	1.00	4.4
H ₂ O	2.00	8.7
O ₂	2.60	11.3
N ₂	$\left\{ \begin{array}{l} 7.52 \\ 9.80 \end{array} \right.$	75.6
Total	22.9	100.0

The salesman's line seems to be rot all the way through.

Step 10 Check the answer.

EXAMPLE 2.9 Material Balance in Drying Pulp

A wet paper pulp is found to contain 71% water. After drying it is found that 60% of the original water has been removed. Calculate the following:

- The composition of the dried pulp
- The mass of water removed per kilogram of wet pulp

Solution

This is a steady steady-state process. No chemical reaction occurs.

Steps 1, 2, 3, and 5 The process is shown in Fig. E2.9.

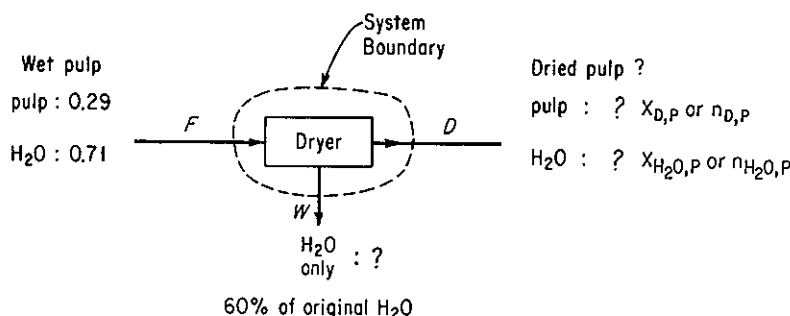


Figure E2.9

Step 4 Pick a convenient basis.

Basis: 1 kg of wet pulp

Step 3 (Repeated) From the statement about the mass of water removed, we can calculate that

$$\text{H}_2\text{O removed} = 0.60(0.71) = 0.426 \text{ kg} = W$$

This is the answer to question (b).

Steps 5 and 6 Only two independent material balances (pulp and H_2O) and one sum of component masses or mass fractions (for the dried pulp) can be written. How many unknowns exist in this problem? Only two, either (a) D and $x_{D,P}$ or $x_{\text{H}_2\text{O},P}$, or (b) D and $m_{D,P}$ or $m_{\text{H}_2\text{O},P}$, or (c) $m_{D,P}$ and $m_{\text{H}_2\text{O},P}$. Keep in mind that

$$\omega_{D,P} + \omega_{\text{H}_2\text{O},P} = 1$$

or

$$m_{D,P} + m_{\text{H}_2\text{O},P} = P$$

are implicit equations relating stream fluid variables. Consequently, a unique solution is possible.

Steps 7, 8, and 9 Formal statement of the component mass balances and the total balance is (in kg)

Balance	F in	W out	D out
Pulp	0.29	0	$m_{p,P}$
H_2O	0.71	0.426	$m_{\text{H}_2\text{O},P}$
Total	1.00	0.426	P

By direct subtraction using the water balance, we can calculate $m_{\text{H}_2\text{O},P}$:

$$m_{\text{H}_2\text{O},P} = 0.71 - 0.426 = 0.284 \text{ kg}$$

or alternatively, we can use the fact that 40% of the original water exits with P :

$$(0.71)(0.40) = 0.284 \text{ kg H}_2\text{O}$$

Of course, all the pulp exits directly in P , so that

$$m_{p,P} = 0.29 \text{ kg}$$

The composition of the dried pulp is

Component	kg	Percent
Pulp (dry)	0.29	50.5
H ₂ O	0.284	49.5
Total	0.574	100.0

Step 10 We check to make sure that total kg in = total kg out

$$\begin{array}{rcl} \text{Wet pulp} & = & \text{Water} + \text{Dried pulp} \\ 1.00 & = & 0.426 + 0.574 \end{array}$$

EXAMPLE 2.10 Crystallization

A tank holds 10,000 kg of a saturated solution of NaHCO_3 at 60°C . You want to crystallize 500 kg of NaHCO_3 without any accompanying water from this solution. To what temperature must the solution be cooled?

Solution

No reaction occurs in this problem. It is best viewed as an unsteady-state problem in which the material in the tank changes and some material is removed from the tank. (The problem could be forced to be a steady-state problem if material is placed into the tank and two types of material are removed, crystals and saturated solution at the unknown temperature.) The major difficulty posed in this problem is to get all the necessary information about the compositions of the solutions and solids.

Step 1 Figure E2.10a is a diagram of the process.

Steps 2 and 3 We do not have at this stage any of the compositions of the saturated solutions in the tank, hence we need to collect some additional data. It is not worth the trouble to label the unknown variables in Fig. 2.10a yet—there are too many of them. Both the initial and final solutions in the tank are saturated solutions; hence if we can find the final concentration of the NaHCO_3 in the tank, we can look up the corresponding temperature in a handbook containing solubility data.

Thus, additional data are needed on the solubility of NaHCO_3 as a function of temperature. From any handbook you can find

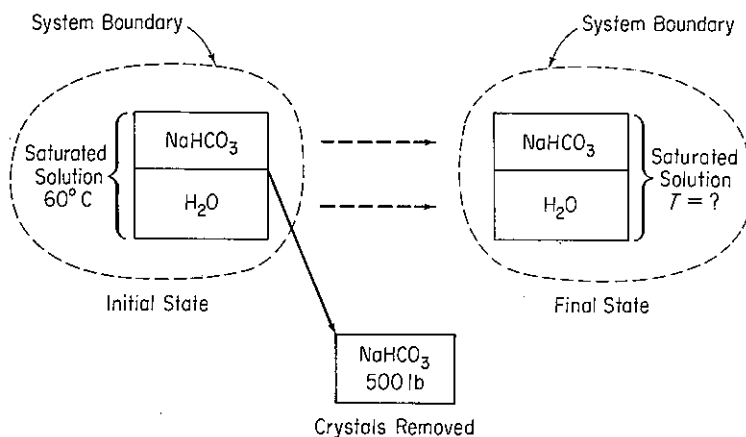


Figure E2.10a

Temp. (°C)	Solubility (g NaHCO ₃ /100 g H ₂ O)
60	16.4
50	14.45
40	12.7
30	11.1
20	9.6
10	8.15

Because the initial solution is saturated at 60°C, we can calculate the composition of the initial solution at this stage:

$$\frac{16.4 \text{ g NaHCO}_3}{16.4 \text{ g NaHCO}_3 + 100 \text{ g H}_2\text{O}} = 0.141 \quad \text{or} \quad 14.1\% \text{ NaHCO}_3$$

The remainder of the solution is water, or 85.9%

Step 4 Take a basis.

Basis: 10,000 kg of saturated solution at 60°C

Steps 2 and 3 (Repeated) We can now enter the known data concerning the compositions on the diagram as shown in Fig. E2.10b.

Steps 5 and 6 We are missing the mass of the final solution and its composition, but can make two independent mass balances and one summation of masses or mass fractions, so a unique solution is possible.

Steps 7, 8, and 9 A direct solution is possible by subtraction. The accumulation term in Eq. (2.1) is negative (depletion) and the generation and consumption terms are zero. Also, only transport out occurs; transport in is zero. The component and total balances in kilograms are (only two are independent)

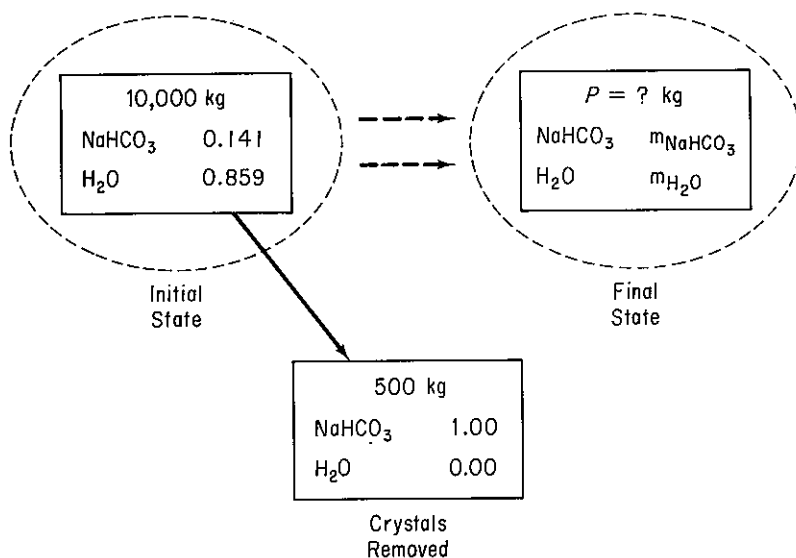


Figure E2.10b

Accumulation in Tank

	<i>Final</i>	<i>Initial</i>	<i>Transport out</i>
NaHCO_3	m_{NaHCO_3}	$10,000 (0.141)$	$= -500(1.00)$
H_2O	$m_{\text{H}_2\text{O}}$	$10,000 (0.859)$	$= -0$
Total	P	$10,000$	$= -500$

The composition of the final solution is

Component	kg	Percent
m_{NaHCO_3}	910	9.6
$m_{\text{H}_2\text{O}}$	8590	90.4
Total	9500	100.0

Step 10 Check on total:

$$9500 + 500 = 10,000$$

To find the temperature of the final solution, calculate the composition of the final solution in terms of grams of NaHCO_3 /100 gram of H_2O .

$$\frac{910 \text{ g NaHCO}_3}{8590 \text{ g H}_2\text{O}} = \frac{10.6 \text{ g NaHCO}_3}{100 \text{ g H}_2\text{O}}$$

Thus the temperature to which the solution must be cooled is (using linear interpolation)

$$30^{\circ}\text{C} - \frac{11.1 - 10.6}{11.1 - 9.6}(10.0^{\circ}\text{C}) = 27^{\circ}\text{C}$$

EXAMPLE 2.11 Continuous Distillation

A novice manufacturer of alcohol for gasohol is having a bit of difficulty with a distillation column. The operation is shown in Fig. E2.11. Too much alcohol is lost in the bottoms (waste). Calculate the composition of the bottoms and the weight of alcohol lost in the bottoms.

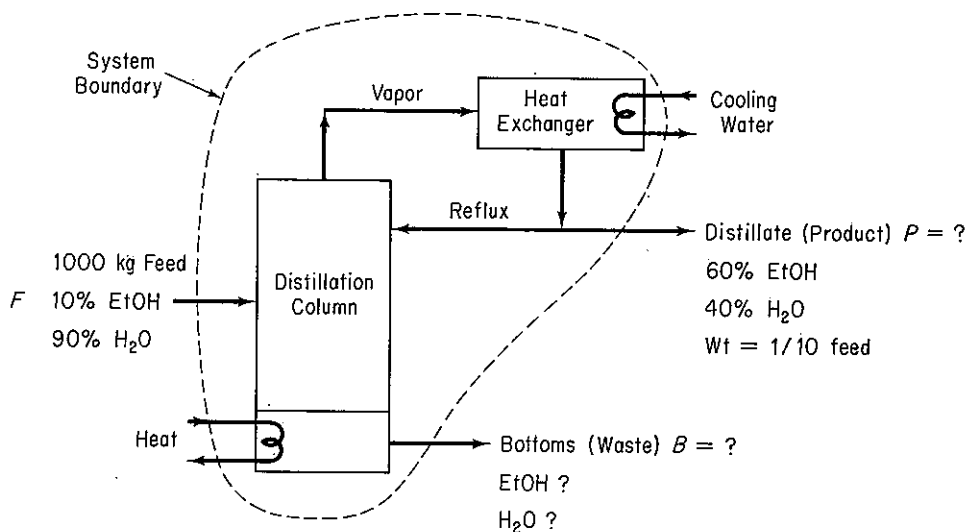


Figure E2.11

Solution

Steps 1, 2, and 3 See Fig. E2.11.

Step 4 Select as the basis the given feed

Basis: 1000 kg of feed

Step 3 (Continued) We are given that P is $\frac{1}{10}$ of F , so that

$$P = 0.1(1000) = 100 \text{ kg}$$

Steps 7, 8, and 9 Calculate B by direct subtraction using the total mass balance

$$B = 1000 - 100 = 900 \text{ kg}$$

Steps 5, 6, 7, 8, and 9 The unknown quantities are the bottoms compositions. We can make two component mass balances, or one sum of masses or mass fractions of the components in B plus one component mass balance, so that the problem has a unique solution.

The solution can be computed directly by subtraction.

	<u>kg feed in</u>	<u>- kg distillate out</u>	<u>= kg bottoms out</u>	<u>percent</u>
EtOH balance:	0.10(1000)	- 0.60(100)	= 40	4.4
H ₂ O balance:	0.90(1000)	- 0.40(100)	= <u>860</u>	<u>95.6</u>
			900	100.0

If we use the total balance to calculate B , all we need to do is make one component balance because

$$\text{mass H}_2\text{O in } B = 900 - 40 = 860 \text{ kg}$$

Step 10 Check: $900 \text{ kg } B + 100 \text{ kg } P = 1000 \text{ kg } F$.

EXAMPLE 2.12 Combustion

Ethane is initially mixed with oxygen to obtain a gas containing 80% C₂H₆ and 20% O₂ that is then burned in an engine with 200% excess air. Eighty percent of the ethane goes to CO₂, 10% goes to CO, and 10% remains unburned. Calculate the composition of the exhaust gas on a wet basis.

In this problem to save space we do not explicitly outline the steps in the analysis and solution of the problem. Use your mental checklist, nevertheless, to ascertain that each step is indeed taken into account.

Solution

We know the composition of the air and fuel gas; if a weight of fuel gas is chosen as the basis, the weight of air can easily be calculated. However, it is wasted effort to convert to a weight basis for this type of problem because all the compositions are expressed in moles or mole percent for gases. The total moles entering and leaving the boiler are not equal, but if we look at any one component and employ the stoichiometric principles discussed in Chap. 1, together with Eq. (2.1), we can easily obtain the composition of the stack gas. The net generation-consumption terms in Eq. (2.1) can be evaluated from the stoichiometric equations listed below. The problem can be worked in the simplest fashion by choosing a basis of 100 moles of entering gas. See Fig. E2.12.

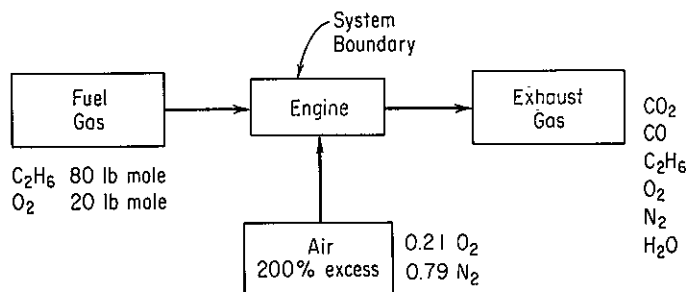
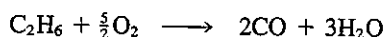
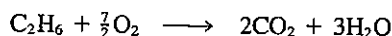


Figure E2.12

Basis: 100 lb mol of fuel



The total O_2 entering is 3.00 times the required O_2 (100% required plus 200% excess). Let us calculate the required oxygen:

O_2 (for complete combustion):

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{3.5 \text{ lb mol O}_2}{1 \text{ lb mol C}_2\text{H}_6} \right| = 280 \text{ lb mol O}_2$$

Required O_2 :

$$280 - 20 = 260 \text{ lb mol O}_2$$

(Note: The oxygen used to completely burn the fuel is reduced by the oxygen already present in the fuel to obtain the oxygen required in the entering air.)

Next we calculate the input of O_2 and N_2 to the system:

O_2 entering with air:

$$3(260 \text{ lb mol O}_2) = 780 \text{ lb mol O}_2$$

N_2 entering with air:

$$\frac{780 \text{ mol O}_2}{21 \text{ lb mol O}_2} \left| \frac{79 \text{ lb mol N}_2}{21 \text{ lb mol O}_2} \right| = 2934 \text{ lb mol N}_2$$

Now we apply our stoichiometric relations to find the components generated by the system:

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{2 \text{ lb mol CO}_2}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.8}{1} \right| = 128 \text{ lb mol CO}_2$$

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{3 \text{ lb mol H}_2\text{O}}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.8}{1} \right| = 192 \text{ lb mol H}_2\text{O}$$

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{2 \text{ lb mol CO}}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.1}{1} \right| = 16 \text{ lb mol CO}$$

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{3 \text{ lb mol H}_2\text{O}}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.1}{1} \right| = 24 \text{ lb mol H}_2\text{O}$$

To determine the O_2 remaining in the exhaust gas, we have to find how much of the available (800 lb mol) O_2 combines with the C and H.

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{3.5 \text{ lb mol O}_2}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.8}{1} \right| = 224 \text{ lb mol O}_2 \text{ to burn to CO}_2 \text{ and H}_2\text{O}$$

$$\frac{80 \text{ lb mol C}_2\text{H}_6}{1 \text{ lb mol C}_2\text{H}_6} \left| \frac{2.5 \text{ lb mol O}_2}{1 \text{ lb mol C}_2\text{H}_6} \right| \left| \frac{0.1}{1} \right| = 20 \text{ lb mol O}_2 \text{ to burn to CO and H}_2\text{O}$$

244 lb mol O_2 total "used up" by reaction

By an oxygen (O_2) balance we get

$$\text{O}_2 \text{ out} = 780 \text{ lb mol} + 20 \text{ lb mol} - 244 \text{ lb mol} = 556 \text{ lb mol O}_2$$

The water exiting is

$$\text{H}_2\text{O out} = 192 \text{ lb mol} + 24 \text{ lb mol} = 216 \text{ lb mol H}_2\text{O}$$

The balances on the other compounds— C_2H_6 , CO_2 , CO , N_2 —are too simple to be formally listed here.

Summarizing these calculations, we have:

Component	lb mol			Percent in exhaust gas
	Fuel	Air	Exhaust gas	
C_2H_6	80	—	8	0.21
O_2	20	780	556	14.41
N_2	—	2934	2934	76.05
CO_2	—	—	128	3.32
CO	—	—	16	0.41
H_2O	—	—	216	5.60
Total	100	3714	3858	100.00

On a dry basis we would have (the water is omitted in the exhaust gas):

Component	lb mol	Percent
C_2H_6	8	0.22
O_2	556	15.27
N_2	2934	80.56
CO_2	128	3.51
CO	16	0.44
Total	3642	100.00

Do you know how to solve this problem using the element balances C, H, O, etc.?

Summary of Material Balances

Total mol in $\neq$ total mol out	
Total lb in = total lb out:	
Balances on atomic species	lb mol C in = lb mol C out: $2(80) = 128 + 16 + 2(8)$
	lb mol H_2 in = lb mol H_2 out: $3(80) = 216 + 3(8)$
	lb mol O_2 in = lb mol O_2 out: $20 + 780 = 128 + \frac{1}{2}(216) + \frac{1}{2}(16) + 556$
	lb mol N_2 in = lb mol N_2 out: $2934 = 2934$
Balances on compounds (Eq. 2.1)	lb mol C_2H_6 in = lb mol C_2H_6 out + lb mol C_2H_6 consumed: $80 = 8 + 72$
	lb mol CO_2 in = lb mol CO_2 out - lb mol CO_2 generated: $0 = 128 - 128$
	lb mol CO in = lb mol CO out - lb mol CO generated: $0 = 16 - 16$
	lb mol H_2O in = lb mol H_2O out - lb mol H_2O generated: $0 = 216 - 216$
	lb mol N_2 in = lb mol N_2 out + lb mol N_2 consumed: $2934 = 2934 + 0$
	lb mol O_2 in = lb mol O_2 out + lb mol O_2 consumed: $800 = 556 + 244$

In the next section we discuss problems that require the simultaneous solution of coupled equations.

Self-Assessment Test

1. Explain the difference between flue gas analysis and Orsat analysis; wet basis and dry basis for a gas.
2. What does "SO₂ free basis" mean?
3. Write down the equations relating percent excess air to required air and entering air.
4. Will the percent excess air always be the same as the percent excess oxygen in combustion (by oxygen)?
5. In a combustion process in which a specified percentage of excess air is used, and in which CO is one of the products of combustion, will the analysis of the resulting exit gases contain more or less oxygen than if all the carbon had gone to CO₂?

In solving the following problems, be sure to employ the 10 steps listed in Table 2.4.

6. Pure carbon is burned in oxygen. The flue-gas analysis is:

CO ₂	75 mol %
CO	14 mol %
O ₂	11 mol %

What was the percent excess oxygen used?

7. Toluene, C₇H₈, is burned with 30% excess air. A bad burner causes 15% of the carbon to form soot (pure C) deposited on the walls of the furnace. What is the Orsat analysis of the gases leaving the furnace?
8. A cereal product containing 55% water is made at the rate of 500 kg/hr. You need to dry the product so that it contains only 30% water. How much water has to be evaporated per hour?
9. If 100 g of Na₂SO₄ is dissolved in 200 g of H₂O and the solution is cooled until 100 g of Na₂SO₄ · 10H₂O crystallizes out, find
 - (a) The composition of the remaining solution (*mother liquor*)
 - (b) The grams of crystals recovered per 100 g of initial solution
10. A synthesis gas analyzing CO₂: 6.4%, O₂: 0.2%, CO: 40.0%, and H₂: 50.8% (the balance is N₂) is burned with excess dry air. The problem is to determine composition of the flue gas. How many degrees of freedom exist in this problem, that is, how many additional variables have to have their values specified?

Thought Problems

1. In a small pharmaceutical plant, it had not been possible for a period of two months to get more than 80% of rated output from a boiler rated at 120,000 lb of steam per hour. The boiler had complete flow metering and combustion control equipment, but the steam flow could not be brought to more than 100,000 lb/hr.

What would you recommend be done to find the cause(s) of the problem and alleviate it?
2. Although modern counterfeiters have mastered the duplication of the outside appearance of precious metals, some simple chemical/physical testing can determine their authenticity. Consult a reference book and determine the densities of gold, silver, copper, lead, iron, nickel, and zinc.

- (a) Could the density of pure gold be duplicated by using any of these metals?
- (b) Could the density of pure silver be duplicated by using any of these metals?
- (c) Assume that the volumes are conserved on mixing of the metals. What is the weight percent composition of a lead/tin alloy that could be used to counterfeit silver? What physical property makes this alloy an unlikely candidate?

2.4 SOLVING MATERIAL BALANCE PROBLEMS INVOLVING SIMULTANEOUS EQUATIONS

Your objectives in studying this section are to be able to:

1. Write a set of independent material balance equations for a process.
2. Solve one or two simultaneous nonlinear equations.
3. Solve a set of linear equations.
4. Apply the 10-step strategy to solve steady-state problems (with or without chemical reaction) that require the solution of simultaneous equations.

In Sec. 2.3 you learned how to formulate uncoupled material balances that could be solved one equation at a time for the values of the unknown variables. All the equations were quite simple, and you did not have to solve two or more equations (called *irreducible equations* because of the nature of their coupling) simultaneously. You should have observed in Sec. 2.3 that the information provided with respect to the problems such as flows and concentrations was of a special structure that led to the resulting material balances being uncoupled.

Many of the problems that you will work with will lead to sets of equations that do have to be solved simultaneously. The purpose of this section is to show you that the formulation of such problems follows the same strategy as used in Sec. 2.3, the strategy outlined in Table 2.4. If only two or three coupled linear material balances are written for a problem, the unknown variables can be solved for by substitution of one equation into another. If the material balances consist of large sets of linear equations, you will find suggestions for solving them in Appendix L, and Fortran computer codes to solve them on the disk in the pocket in the back of this book. A single material balance that is a nonlinear equation of the form $f(x_1, \dots, x_n) = 0$ can be plotted by hand or by using a computer routine, and the root(s), that is, the crossing(s) of the horizontal axis, located; examine Fig. 2.10. If two nonlinear material balances have to be solved, you can plot them and see where they intersect, as in Fig. 2.11. Or, you can use one of the computer codes on the disk accompanying this book to obtain a solution(s) (refer to Sec. L.2 for details). All of the discussion in Sec. 2.2 concerning writing independent equations applies in this section as well. By making a balance for each component or atomic species in the sys-

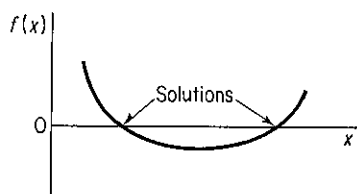


Figure 2.10 Solution(s) of a nonlinear equation by plotting.

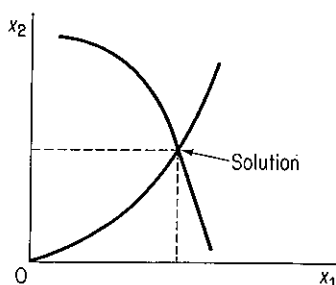


Figure 2.11 Solution of two simultaneous nonlinear equations by plotting.

tem defined, in most instances you can obtain a set of independent equations whether linear or nonlinear. A total mass balance may be substituted for one of the component mass balances. By following these rules of thumb, you should encounter no difficulty in generating sets of independent material balances for any process.

A special case of information structure occurs in some problems that leads to a particularly simple material balance, one that involves only two variables. For example, refer to Fig. 2.12. You note that no reaction takes place and the process presumably is in the steady state. Two components are present, hence you can write two independent mass balances:

$$\text{EtOH:} \quad 0.50 = W(0) + P(0.91)$$

$$\text{H}_2\text{O:} \quad 0.50 = W(1.00) + P(0.09)$$

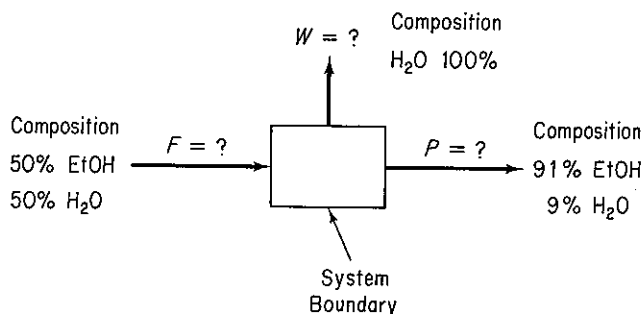


Figure 2.12 Ethanol is the tie component.

The ethanol (EtOH) balance directly relates F and P ; W is not involved, as there is no ethanol in the W stream. A material or component that goes directly from one stream into another without changing in any respect or having like material added to it or lost from it is called a **tie component (element)**. If a tie component exists in a problem, in effect you can write a material balance that involves *only two* streams. To detect a tie component, ask yourself the question: What component or element passes from one stream to another unchanged with constant mass? The answer is the tie component. Frequently, several components pass through a process with continuity so that more than one stream can be connected directly to its respective companion stream. Sometimes a minor constituent passes through with continuity, but if the

percentage error for the analysis of this component is large, you should not use it as a tie component. A tie component can be useful even if you do not know all the compositions and weights in any given problem because the tie component enables you to put two streams on the same basis, thus effecting a *partial solution* even if the entire problem is not resolved.

Several examples follow in which two or more material balance equations have to be solved simultaneously.

EXAMPLE 2.13 Mixing

Dilute sulfuric acid has to be added to dry charged batteries at service stations to activate a battery. You are asked to prepare a batch of new acid as follows. A tank of old weak battery acid (H_2SO_4) solution contains 12.43% H_2SO_4 (the remainder is pure water). If 200 kg of 77.7% H_2SO_4 is added to the tank, and the final solution is 18.63% H_2SO_4 , how many kilograms of battery acid have been made? See Fig. E2.13.

Added Solution 200 kg = A

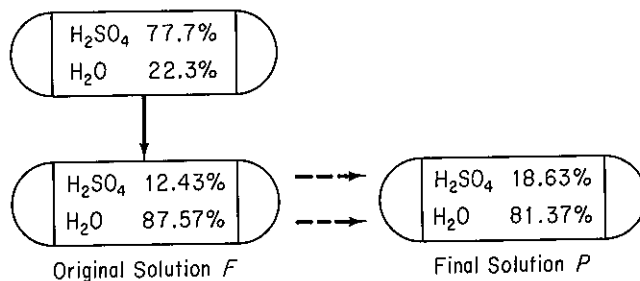


Figure E2.13

Solution

Steps 1, 2, and 3 All the data have been entered on Fig. E2.13. The unknown quantities are F and P .

Step 4 Take 200 kg of A as the basis.

Steps 5 and 6 Two components are present: H_2SO_4 and H_2O , hence you can write two independent mass balances. A unique solution exists.

Steps 7 and 8 The balances will be in kilograms.

Type of Balance	Accumulation in Tank		A in
	Final	Initial	
H_2SO_4	$P(0.1863) - F(0.1243) = 200(0.777)$		
H_2O	$P(0.8137) - F(0.8757) = 200(0.223)$		
Total	P	$- F$	$= 200$

Note that any pair of the three equations are independent, but coupled, hence must be solved simultaneously to get F and P .

Step 9 Because the equations are linear and only two occur, you can take the total mass balance, solve it for F , and substitute for F in the H_2SO_4 balance to calculate P .

$$(P - 200)(0.1243) + 200(0.777) = P(0.1863)$$

$$P = 2110 \text{ kg acid}$$

$$F = 1910 \text{ kg acid}$$

Step 10 Check the answer using the H_2O balance.

EXAMPLE 2.14 Distillation

A typical distillation column is shown in Fig. E2.14 together with the known information for each stream. Calculate the kilograms of distillate per kilogram of feed and per kilogram of waste.

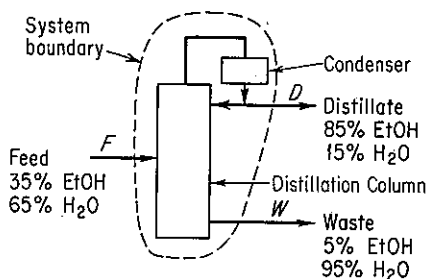


Figure E2.14

Solution

Superficially, the problem in this example looks to be similar to the problem in Example 2.11, but a different set of stream flow values are known and unknown in this example. Consequently, a set of coupled mass balances results. Compare Fig. E2.14 with Fig. E2.11.

Steps 1, 2, and 3 Figure E2.14 already has inscribed on it the preliminary information.

Step 4 Select a basis.

$$\text{Basis: } 1.00 \text{ kg of feed} = F$$

Step 5 The unknown quantities are D and W .

Step 6 Two independent material balances can be written so that a unique solution is possible.

Step 7

Type of balance	In	=	Out
Total	1.00	=	$D + W$
EtOH	$1.00(0.35)$	=	$D(0.85) + W(0.05)$
H_2O	$1.00(0.65)$	=	$D(0.15) + W(0.95)$

Step 8 Two equations must be solved simultaneously.

Step 9 Solve for $W = (1.00 - D)$ using the total balance. Then

$$1.00(0.35) = D(0.85) + (1.00 - D)(0.05)$$

$$D = 0.375 \text{ kg/kg feed}$$

Since $W = 1 - 0.375 = 0.625$ lb,

$$\frac{D}{W} = \frac{0.375}{0.625} = \frac{0.60 \text{ kg}}{\text{kg}}$$

Step 10 Check your solution by using the H_2O balance.

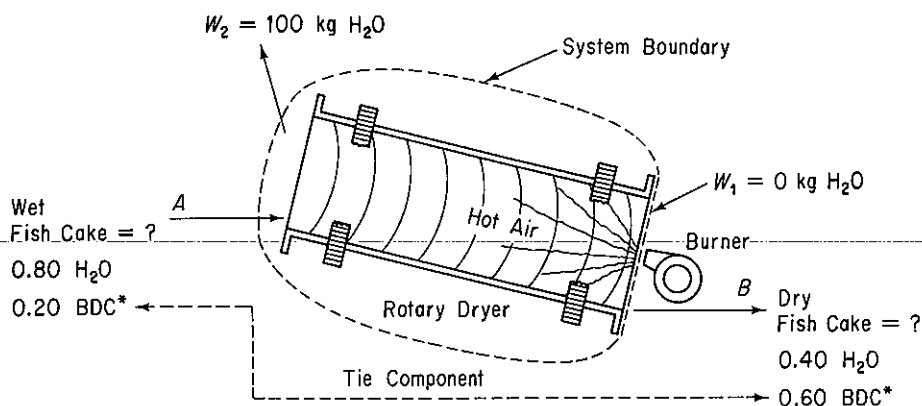
EXAMPLE 2.15 Drying

Fish caught by human beings can be turned into fish meal, and the fish meal can be used as feed to produce meat for human beings or used directly as food. The direct use of fish meal significantly increases the efficiency of the food chain. However, fish-protein concentrate, primarily for aesthetic reasons, is used as a supplementary protein food. As such, it competes with soy and other oilseed proteins.

In the processing of the fish, after the oil is extracted, the fish cake is dried in rotary drum dryers, finely ground, and packed. The resulting product contains 65% protein. In a given batch of fish cake that contains 80% water (the remainder is dry cake), 100 kg of water is removed, and it is found that the fish cake is then 40% water. Calculate the weight of the fish cake originally put into the dryer.

Solution

Steps 1, 2, and 3 Figure E2.15 is a diagram of the process.



*Bone Dry Cake

Figure E2.15

Step 4 Take a basis of what is given.

Basis: 100 kg of water evaporated = W

Step 5 The unknown stream flows are two: A and B. All the compositions are known.

Step 6 Two independent balances can be written so that a unique solution exists.

Steps 7 and 8 To solve the problem requires the use of two independent mass balance equations.

	<u>In</u>	<u>Out</u>	
Total balance:	$A = B + W = B + 100$		} mass balances
BDC balance:	$0.20A = 0.60B$		

Step 9 Notice that because the bone dry cake balance involves only two streams, you can set up a direct ratio of A to B , which is the essence of the tie component:

$$B = \frac{0.20 A}{0.60} = \frac{1}{3}A$$

Introduction of this ratio into the total balance gives

$$A = \frac{1}{3}A + 100$$

$$\frac{2}{3}A = 100$$

$$A = 150 \text{ kg initial cake}$$

We did not use the water balance

$$0.80A = 0.40B + 100$$

because the BDC balance was slightly easier to use. The water balance can be used as a check on the calculations.

Step 10 Water balance:

$$0.80(150) \stackrel{?}{=} 0.40(150)(1/3) + 100$$

$$120 = 120$$

EXAMPLE 2.16 Crystallization

The solubility of barium nitrate at 100°C is 34 g/100 g of H₂O and at 0°C is 5.0 g/100 g of H₂O. If you start with 100 g of Ba(NO₃)₂ and make a saturated solution in water at 100°C, how much water is required? If the saturated solution is cooled to 0°C, how much Ba(NO₃)₂ is precipitated out of solution? The precipitated crystals carry along with them on their surface 4 g of H₂O per 100 g of crystals.

Solution

Steps 1, 2, and 3 The process can be viewed as an unsteady process without reaction in a tank as shown in Fig. E2.16. We first must calculate all the compositions for which data are given.

Step 4 Take as a basis 100 g of Ba(NO₃)₂.

Step 3 (Continued) The maximum solubility of Ba(NO₃)₂ in H₂O at 100°C is a saturated solution, 34 g/100 g of H₂O. Thus the amount of water required at 100°C is

$$\frac{100 \text{ g H}_2\text{O}}{34 \text{ g Ba(NO}_3)_2} \bigg| \frac{100 \text{ g Ba(NO}_3)_2}{1} = 294.1 \text{ g H}_2\text{O}$$

If the 100°C solution is cooled to 0°C, the Ba(NO₃)₂ solution will still be saturated so that the

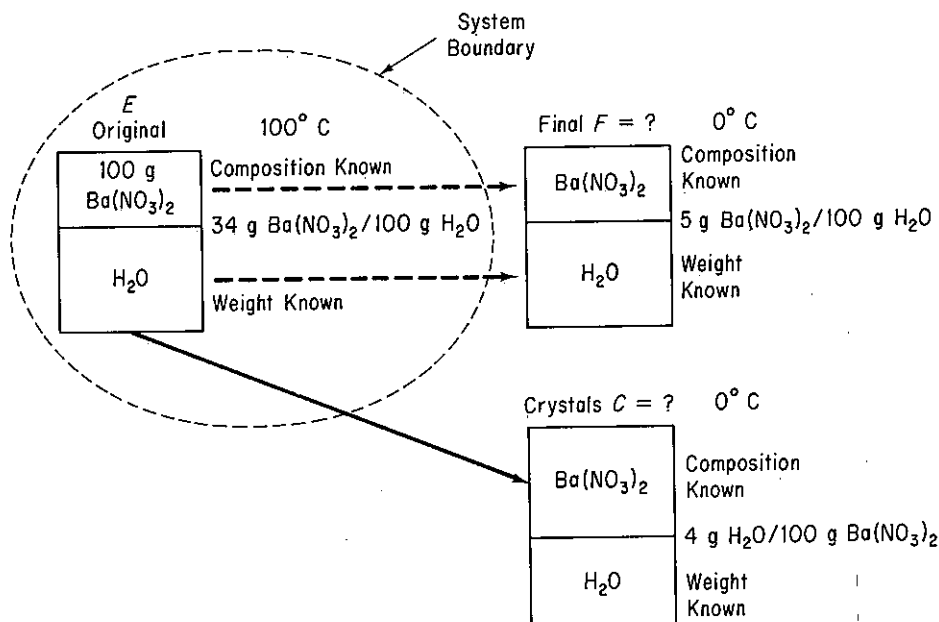


Figure E2.16

composition of the final solution is

$$\text{Ba(NO}_3)_2: \frac{5}{100 + 5} = 0.0476$$

$$\text{H}_2\text{O}: \frac{100}{100 + 5} = 0.9524$$

The composition of the crystals is

$$\text{Ba(NO}_3)_2: \frac{100}{100 + 4} = 0.9615$$

$$\text{H}_2\text{O}: \frac{4}{100 + 4} = 0.0385$$

Steps 5 and 6 We have two unknowns, F and C , and can make two independent mass balances so that the problem has a unique solution.

Steps 7, 8, and 9

<i>Balance</i>	<i>Final solution</i>	<i>Initial solution</i>	<i>Transport through boundary (out)</i>
Ba(NO ₃) ₂ :	$F(0.0476)$	$- 100$	$= -C(0.9615)$
H ₂ O:	$F(0.9524)$	$- 294.1$	$= -C(0.0385)$
Total:	F	$- (100 + 294.1)$	$= -C$

Solve the $\text{Ba}(\text{NO}_3)_2$ and total balance to get

$$F = 305.2 \text{ g} \quad C = 88.89 \text{ g}$$

Step 10 Check using the water balance

$$305.2(0.9524) - 294.1 \stackrel{?}{=} -88.89(0.0385)$$

$$-3.42 = -3.42$$

The $\text{Ba}(\text{NO}_3)_2$ that precipitates out on a dry basis is

$$\frac{88.89 \text{ g C}}{1 \text{ g C}} \times \frac{0.9615 \text{ g Ba}(\text{NO}_3)_2}{1 \text{ g C}} = 85.5 \text{ g Ba}(\text{NO}_3)_2$$

EXAMPLE 2.17 Combustion

The main advantage of catalytic incineration of odorous gases or other obnoxious substances over direct combustion is the lower cost. Catalytic incinerators operate at lower temperatures—500 to 900°C compared with 1100 to 1500°C for thermal incinerators—and use substantially less fuel. Because of the lower operating temperatures, materials of construction do not need to be as heat resistant, reducing installation and construction costs.

In a test run, a liquid having the composition 88% C and 12% H_2 is vaporized and burned with dry air to a flue gas (fg) of the following composition on a dry basis:

CO_2	13.4%
O_2	3.6%
N_2	<u>83.0%</u>
	100.0%

To compute the volume of the continuous steady state combustion device, determine how many kilogram moles of dry fg are produced per 100 kg of liquid feed. What was the percentage of excess air used?

Solution

Steps 1, 2, and 3 The necessary data are placed in Fig. E2.17. Do not forget the water vapor! And that a reaction occurs!

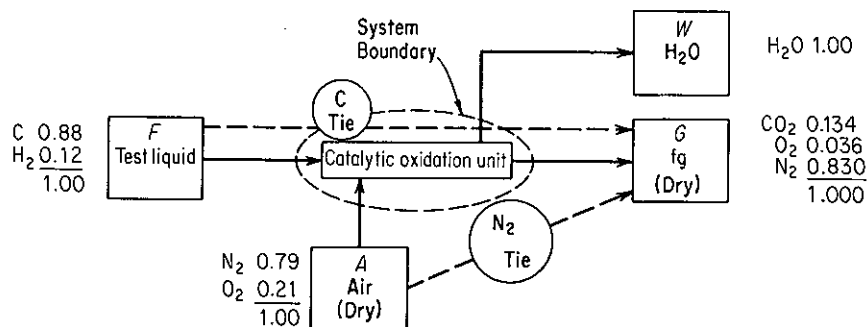


Figure E2.17

Step 4 Select 100 kg mol of dry fg as a convenient basis.

Basis: 100 kg mol dry fg = G

Steps 5 and 6 F , A , and W are all unknown values; all the compositions are known. Four material balances on the elements can be made; hence the problem has a unique solution, and one equation will not be independent. Can you show that the rank of the matrix of the coefficients of F , A , and W is only 3?

Step 7 The atomic species balances (in moles) are (F is in kg; A , W , and G are in kg moles):

$$\begin{array}{rclcl}
 & \frac{F}{12} & + & 0 & = & 0 & + & 100(0.134) \\
 \text{C:} & \frac{F(0.88)}{12} & + & 0 & = & 0 & + & 100(0.134) \\
 & \frac{F(0.12)}{2.016} & + & 0 & = & W & + & 0 \\
 \text{H}_2: & \frac{F(0.12)}{2.016} & + & 0 & = & W & + & 0 \\
 & 0 & + & A(0.21) & = & W(\frac{1}{2}) & + & 100(0.134 + 0.036) \\
 \text{O}_2: & 0 & + & A(0.21) & = & W(\frac{1}{2}) & + & 100(0.134 + 0.036) \\
 & 0 & + & A(0.79) & = & 0 & + & 100(0.830) \\
 \text{N}_2: & 0 & + & A(0.79) & = & 0 & + & 100(0.830)
 \end{array}$$

Steps 8 and 9 The equations can be solved sequentially starting with C, then N_2 , and finally H_2 (to get W , although W is not asked for).

$$\text{From C: } F = \frac{13.4(12)}{0.88} = 182.73 \text{ kg}$$

$$\text{From } N_2: A = \frac{83.0}{0.79} = 105.06 \text{ kg mol}$$

$$\text{From } H_2: W = \frac{182.73(0.12)}{2.016} = 10.88 \text{ kg mol}$$

Step 10 Use the oxygen balance as a check.

$$\begin{array}{rcl}
 105.06(0.21) & \stackrel{?}{=} & 10.88(\frac{1}{2}) + 17.00 \\
 22.06 & \stackrel{?}{=} & 22.44
 \end{array}$$

An exact balance does not occur, but the answers agree reasonably well here. In many combustion problems, slight errors in the data will cause large differences in the calculated flows and percentage of excess air. Assuming that no mathematical mistakes have been made (it is wise to check), the better solution is the one involving the use of the most precise data.

We can now answer the requested questions.

$$\frac{G}{F} = \frac{100 \text{ kg mol}}{182.73 \text{ kg}} = \frac{54.73 \text{ kg mol } G}{100 \text{ kg } F}$$

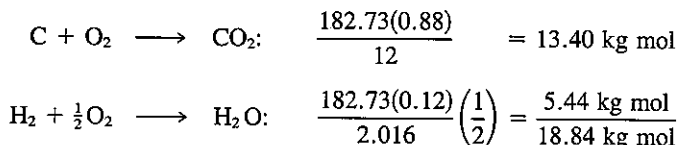
The percent excess air can be calculated via two routes:

$$\% \text{ excess air} = 100 \times \frac{\text{excess } O_2}{O_2 \text{ entering} - \text{excess } O_2} \quad (a)$$

or

$$\% \text{ excess air} = 100 \times \frac{\text{excess O}_2}{\text{required O}_2} \quad (\text{b})$$

The excess O₂ is 3.6 mol; the O₂ entering is [13.4 + 3.6 + 10.88(0.5)] = 22.44 kg mol; and the required O₂ is



By Eq. (a),

$$\% \text{ excess air} = 100 \frac{3.6}{22.44 - 3.6} = 19.1\%$$

By Eq. (b),

$$\% \text{ excess air} = 100 \frac{3.6}{18.84} = 19.1\%$$

You can make use of the tie components as an alternative way to solve the problem in this example as follows.

Because we do not have to answer any questions about the water in the exit flue-gas stream, a tie element to relate the test fluid to the dry flue gas and one to relate the air to the dry flue gas would be sufficient to solve the problem. In examining the data to determine whether a tie element exists, we see that the carbon goes directly from the test fluid to the dry flue gas, and nowhere else, so carbon will serve as one tie component. The N₂ in the air all shows up in the dry flue gas, so N₂ can be used as another tie component.

Let us start to work the problem on a different basis.

Basis: 100 kg of test fluid

$$\frac{88 \text{ kg C}}{12 \text{ kg C}} \left| \frac{1 \text{ kg mol C}}{12 \text{ kg C}} \right. = 7.33 \text{ kg mol C}$$

$$\frac{12 \text{ kg H}_2}{2.016 \text{ kg H}_2} \left| \frac{1 \text{ kg mol H}_2}{2.016 \text{ kg H}_2} \right. = 5.95 \text{ kg mol H}_2$$

(a) By using kilogram moles of C as the tie component, we obtain

$$\frac{100 \text{ kg mol dry fg}}{13.4 \text{ kg mol C}} \left| \frac{7.33 \text{ kg mol C}}{100 \text{ kg test fluid}} \right. = \frac{54.7 \text{ kg mol dry fg}}{100 \text{ kg test fluid}}$$

(b) The N₂ serves as the tie component to tie the air to the dry flue gas, and since we know the excess O₂ in the dry flue gas, the percentage of excess air can be computed.

Basis: 100 kg mol of dry fg

$$\frac{83.0 \text{ kg mol N}_2}{0.79 \text{ kg mol N}_2} \left| \frac{1.00 \text{ kg mol air}}{100 \text{ kg mol dry fg}} \right. = \frac{105.06 \text{ kg mol air}}{100 \text{ kg mol dry fg}}$$

$$(105.06)(0.21) = \text{O}_2 \text{ entering} = 22.06 \text{ kg mol O}_2$$

Or, saving one step,

$$\frac{83.0 \text{ kg mol N}_2}{0.79 \text{ kg mol N}_2 \text{ in air}} \left| \frac{0.21 \text{ kg mol O}_2 \text{ in air}}{0.79 \text{ kg mol N}_2 \text{ in air}} \right| = \frac{22.06 \text{ kg mol O}_2 \text{ entering}}{100 \text{ kg mol dry fg}}$$

$$\% \text{ excess air} = 100 \frac{\text{excess O}_2}{\text{O}_2 \text{ entering} - \text{excess O}_2} = \frac{(100)3.6}{22.06 - 3.6} = 19.5\%$$

If a dry flue gas analysis shows some CO, as in the following hypothetical analysis:

CO ₂	11.9%
CO	1.6%
O ₂	4.1%
N ₂	82.4%

then, on the basis of 100 moles of dry flue gas, we would calculate the percent excess air as follows:

O₂ entering with air:

$$\frac{82.4 \text{ kg mol N}_2}{0.79 \text{ kg mol N}_2} \left| \frac{0.21 \text{ kg mol O}_2}{0.79 \text{ kg mol N}_2} \right| = 21.9 \text{ kg mol O}_2$$

Excess O₂:

$$4.1 - \frac{1.6}{2} = 3.3 \text{ kg mol}$$

$$\% \text{ excess air} = 100 \frac{3.3}{21.9 - 3.3} = 17.7\%$$

Note that to get the true excess oxygen, the apparent excess oxygen in dry flue gas, 4.1 kg moles, has to be reduced by the amount of the theoretical oxygen not combining with the CO. According to the reactions



for each mole of CO in the dry flue gas, $\frac{1}{2}$ mole of O₂ which should have combined with the carbon to form CO₂ did not do so. This $\frac{1}{2}$ mole carried over into the flue gas and inflated the value of the true excess oxygen expected to be in the flue gas. For this analysis, 1.6 kg mol of CO are in the flue gas, so that (1.6/2) kg mol of theoretical oxygen are found in the flue gas in addition to the true excess oxygen. Always keep in mind that the definition of excess oxygen is based on complete combustion.

EXAMPLE 2.18 Combustion of Coal

A local utility burns coal having the following composition on a dry basis.

Component	Percent
C	83.05
H	4.45
O	3.36
N	1.08
S	0.70
Ash	7.36
Total	100.0

The average Orsat analysis of gas from the stack during a 24-hr test was

Component	Percent
CO ₂ + SO ₂	15.4
CO	0.0
O ₂	4.0
N ₂	80.6
Total	100.0

Moisture in the fuel was 3.90%, and the air on the average contained 0.0048 lb H₂O/lb dry air. The refuse showed 14.0% unburned coal, with the remainder being ash.

You are asked to check the consistency of the data before they are stored in a data base. Is the consistency satisfactory? What was the average percent excess air used?

Solution

This is a steady-state problem with reaction.

Steps 1, 2, and 3 The process is shown in Fig. E2.18. In making the diagram, do not forget the water in the stack!

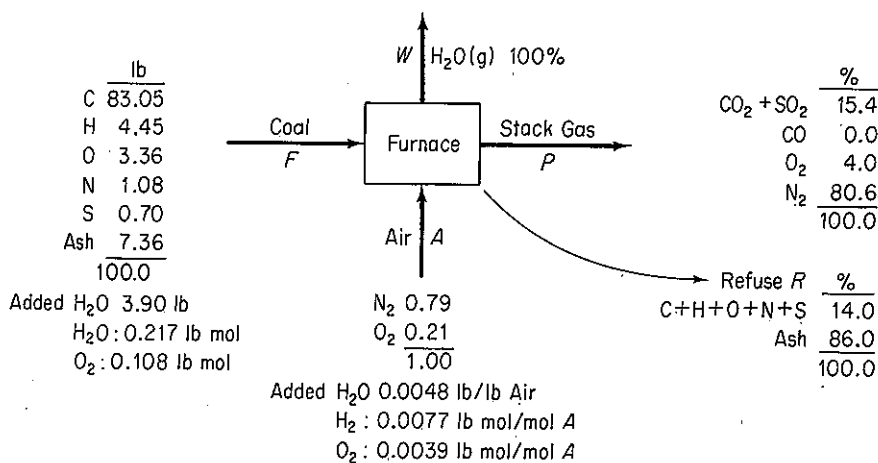


Figure E2.18

Step 4 Pick as a basis $F = 100$ lb.

Step 3, 7, and 8 We need first to get all the compositions possible and put them on the diagram. We calculate the H as H_2 and O as O_2 in the H_2O in both F and A , and the composition of R by use of an ash balance (ash is the tie component).

In coal:

$$\frac{3.90 \text{ lb } H_2O}{18 \text{ lb } H_2O} \left| \frac{1 \text{ lb mol } H_2O}{1 \text{ lb mol } H_2O} \right| \frac{1 \text{ lb mol } H_2}{1 \text{ lb mol } H_2O} = 0.217 \text{ lb mol } H_2$$

$$(0.108 \text{ lb mol } O_2)$$

In air:

$$\frac{0.0048 \text{ lb } H_2O}{\text{lb air}} \left| \frac{29 \text{ lb air}}{1 \text{ lb mol air}} \right| \frac{1 \text{ lb mol } H_2O}{18 \text{ lb } H_2O} = 0.0077 \frac{\text{lb mol } H_2O}{\text{lb mol air}}$$

$$(0.0077 \text{ lb mol } H_2/\text{lb mol } A)$$

$$(0.0039 \text{ lb mol } O_2/\text{lb mol } A)$$

We might neglect the C, H, O, N, and S in the refuse but will include the amounts to show what calculations are necessary if the amounts of the elements are significant. The ash balance is (ash is a tie component)

$$7.36 = R(0.86)$$

$$R = 8.56 \text{ lb}$$

The unburned coal in the refuse is

$$8.56(0.14) = 1.20 \text{ lb}$$

If we assume that the combustibles in the refuse occur in the same proportions as they do in the coal (which may not be true), the quantities of the combustibles in R on an ash-free basis are:

Component	wt %	lb	lb mol
C	89.65	1.076	0.0897
H	4.80	0.058	0.0537
O	3.63	0.0436	0.0027
N	1.17	0.014	0.0010
S	0.76	0.009	0.0003
	100.00	1.20	0.1474

Steps 5 and 6 We can make four additional balances (the C and S must be combined because of the stack gas analysis), and have only three unknown variables: A , W , and P ; hence one balance can be used to check the data as it is redundant. The balances (all in lb mol) are:

	In		Out		
	in F	in A	in W	in P	in R
C + S:	$\frac{83.05}{12} + \frac{0.70}{32}$	+	0	= 0 +	$P(0.154) + 0.0897 + 0.0003$

	<i>In</i>		<i>Out</i>			
	in <i>F</i>	in <i>A</i>	in <i>W</i>	in <i>P</i>	in <i>R</i>	
H ₂ :	$\frac{4.45}{2.016} + 0.217 +$	$0.0077A$	$= W +$	0	$+$	$\frac{0.0537}{2}$
O ₂ :	$\frac{3.36}{32} + 0.108 + 0.21A + 0.0039A =$	$\frac{W}{2} + P(0.154 + 0.040) +$				$\frac{0.0027}{2}$
N ₂ :	$\frac{1.08}{28} +$	$0.79A =$	$0 +$	$P(0.806)$	$+$	$\frac{0.001}{2}$

Solve the C + S balance to get $P = 44.99$. Then solve the N₂ balance to get $A = 45.35$. Next, solve the H₂ balance to get $W = 2.747$. Finally, use the O₂ balance to serve as a check: $9.914 \approx 10.009$. The difference is about 1%. Inasmuch as the data provided are actual measurements, in view of the random and possibly biased errors in the data, the round-off error introduced in the calculations, and possible leaks in the furnace, the data seem to be quite satisfactory. Try calculating W , a small number, from both the H₂ and O₂ balances. What size error do you find?

To calculate the excess air, because of the oxygen in the coal and the existence of unburned combustibles, we will calculate the total oxygen in and the required oxygen:

$$\% \text{ excess air} = 100 \times \frac{\text{O}_2 \text{ entering} - \text{O}_2 \text{ required}}{\text{O}_2 \text{ required}}$$

Assume that no oxygen is required by the ash. The required O₂ is

Component	Reaction	lb		lb mol	Required O ₂ (lb mol)
C	$C + O_2 \rightarrow CO_2$	83.05	C	6.921	6.921
H	$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$	4.45	H ₂	2.207	1.104
O	—	3.36	O ₂	0.105	(0.105)
N	—	—	—	—	—
S	$S + O_2 \rightarrow SO_2$	0.70	S	0.022	0.022
					7.942

and the oxygen in the air is $(45.35)(0.21) = 9.524$ lb mol.

$$\% \text{ excess air} = 100 \times \frac{9.524 - 7.942}{7.942} = 19.9\%$$

If you (incorrectly) calculated the % excess air from just the wet stack gas analysis, you would get

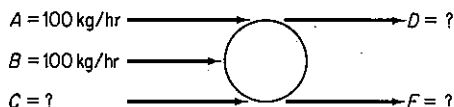
$$100 \times \frac{4.0}{15.4 + 2.747/2} = 23.8\%$$

In the next section we turn to the solution of material balance problems involving multiple units.

Self-Assessment Test

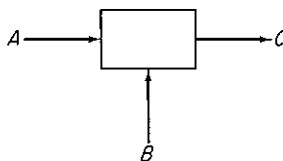
1. Write a set of independent material balance equations for the system shown. The liquid streams are A, B, C, D, and E. The components are 1, 2, and 3. The table lists the known mass fractions.

	A	B	C	D
1:	0.5	0.3	0.6	0.8
2:	0.3	0.2	0.4	0.2
3:	0.2	0.5		



2. In the figure the compositions are (no reaction occurs):

	A	B	C
MgCO ₃ :	0.10	0.05	0.0675
Na ₂ CO ₃ :	0.20	0.10	0.1350
H ₂ O:	0.70	0.85	0.7975



Write down the mass balances and solve for A and B given that C = 100 kg/hr.

3. Solve this equation for positive-real roots:

$$X^4 - X^3 - 10X^2 - X + 1 = 0$$

4. Solve these equations for a set of (x, y) that are real:

$$4.20x^2 + 8.80y^2 = 1.42$$

$$(x - 1.2)^2 + (y - 0.6)^2 = 1$$

In solving the following problems, be sure to employ the 10 steps listed in Table 2.4.

5. A coal analyzing 65.4% C, 5.3% H, 0.6% S, 1.1% N, 18.5% O, and 9.1% ash is burned so that all combustible is burnt out of the ash. The dry flue gas analyzes 13.00% CO₂, 0.76% CO, 6.17% O₂, 0.87% H₂, and 79.20% N₂. All the sulfur burns to SO₂, which is included in the CO₂ figure in the gas analysis (i.e., CO₂ + SO₂ = 13.00%). Calculate:
- Pounds of coal fired per 100 lb mol of dry flue gas as analyzed
 - Ratio of moles total combustion gases to moles of dry air supplied
 - Total moles of water vapor in the stack gas per 100 lb of coal if the air is dry
 - Percent excess air

6. A cellulose solution contains 5.2% cellulose by weight in water. How many kilograms of 1.2% solution are required to dilute 100 kg of 5.2% solution to 4.2%?
7. Hydrofluoric acid (HF) can be manufactured by treating calcium fluoride (CaF_2) with sulfuric acid (H_2SO_4). The other product of the reaction is calcium sulfate (CaSO_4). A sample of fluorospar (the raw material) contains 75% by weight CaF_2 and 25% inert (nonreacting) materials. The pure sulfuric acid used in the process is in 30% excess of that theoretically required. Most of the manufactured HF leaves the reaction chamber as a gas, but a solid cake is also removed from the reaction chamber that contains 5% of all the HF formed, plus CaSO_4 , inerts, and unreacted sulfuric acid. How many kilograms of cake are produced per 100 kg of fluorospar charged to the process?
8. A solution contains 60% $\text{Na}_2\text{S}_2\text{O}_5$ and 1% soluble impurity in water. Upon cooling to 10°C , $\text{Na}_2\text{S}_2\text{O}_5 \cdot 5\text{H}_2\text{O}$ crystallizes out. The solubility of this hydrate is 1.4 lb $\text{Na}_2\text{S}_2\text{O}_5 \cdot 5\text{H}_2\text{O}$ /lb free water. The crystals removed carry as adhering solution 0.06 lb solution/lb crystals. These are dried to remove the remaining water (but not the water of hydration). The final dry $\text{Na}_2\text{S}_2\text{O}_5 \cdot 5\text{H}_2\text{O}$ crystals must not contain more than 0.1% by weight of impurity. In order to meet this specification calculate:
 - (a) The amount of water added before cooling
 - (b) The percentage recovery of the $\text{Na}_2\text{S}_2\text{O}_5$ in the crystals.
9. Salt in crude oil must be removed before the oil undergoes processing in a refinery. The crude oil is fed to a washing unit where freshwater feed to the unit mixes with the oil and dissolves a portion of the salt contained in the oil. The oil (containing some salt but no water), being less dense than the water, can be removed at the top of the washer. The "spent" wash water (containing salt but no oil) is removed at the bottom of the washer. If the "spent" wash water contains 15% salt and the crude contains 5% salt, determine the concentration of salt in the "washed" oil product if the ratio of crude oil (with salt) to water used is 4:1.
10. A hydrocarbon fuel is burnt with excess air. The Orsat analysis of the flue gas shows 10.2% CO_2 , 1.0% CO , 8.4% O_2 , and 80.4% N_2 . What is the atomic ratio of H to C in the fuel?

2.5 SOLVING MATERIAL BALANCE PROBLEMS INVOLVING MULTIPLE SUBSYSTEMS

Your objectives in studying this section are to be able to:

1. Write a set of independent material balances for a complex process involving more than one unit.
2. Solve problems involving several connected units by applying the 10-step strategy.

Chemical plants such as illustrated by the flowsheet in Fig. 2.4 are comprised of many interconnected units. However, you can easily apply the same techniques dis-

cussed so far to such complex processes. **All you have to do is divide the plant up into subsystems that you analyze one at a time.** If more than one piece of equipment or more than one junction point is involved in the problem to be solved, you can write material balances for each piece of equipment and a balance around the whole process. However, since the overall balance is nothing more than the sum of the balances about each piece of equipment, not all the balances you might write will be independent. Appendix L discusses how you can determine whether linear equations are independent or not.

Under some circumstances, particularly if you split a big problem into smaller parts to make the calculations easier, you may want to make a material balance about a **mixing point**. As illustrated in Fig. 2.13, a mixing point is nothing more than a junction of three or more streams and can be designated as a system in exactly the same fashion as any other piece of equipment. Examine Fig. 2.14. Which streams have the same composition? Is the composition of stream 5 the same as the composition inside the unit? It will be if the contents of the unit are **well mixed**, the assumption in this text. Streams 5, 6, and 7 must have the same composition, and presumably if no reaction takes place, the output composition is the properly weighted average of the input compositions 3 and 4.

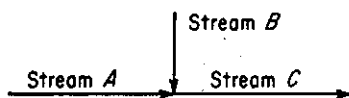


Figure 2.13 Mixing point.

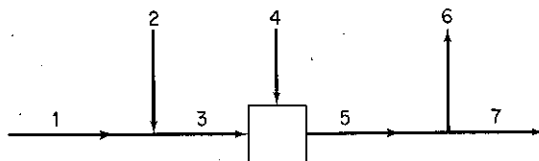


Figure 2.14 Multiple junction points.

Now let us turn to the analysis of simple combinations of units. Suppose that a system is comprised of three subsystems as indicated in Fig. 2.15. You can make material balances for the subsystems and overall system—just make sure that the balances selected for your solution are independent! How many values of the variables are unknown? There will be seven in all: W , P , A , B , C , $\omega_{\text{KCl},A}$ and $\omega_{\text{H}_2\text{O},A}$.

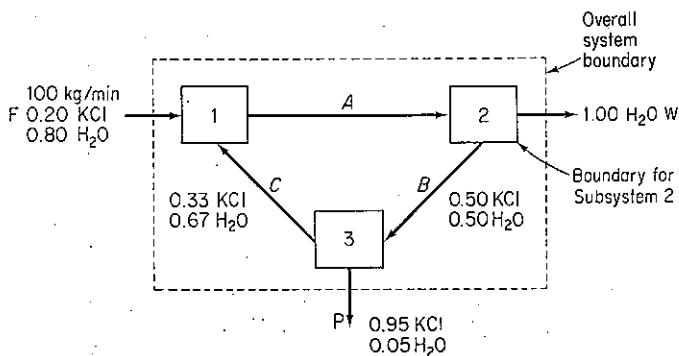


Figure 2.15 Flow diagram of a system comprised of three subsystems. No reaction takes place. The subsystem boundaries are the boxes.

How many independent equations must be written to obtain a unique solution? Seven. How many independent equations can you write? Two for each subsystem plus the sum of mass fractions for stream A. What are the names of such a set of equations? One set might be:

$$\begin{aligned} \text{Unit 1, total:} \quad & 100 + C = A \\ \text{Unit 1, KCl:} \quad & (0.20)(100) + (0.33)(C) = (\omega_{\text{KCl},A})(A) \\ \text{Unit 2, total:} \quad & A = W + B \\ \text{Unit 2, KCl:} \quad & (\omega_{\text{KCl},A})(A) = (0.50)B \\ \text{Overall, total:} \quad & 100 = W + P \\ \text{Overall, KCl:} \quad & (0.20)100 = (0.95)P \\ \Sigma \omega_i = 1: \quad & \omega_{\text{KCl},A} + \omega_{\text{H}_2\text{O},A} = 1 \end{aligned}$$

Other sets are possible. Write down a different set. Did you note that the set we have used has been chosen so as to include as few of the unknown variables as possible in a given equation; that is, we made component balances on KCl and not on H_2O . Keep in mind that the number of degrees of freedom summed for all the subsystems must be equal to zero to have a unique solution for the equations. In general, you can make the same number of independent balances as there are components or atomic species in each subsystem you define, **but not always**. Read the following example.

EXAMPLE 2.19 Independent Material Balances

Examine Fig. E2.19. The composition of each stream is as follows:

- (1) Pure A
- (2) Pure B
- (3) A and B, concentrations known
- (4) Pure C
- (5) A, B, and C, concentrations known
- (6) Pure D
- (7) A and D, concentrations known
- (8) B and C, concentrations known

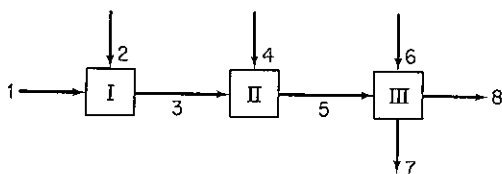


Figure E2.19

What is the maximum number of independent material balances that could be generated to solve this problem?

Solution

	<u>Number of component balances</u>
At unit I, two components are involved	2
At unit II, three components are involved	3
At unit III, four components are involved	4
Total	9

However, not all of the balances are independent. In the following list, all the concentrations are known values, and F represents the stream flow designated by the subscript.

Subsystem 1

$$\text{Balances } \begin{cases} A: F_1(1.00) + F_2(0) = F_3(\omega_{F_3,A}) \\ B: F_1(0) + F_2(1.00) = F_3(\omega_{F_3,B}) \end{cases}$$

Subsystem 2

$$\text{Balances } \begin{cases} A: F_3(\omega_{F_3,A}) + F_4(0) = F_5(\omega_{F_5,A}) \\ B: F_3(\omega_{F_3,B}) + F_4(0) = F_5(\omega_{F_5,B}) \\ C: F_3(0) + F_4(1.00) = F_5(\omega_{F_5,C}) \end{cases}$$

Subsystem 3:

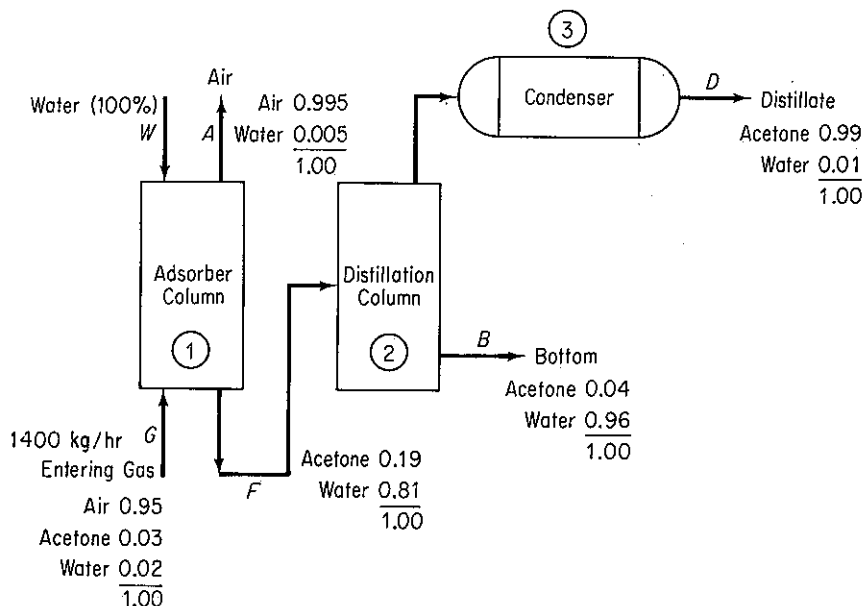
$$\text{Balances } \begin{cases} A: F_5(\omega_{F_5,A}) + F_6(0) = F_7(\omega_{F_7,A}) + F_8(0) \\ B: F_5(\omega_{F_5,B}) + F_6(0) = F_7(0) + F_8(\omega_{F_8,B}) \\ C: F_5(\omega_{F_5,C}) + F_6(0) = F_7(0) + F_8(\omega_{F_8,C}) \\ D: F_5(0) + F_6(1.00) = F_7(\omega_{F_7,D}) + F_8(0) \end{cases}$$

If you take as a basis F_1 , seven values of F_i are unknown, hence only seven independent equations need be written. The B balance in subsystem II and the C balance in subsystem III prove to be redundant (can you show why?), so that a unique solution can be obtained.

Can you show that one or more component balances around systems I and II, or II and III in Example 2.19, or the entire set of three units, will add *no* additional independent balances to the set of component material balances made on each individual unit? Can you show that a total balance on each unit, or on units I + II, or II + III, or around the entire system of three units, will add *no* additional independent balances? Can you substitute one of the indicated alternative material balances for an independent component balance? Yes (as long as the precision of the balance is about the same).

EXAMPLE 2.20 Multiple Units in Which No Reaction Occurs

Acetone is used in the manufacture of many chemicals and also as a solvent. In its latter role, many restrictions are placed on the release of acetone vapor to the environment. You are asked to design an acetone recovery system having the flowsheet illustrated in Fig. E2.20. As a starting point, make a material balance for the system shown in Fig. E2.20. All the concentrations shown in Fig. E2.20 of both gases and liquids are specified in *weight percent* to make the calculations simpler.


Figure E2.20
Solution

Steps 1, 2, and 3 All the stream compositions are given. No reaction occurs, and we assume that the process is in the steady state. All of the unknown stream flows are designated by symbols in the figure.

Step 4 Pick 1 hr as a basis so that $G = 1400$ kg.

Steps 5 and 6 We start the analysis with column 1. Three components exist in column 1, and three values of the streams are unknown: W , A , and F ; hence a unique solution can be obtained for column 1.

Steps 7 and 8 The mass balances for column 1 are as follows.

	<i>In</i>	<i>Out</i>	
Air:	$1400 (0.95)$	$= A (0.995)$	(a)
Acetone:	$1400 (0.03)$	$= F (0.19)$	(b)
Water:	$1400 (0.02) + W (1.00)$	$= F (0.81) + A (0.005)$	(c)

The air and acetone are tie components. (Check to make sure that the equations are independent.)

Step 9 Solve Eqs. (a), (b), and (c) to get

$$A = 1336.7 \text{ kg/hr}$$

$$F = 221.05 \text{ kg/hr}$$

$$W = 157.7 \text{ kg/hr}$$

Step 10 (Check) Use the total balance.

$$G + W = A + F$$

$$\begin{array}{r} 1400 \quad 1336 \\ 157.7 \quad 221.05 \\ \hline 1557.7 \approx 1557.1 \end{array}$$

Steps 5 and 6 Applied to Units 2 and 3 Combined The subsystem boundary for analysis encompasses the distillation column and the condenser. Two components exist, and two values of the flow streams are unknown, D and B ; hence a unique solution exists (if the mass balances are independent, as they are).

Steps 7 and 8 The mass balances are

$$\text{Acetone: } 221.05(0.19) = D(0.99) + B(0.04) \quad (d)$$

$$\text{Water: } 221.05(0.81) = D(0.01) + B(0.96) \quad (e)$$

Step 9 Solve Eqs. (d) and (e) to get

$$D = 34.91 \text{ kg/hr}$$

$$B = 186.1 \text{ kg/hr}$$

Step 10 (Check) Use the total balance

$$F = D + B \text{ or } 221.05 \approx 34.91 + 186.1 = 221.01$$

As a matter of interest, what other mass balances could be written for the system and substituted for any one of the Eqs. (a)–(e)? Typical balances would be the overall balances.

	<u>In</u>	<u>Out</u>	
Air:	$G(0.95)$	$A(0.995)$	(f)
Acetone:	$G(0.03)$	$D(0.99) + B(0.04)$	(g)
Water:	$G(0.02)$	$A(0.005) + D(0.01) + B(0.96)$	(h)
Total:	$G + W$	$A + D + B$	(i)

Equations (f)–(i) do not add any extra information to the problem; the degrees of freedom are still zero. But any one of the equations can be substituted for one of Eqs. (a)–(e) as long as you make sure that the resulting set of equations is independent.

EXAMPLE 2.21 Multiple Units in Which a Reaction Occurs

In the face of higher fuel costs and the uncertainty of the supply of a particular fuel, many companies operate two furnaces, one with natural gas and the other with fuel oil. In the

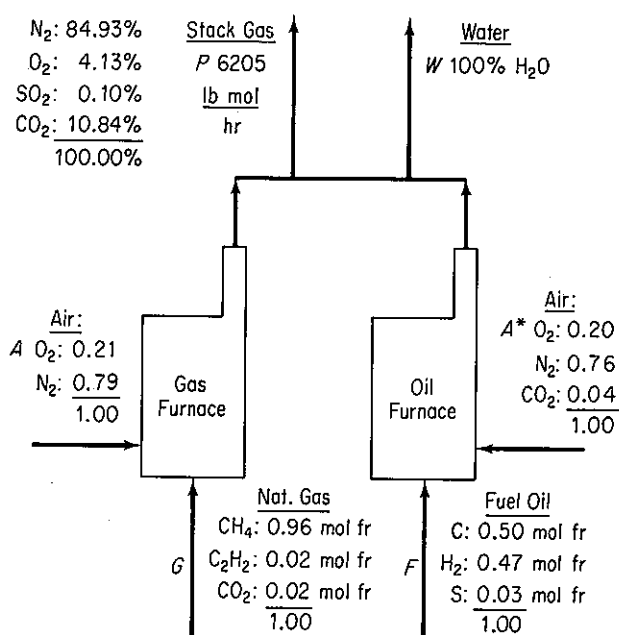


Figure E2.21

RAMAD Corp., each furnace had its own supply of oxygen; the gas furnace used air, and the fuel oil furnace used a gas stream that analyzed: O_2 , 20%; N_2 , 76%; and CO_2 , 4%, but the stack gases went up a common stack. See Fig. E2.21.

During one blizzard, all transportation to the RAMAD Corp. was cut off, and officials were worried about the dwindling reserves of fuel oil because the natural gas supply was being used at its maximum rate possible. The reserve of fuel oil was only 1000 bbl. How many hours could the company operate before shutting down if no additional fuel oil was attainable? How many lb mol/hr of natural gas were being consumed? The minimum heating load for the company when translated into the stack gas output was 6205 lb mol/hr of dry stack gas. Analysis of the fuels and stack gas at this time were:

Natural gas		Fuel oil (API gravity = 24.0) (Mol %)		Stack gas (Orsat analysis)	
CH_4	96%	C	50	N_2	84.93%
C_2H_2	2%	H_2	47	O_2	4.13%
CO_2	2%	S	3	CO_2	10.84%
				SO_2	0.10%

Solution

We want to calculate F and G in lb mol/hr and then F in bbl/hr.

Steps 1, 2, and 3 The molecular weight of the fuel oil is 7.91 lb/lb mol, and the density is 7.578 lb/gal.

Step 4

Basis: 1 hr, so that $P = 6205$ lb mol

Steps 5 and 6 We have five atomic components in this problem and five streams whose values are unknown, A , G , F , A^* , and W ; hence, if the elemental mole balances are independent, we can obtain a unique solution for the problem.

Step 7 The overall mole balances for the elements are

	<i>In</i>	<i>Out</i>
H ₂ :	$G(1.94) + F(0.47)$	$= W(1)$
N ₂ :	$A(0.79) + A^*(0.76)$	$= 6205(0.8493)$
O ₂ :	$A(0.21) + A^*(0.20 + 0.04)$ $+ G(0.02)$	$= 6205(0.0413 + 0.001 + 0.1084)$ $+ W(\frac{1}{2})$
S:	$F(0.03)$	$= 6205(0.001)$
C:	$G(0.96 + (2)(0.02) + 0.02)$ $+ F(0.50) + 0.04A^* = 6205(0.1084)$	

Steps 8 and 9 Solve the S balance for F (inaccuracy in the SO₂ concentration will cause some error in F , unfortunately); the sulfur is a tie component. Then solve the other four balances simultaneously for G .

$$F = 207 \text{ lb mol/hr}$$

$$G = 498 \text{ lb mol/hr}$$

Finally, the fuel oil consumption is

$$\frac{207 \text{ mol}}{\text{hr}} \left| \frac{7.91 \text{ lb}}{\text{mol}} \right| \left| \frac{\text{gal}}{7.578 \text{ lb}} \right| \left| \frac{\text{bbl}}{42 \text{ gal}} \right| = 5.14 \text{ bbl/hr}$$

If fuel oil reserve was only 1000 bbl, this amount could last

$$\frac{1000 \text{ bbl}}{5.14 \frac{\text{bbl}}{\text{hr}}} = 195 \text{ hr}$$

EXAMPLE 2.22 Countercurrent Stagewise Mass Transfer

In many commercial processes such as distillation, extraction, absorption of gases in liquids, and the like, the entering and leaving streams represent two different phases that flow in opposite directions to each other, as shown in Fig. E2.22a. (The figure could just as well be laid on its side.)

This type of operation is known as *countercurrent* operation. If equilibrium is attained between each stream at each stage in the apparatus, calculations can be carried out to relate the flow rates and concentration of products to the size and other design features of the apparatus. We shall illustrate how a material balance can be made for such type of equipment. The letter X stands for the weight concentration of solute in pounds of solute per pound of stream, solute-free. The streams are assumed immiscible as in a liquid-liquid extraction process.

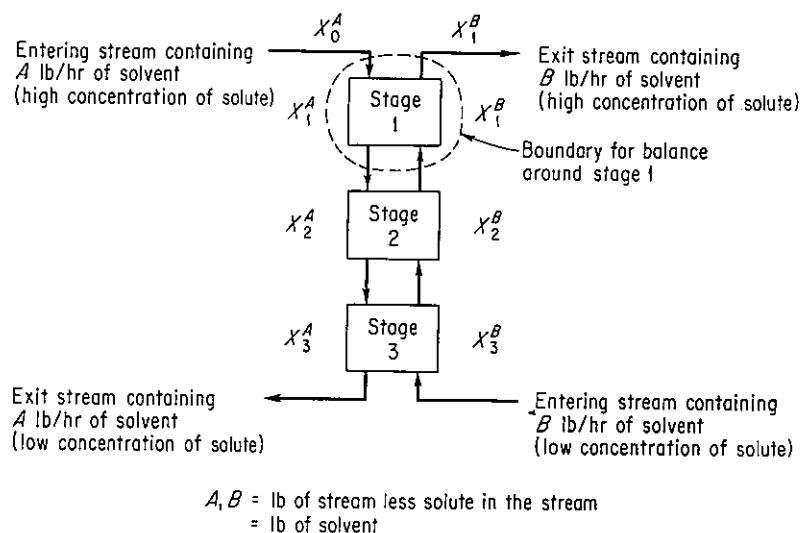


Figure E2.22a

Around stage 1 the solute material balance is (the superscript on X denotes the stream and the subscript the stage)

$$\begin{array}{c} \text{in} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_0^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_2^B \text{ lb}}{\text{lb } B} \end{array} = \begin{array}{c} \text{out} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_1^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_2^B \text{ lb}}{\text{lb } B} \end{array}$$

or

$$A(X_0^A - X_1^A) = B(X_2^B - X_3^B) \quad (\text{a})$$

Around stage 2 the solute material balance is

$$\begin{array}{c} \text{in} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_1^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_3^B \text{ lb}}{\text{lb } B} \end{array} = \begin{array}{c} \text{out} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_2^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_2^B \text{ lb}}{\text{lb } B} \end{array}$$

or

$$A(X_1^A - X_2^A) = B(X_2^B - X_3^B)$$

We could generalize that for any stage number n

$$A(X_{n-1}^A - X_n^A) = B(X_n^B - X_{n+1}^B) \quad (\text{b})$$

Also, a solute material balance could be written about the sum of stage 1 and stage 2 as follows:

$$\begin{array}{c} \text{in} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_0^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_3^B \text{ lb}}{\text{lb } B} \end{array} = \begin{array}{c} \text{out} \\ \hline \frac{A \text{ lb}}{\text{hr}} \mid \frac{X_2^A \text{ lb}}{\text{lb } A} + \frac{B \text{ lb}}{\text{hr}} \mid \frac{X_2^B \text{ lb}}{\text{lb } B} \end{array} \quad (\text{c})$$

or to generalize for an overall solute material balance between the top end and the n th stage,

$$A(X_0^A - X_n^A) = B(X_1^B - X_{n+1}^B)$$